

50^{TOP} CONCEPT

For **JEE**

INORGANIC CHEMISTRY

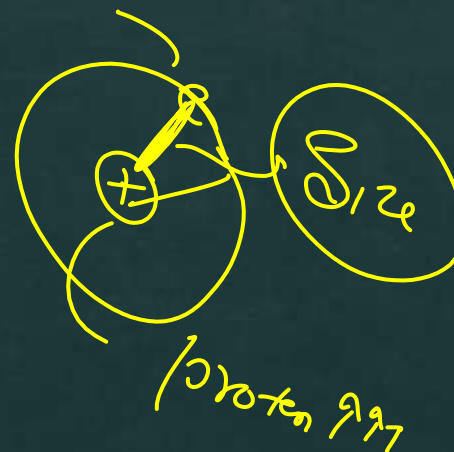
MASTER CLASS

ONE SHOT

Concept :- 1

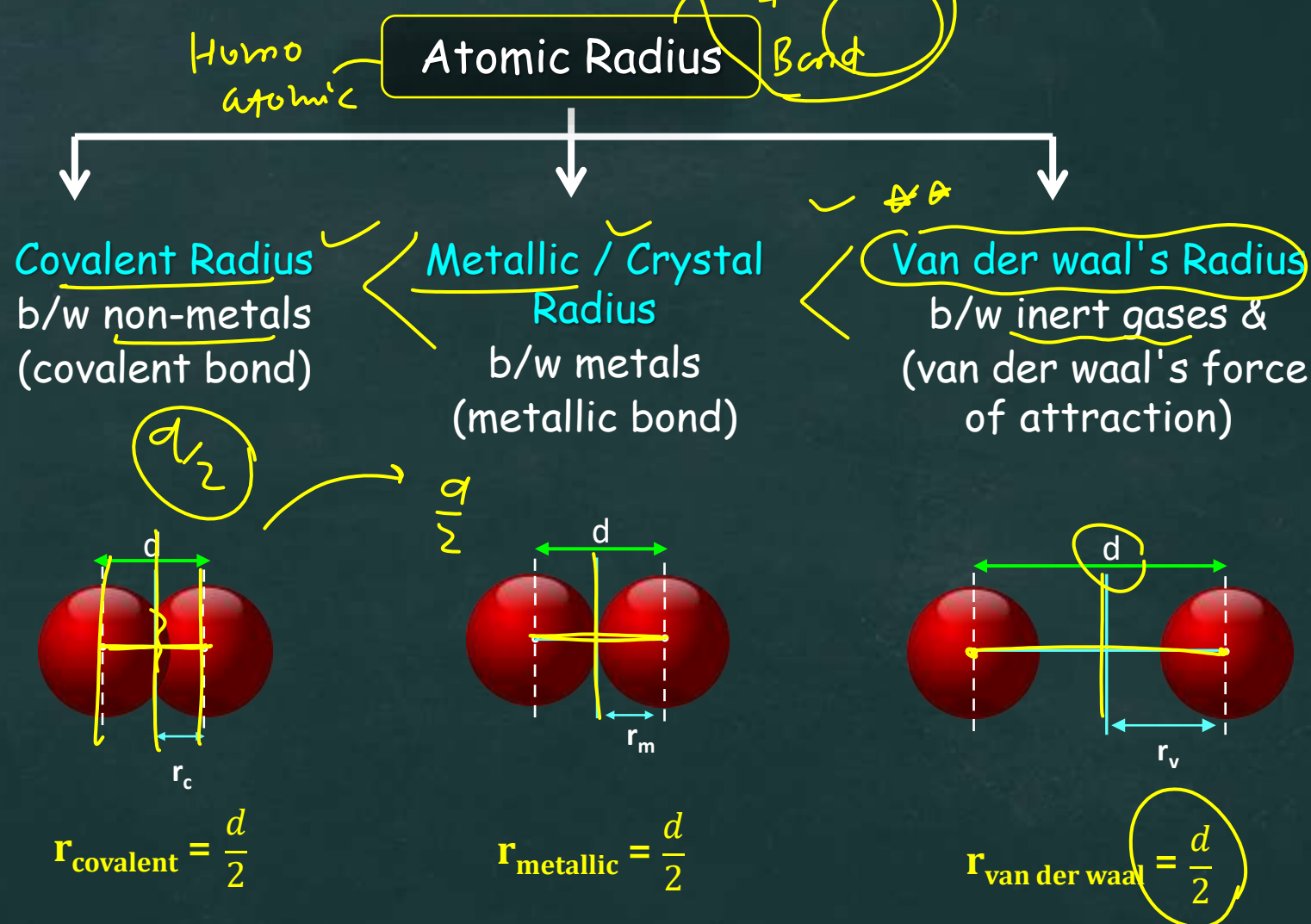
Topic Name :- Atomic radius/size

- It is the distance between the Center of nucleus to outermost shell electron where probability of finding the electron is maximum.
- Since there is a problem in calculating actual size of atom, three types of radii can be defined :



Concept :- 1

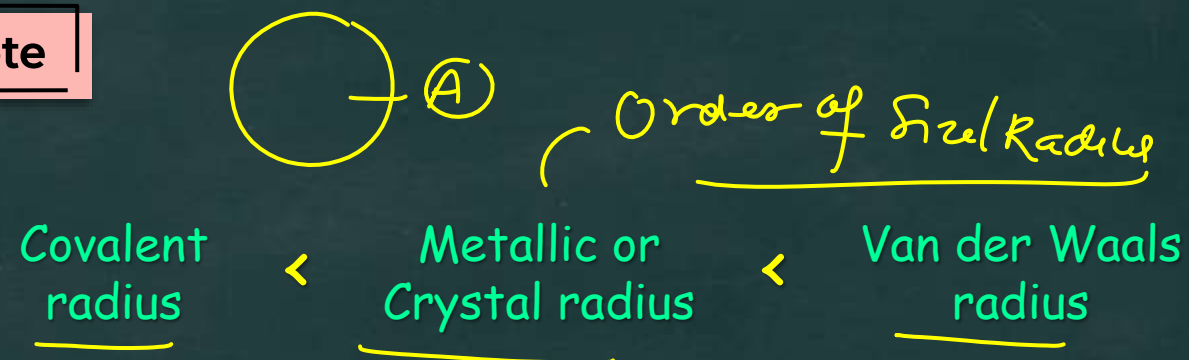
Topic Name :- Atomic radius/size



Concept :- 1

Topic Name :- Atomic radius/size

Note



Concept :- 1

Topic Name :- Atomic radius/size

Periodic Trends of Atomic Radius

(1) Across the period : 2nd Li ————— Ne

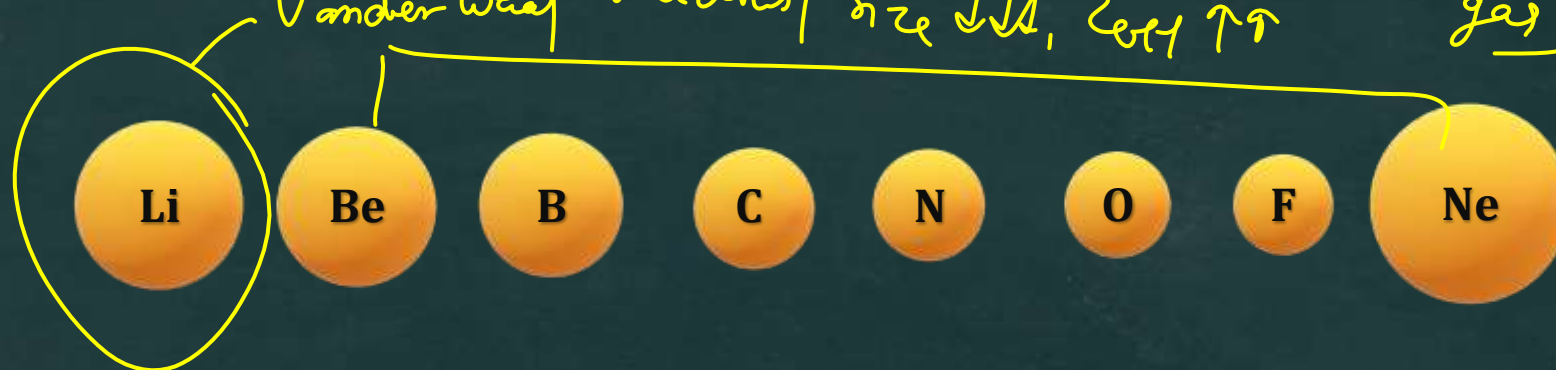
$\rightarrow n \rightarrow \text{Constant}, \text{position} \uparrow \uparrow, Z_{\text{eff}} \uparrow \uparrow$

Generally, Atomic radius decreases from left to right in a period as

Z_{eff} increases.

in alkali metal $r_{\text{Li}} = \text{Noble gas}$

Vander Waal's radius / $r_{\text{Li}} \downarrow \downarrow, Z_{\text{eff}} \uparrow \uparrow$



Concept :- 1

Topic Name :- Atomic radius/size

For p- block elements

due to d-contraction

B < Ga < Al < In < Tl (d-contraction)

C < Si < Ge < Sn < Pb

N < P < As < Sb < Bi

O < S < Se < Te < Po

F < Cl < Br < I

He < Ne < Ar < Kr < Xe < Rn

$\uparrow$, Size $\uparrow$

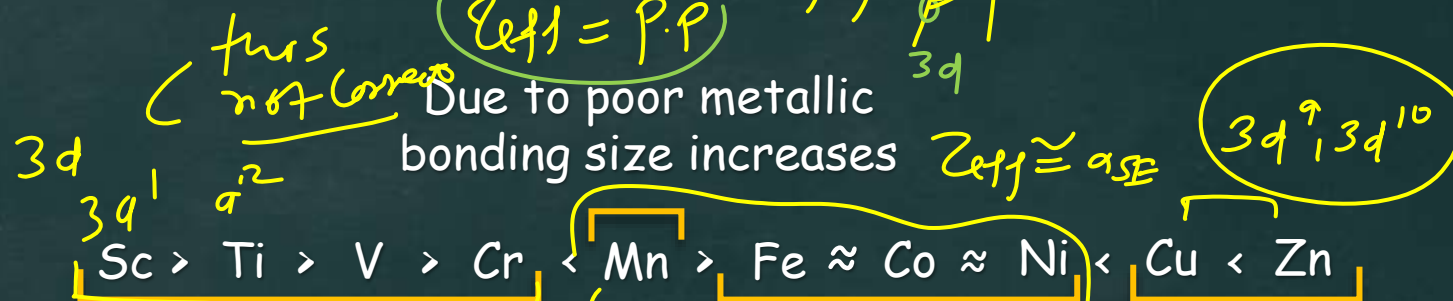
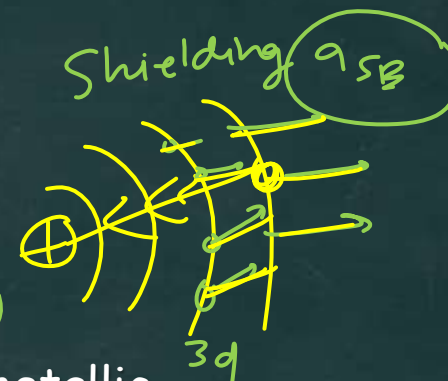
3d

B
Al
Ga
In
Tl
↓
Size $\uparrow$

Concept :- 1

Topic Name :- Atomic radius/size

d-block 3d-series



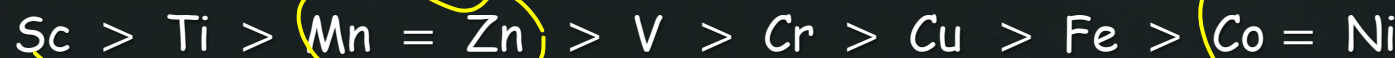
Z_{eff} dominates

Z and σ
balance
each other

σ dominates

Order of a_p $\uparrow$ Atomic Radius $\downarrow$
d-block

$Z_{eff} \downarrow$ Atomic Radius $\uparrow$



largest in 3d

Smallest size

Concept :- 1

Topic Name :- Atomic radius/size

Group $\otimes$ $Sc < Y < La$

$\checkmark$ $Kb = Ta$
 $Cu < Ag \approx Au$
 $Zn < Cd < Hg$

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
144	132	122	119	119	117	116	115	117	125
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
162	145	134	129	125	124	125	128	134	141
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
169	144	134	130	128	126	126	129	134	144

Atomic radii of d-block elements (in pm)

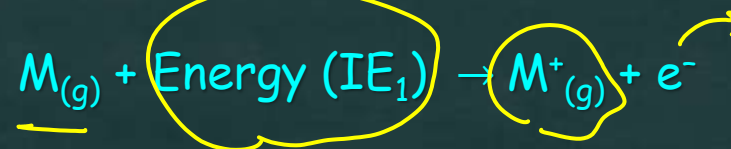
$3d < 4d < 5d$

Concept :- 2

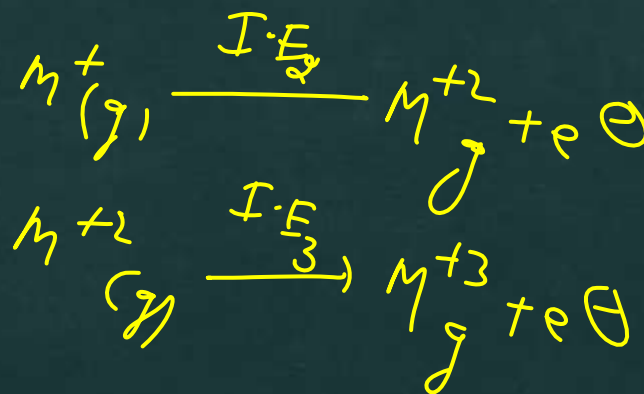
Topic Name :- Ionization Energy

•  Minimum amount of energy required to remove the most loosely bonded electron from an isolated gaseous atom from its ground state.

•  Units : kJ mol^{-1} , kcal mol^{-1} , eV per atom .

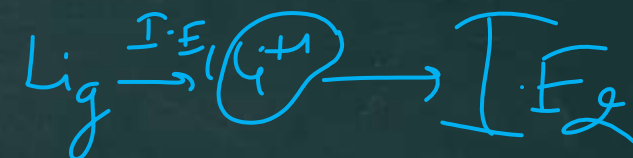


$\Delta H = (+)$ positive
Endo

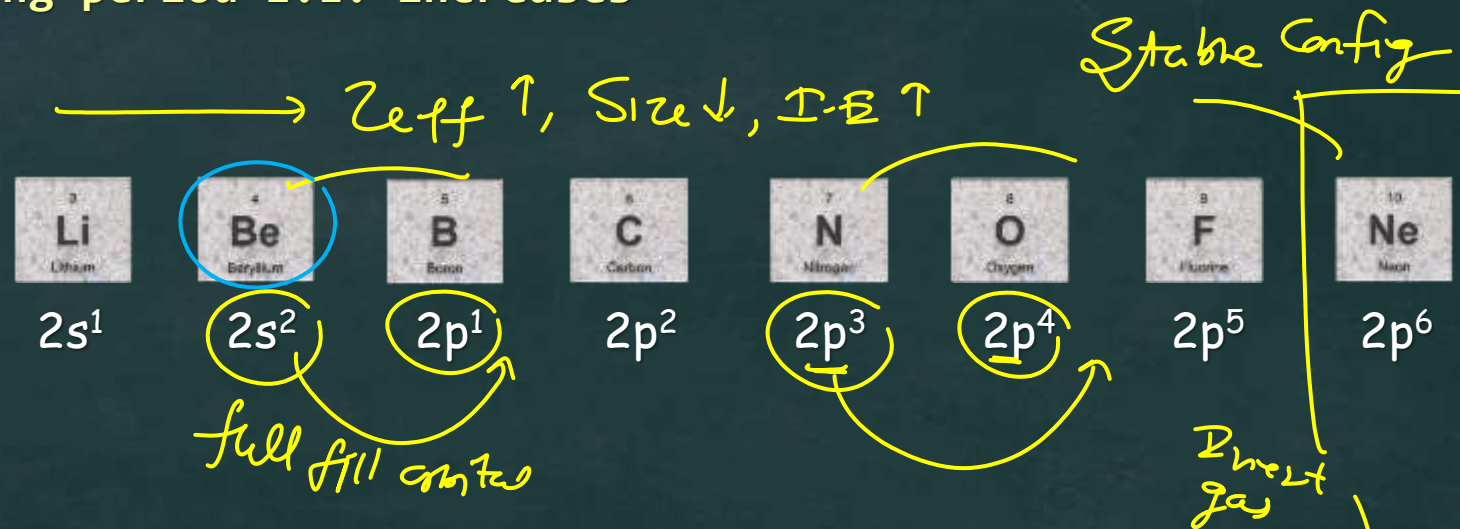


Concept :- 2

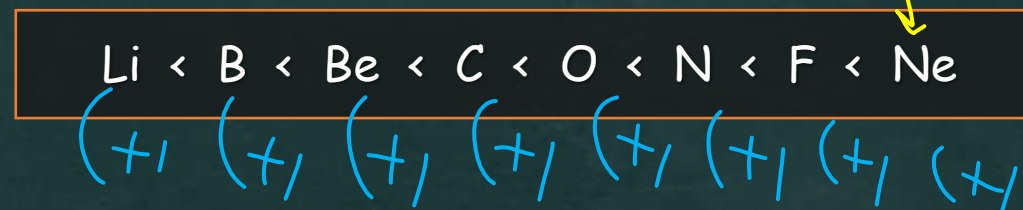
Topic Name :- Ionization Energy



Along period I.E. increases



2nd period IE_1 order :



2nd period IE_2 order :



Concept :- 2

Topic Name :- Ionization Energy

3rd period IE_1 order :

Na < Al < Mg < Si < S < P < Cl < Ar

(+1) (+1)

3rd period IE_2 order : ✓

Concept :- 2

Topic Name :- Ionization Energy

Along group I.E. Decreases ($n \uparrow$)

For s- block elements (IE_1 Order)

✓ $Li > Na > K > Rb > Cs$

✓ $Be > Mg > Ca > Sr > Ba$

Li
 Na
 K
 Rb
 Cs

Si, Ca, π
 $I.E. \downarrow$

Concept :- 2

Topic Name :- Ionization Energy

~~Size~~ $B < Ga < Al < In < Tl$
~~IE~~ $B > Ga > Al > In > Tl$
 $C > Si > Ge > Sn > Pb$

For p- block elements (IE_1 Order)

$B > Tl > Ga > Al > In$

$C > Si > Ge > Pb > Sn$
 zigzag

$IE \propto 1/r_{\text{size}}$
 $N > P > As > Sb > Bi$

$O > S > Se > Te > Po$

$F > Cl > Br > I > At$

$He > Ne > Ar > Kr > Xe > Rn$

Concept :- 2

Topic Name :- Ionization Energy

IE₁ order for d-block elements

Zn > Fe > Co > Cu > Ni > Mn > Ti > Cr > V > Sc

(मन) फिर कोई मध्य नहीं माना रीयर क्लैमी वो।

Science
Tutor

Concept :- 2

Topic Name :- Ionization Energy

From top to bottom in a group:

3d-series	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	∨	^	^	^	∨	∨	∨	^	∨	∨
4d-series	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
	∨	∨	^	^	^	^	^	^	^	^
5d-series	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
	← 4f series →									

f-contraction or Lanthanoid Contraction

3d series < 4d series < 5d series

For Group 4, 5, 6 & 10

4d series < 3d series < 5d series

For Group 7, 8, 9, 11 & 12

Concept :- 3

Topic Name :- Application of Ionization Energy

Metallic character



Ionization Energy $\uparrow$, Metallic character $\downarrow$ $P.E \text{ min}$

Along the period : Ionization Energy $\uparrow$ Metallic character $\downarrow$

Li > Be > B > C > N > O > F

metal

non-metal

Down the group : Ionization Energy $\downarrow$ Metallic character $\uparrow$

Li < Na < K < Rb < Cs

Concept :- 3

Topic Name :- Application of Ionization Energy

Reducing nature



Reducing character of metal $\propto \frac{1}{\text{I.E.}}$ $\uparrow$
(except Li metal)

V.I. $\uparrow$
 $\odot$ 1st group has minimum I.E, so they are strong reducing agent in aqueous medium

~~Ne~~ < Na < K < Rb < Cs < Li Red Agent

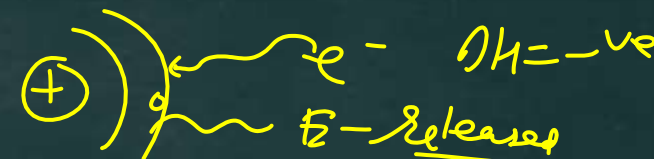
$\odot$ 17th group has maximum I.E, so they are strong oxidizing agent

Concept :- 4

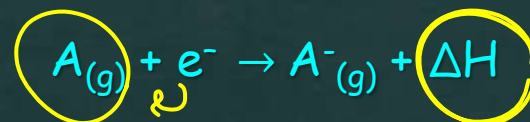
Topic Name :- Electron Gain Enthalpy

Electron Affinity (EA) & Electron Gain Enthalpy (ΔH_{eg})

Electron Gain Enthalpy (ΔH_{eg})



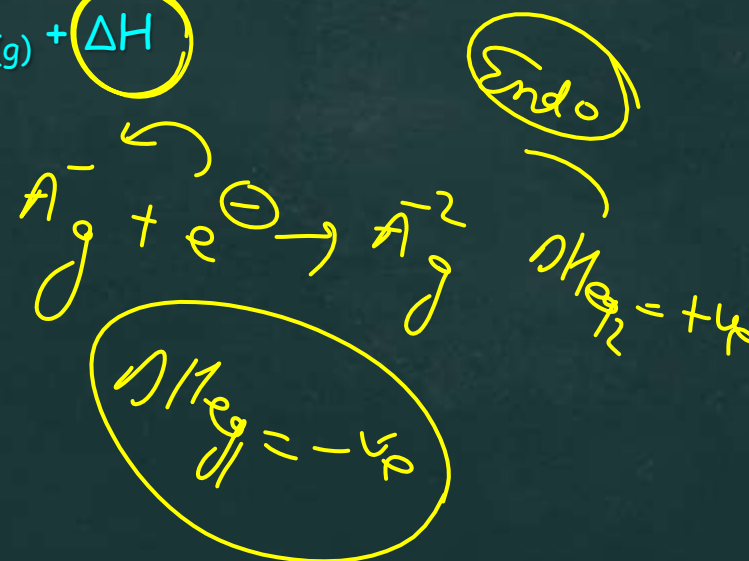
It is the enthalpy change when one mole of electrons are added to one mole of isolated gaseous atoms to convert them into one mole mono-negative gaseous ions.



Note

In general, ΔH_{eg1} = exothermic

$\Delta H_{eg2}, \Delta H_{eg3}$ = endothermic



Concept :- 4

Topic Name :- Electron Gain Enthalpy

2nd, 3rd

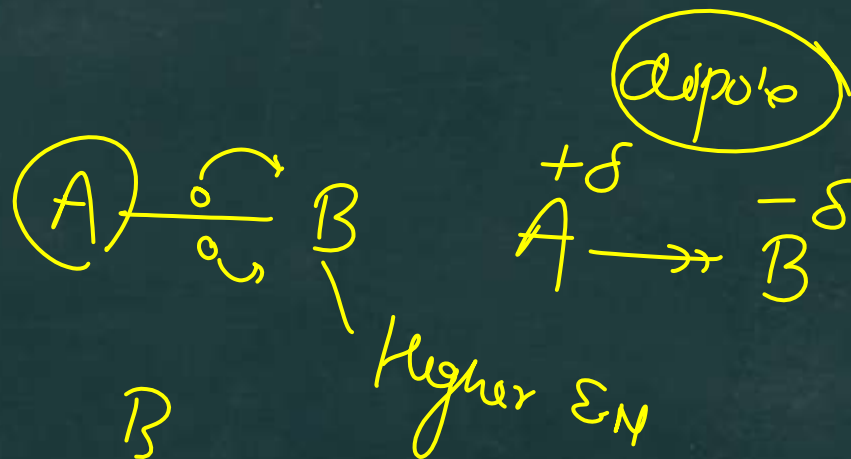
Electron gain enthalpy values for some element (kJ mol ⁻¹)							
H							He
-72							+48
Li	Be	B	C	N	O	F	Ne
-57	+66	-15	-121	+31	-142	-333	+116
Na	Mg	Al	Si	P	S	Cl	Ar
-53	+67	-26	-135	-60	-200	-348	+96
K				As	Se	Br	Kr
-48				-77	-195	-325	+96
Rb				Sb	Te	I	Xe
-47				-101	-190	-295	+77
Cs				Bi	Po	At	Rn
-46				-97	-174	-270	+68

Concept :- 5

Topic Name :- Electronegativity

"Bond property"

• It is the tendency of an element to attract shared pair of electrons towards itself in a covalently bonded molecule.



Concept :- 5

Topic Name :- Electronegativity

General Trends

L - R $\epsilon.n \uparrow$

Top
↓ $\epsilon.n \downarrow$
down

Along a period, electronegativity increases $\uparrow$

Along a group, electronegativity decreases $\downarrow$

Note

$F > Cl$

3rd 2nd
Neg $Cl > F$

Electronegativity of $F > Cl$

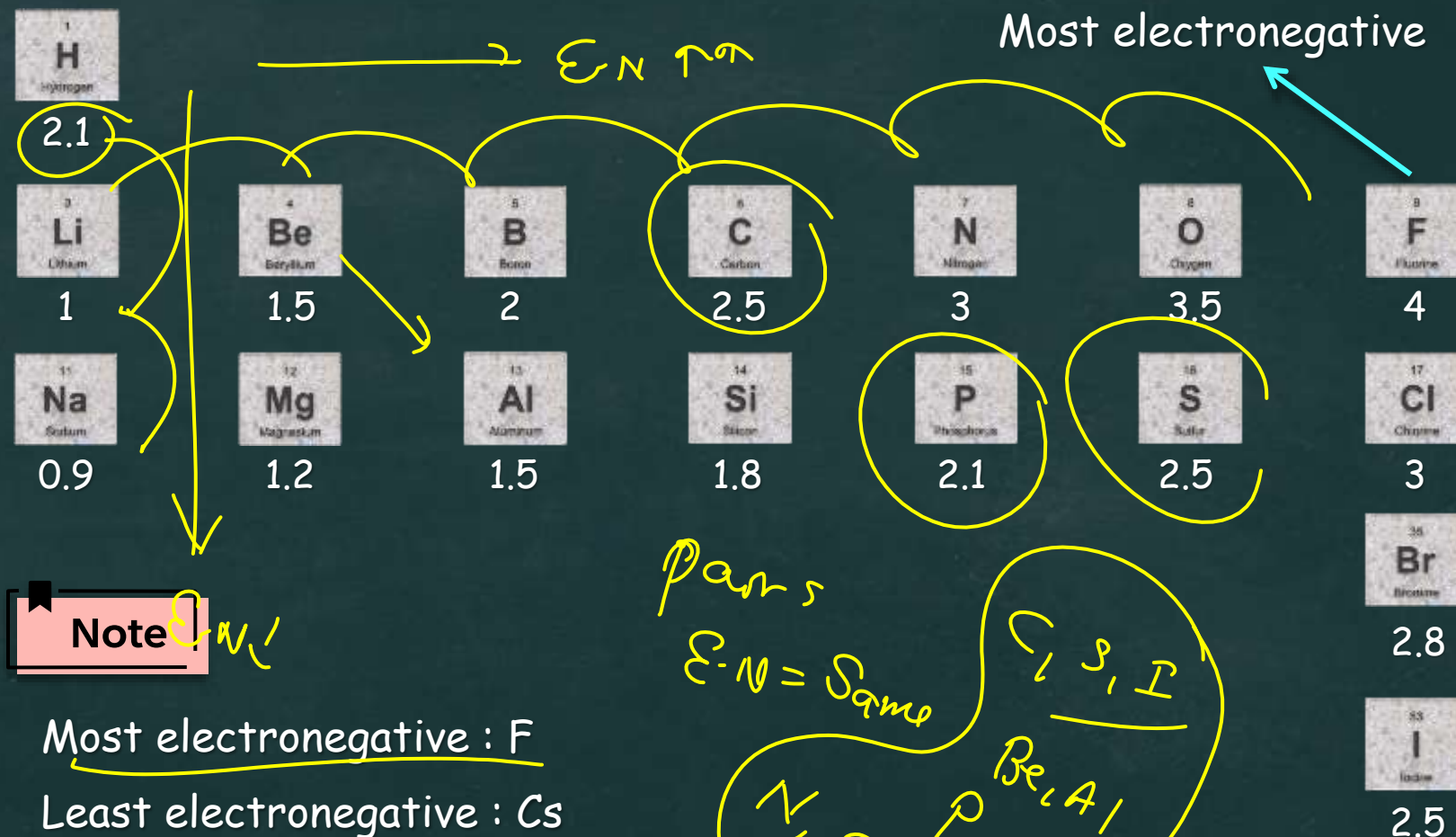
but

Electron affinity of $Cl > F$

Concept :- 5

Topic Name :- Electronegativity

Electronegativity values of some elements according to Pauling Scale



Concept :- 5

Topic Name :- Electronegativity

Important Order

Order of χ

B > Tl > In > Ga > Al

C > Pb > Si = Ge = Sn

N > P > As > Sb = Bi

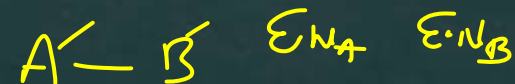
O > S > Se > Te > Po

F > Cl > Br > I

B / Al Ga In Tl
B > Tl > In > Ga > Al

Concept :- 6

Topic Name :- % Ionic Character



$$\% \text{ Ionic character} = 16 | E.N_A - E.N_B | + 3.5 (E.N_A - E.N_B)^2$$

Example

Calculate % ionic character in HCl ($E.N_{Cl} = 3$, $E.N_H = 2.1$)

$$\% I.C. = \frac{16 | 0.9 | + 3.5 (0.9)^2}{11}$$

Concept :- 8

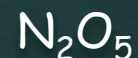
Topic Name :- Types of Oxides

most
imp

Acidic Oxide

• In general, non-metal oxides are acidic in nature.

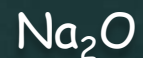
Example



Basic Oxide

• In general, metallic oxides are basic in nature.

Example



Group 1 / 2
Basic

Concept :- 8

Topic Name :- Types of Oxides

Amphoteric Oxide

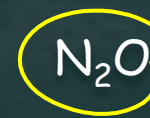
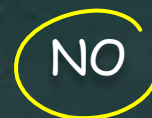
Zn	—	ZnO	Show Rxn With Acid Answer AS Base
Be	—	BeO	
Al	—	Al ₂ O ₃	
Ga	—	Ga ₂ O ₃ , Ga ₂ O	V. Imp
Pb	—	PbO, PbO ₂	
Sn	—	SnO, SnO ₂	
✓ As(III)	—	As ₂ O ₃	
✓ Sb(III)	—	Sb ₂ O ₃	
✓ Cr(III)	—	Cr ₂ O ₃	
✓ Mn(IV)	—	MnO ₂	
✓ V(V)	—	V ₂ O ₅	

Concept :- 8

Topic Name :- Types of Oxides

Neutral Oxide

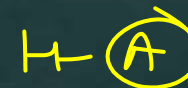
Example



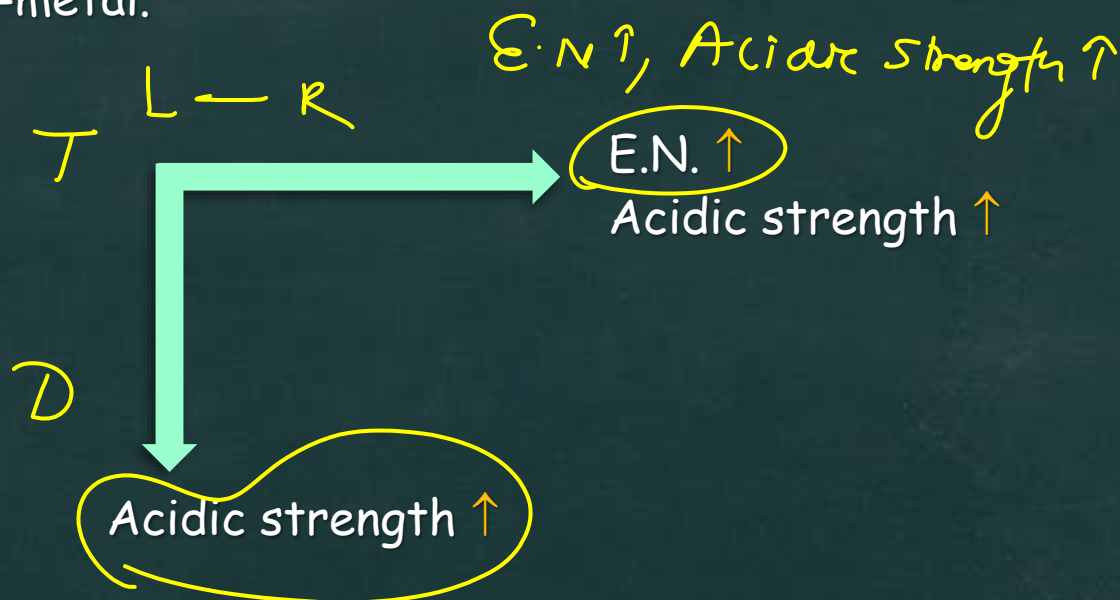
Concept :- 9

Topic Name :- Hydra Acids

Acidic Strength of Hydra Acids



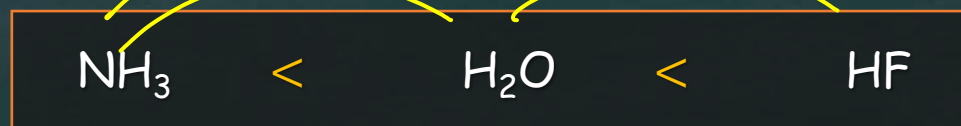
A hydra acid (HA) is a compound which contains hydrogen and a non-metal.



Concept :- 9

Topic Name :- Hydra Acids

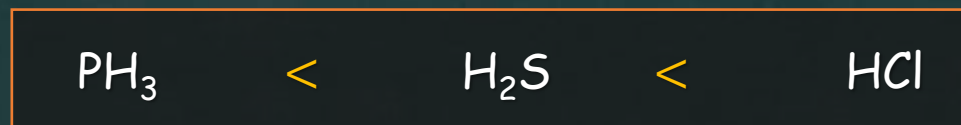
Hydra acid strength order $\rightarrow$ EN of element on



^

^

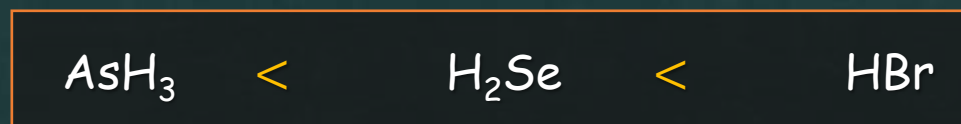
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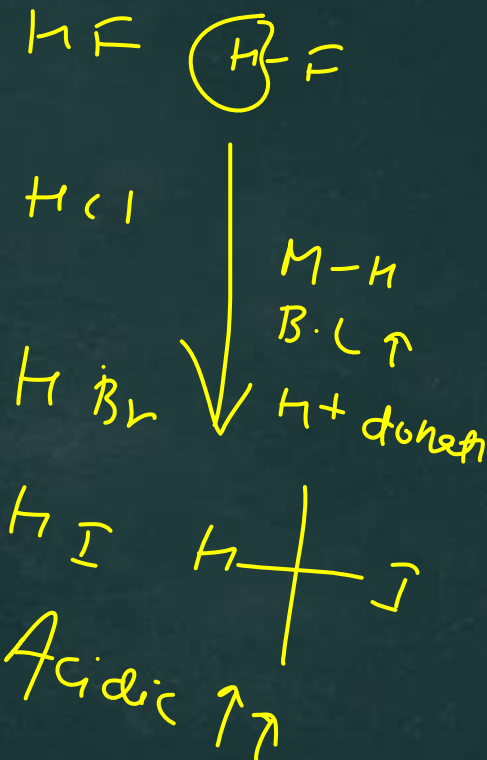
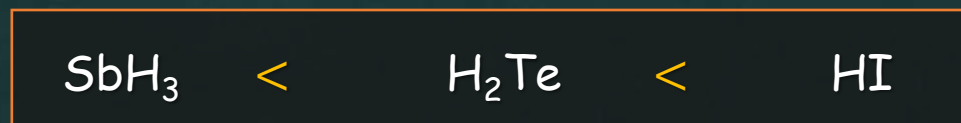
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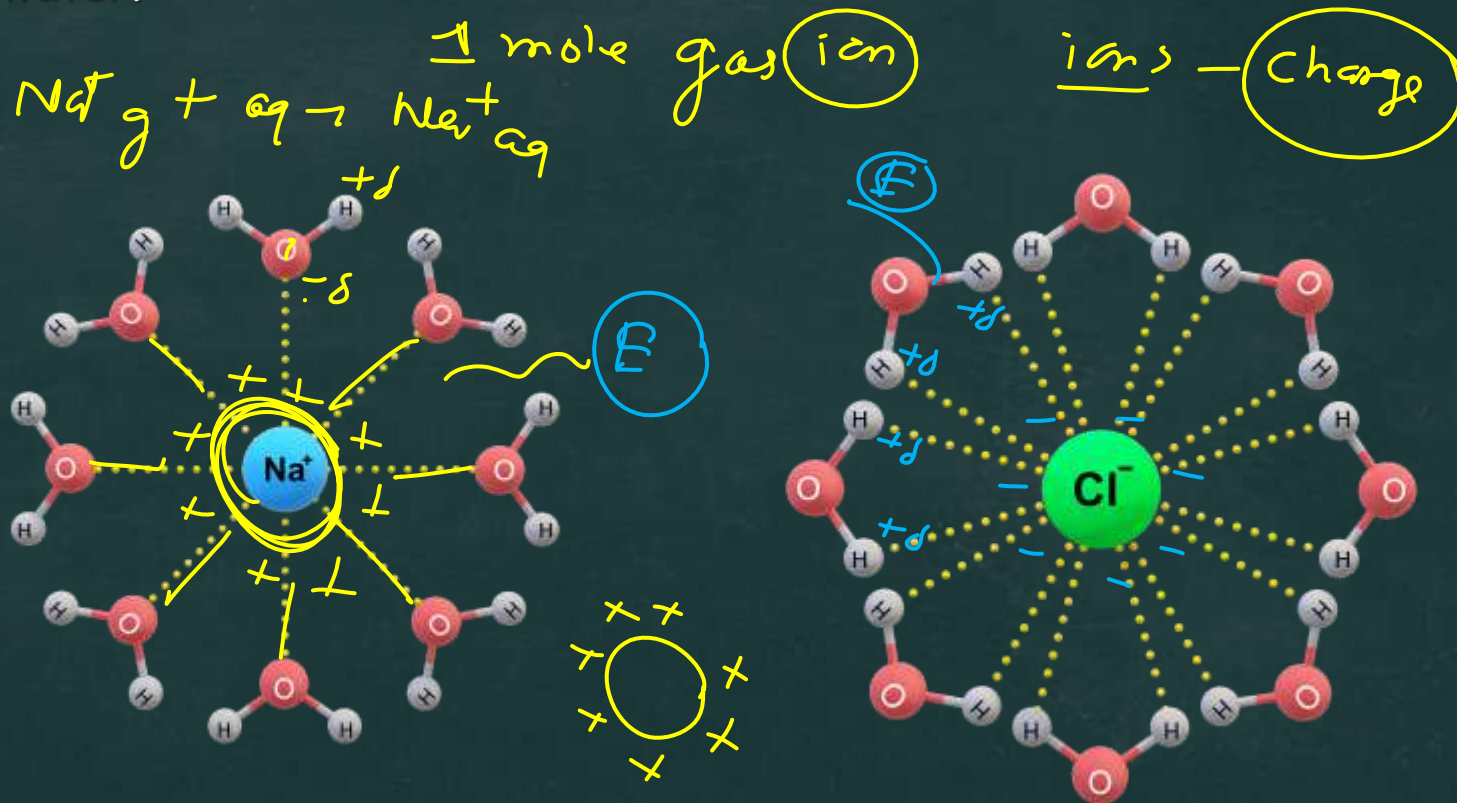


Concept :- 10

Topic Name :- Hydration Energy

-ve

It is the amount of energy released when 1 mol of gaseous ions interact with water.



Concept :- 10

Topic Name :- Hydration Energy

Factors on which hydration energy depends

Radius
Hydrated Radius

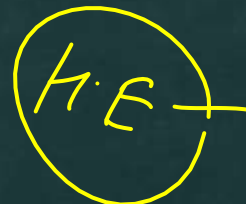
Hydration Energy $\propto$ charge on ions (q)

Charge $\uparrow\uparrow$
more hydration

Hydration Energy $\propto \frac{1}{\text{Size of ion}(r)}$



Hydration Energy $\propto \left(\frac{q}{r}\right)$ (Charge density)

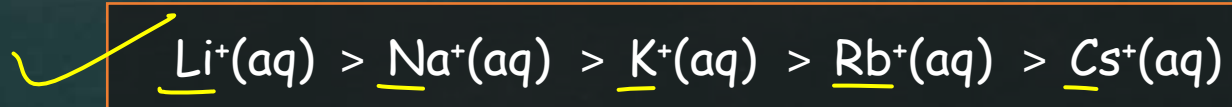


more charge
less size

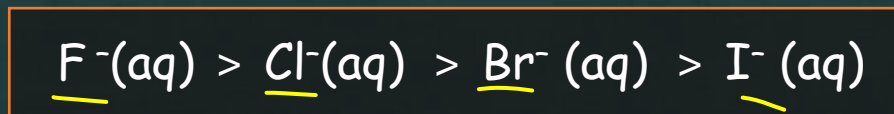
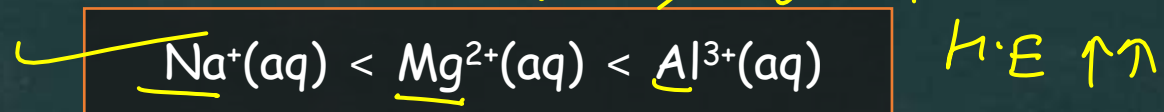
Concept :- 10

Topic Name :- Hydration Energy

hydration energy order



———— charge ↑↑, hydration ↑↑

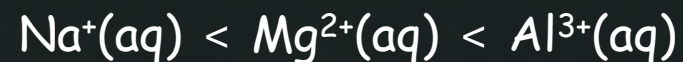
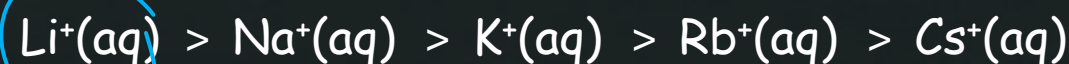


Concept :- 11

Topic Name :- Hydrated Radius

It is the distance from the nucleus of a particular ion to that particular point up to which the associated water molecules are immobile due to its attraction

Hydrated Radius Order :-



Concept :- 12

Topic Name :- Lattice Energy

Ionic Comp / Bond

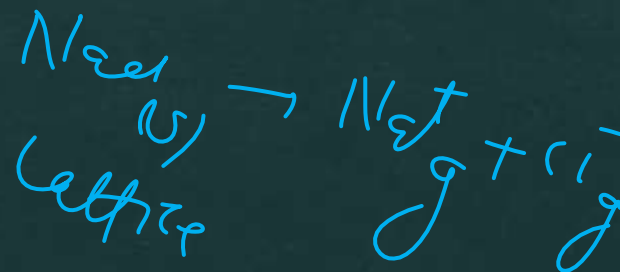
- It is the amount of energy release during the formation of 1 mole of lattice from their respective constituents ions in gaseous state.

NaCl

Or



- It is the amount of energy required to break one mole of lattice into their respective constituent gaseous ions.



Concept :- 12

Topic Name :- Lattice Energy

Factors on which Lattice energy depends

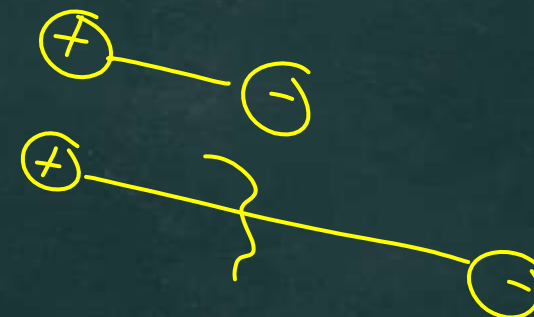
$$L.E \propto q_1 q_2$$

Lattice Energy $\propto$ product of charge on ions ($q_1 \times q_2$)

Lattice Energy $\propto \frac{1}{\text{inter nuclear distance (r)}}$

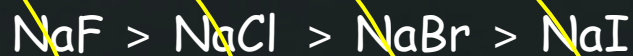
$$\text{Lattice Energy} \propto \frac{q_1 \times q_2}{r}$$

where $r = r_c + r_a$



Concept :- 12

Topic Name :- Lattice Energy



→ $r \uparrow \uparrow$, $LE \downarrow \downarrow$



→ Charge $\uparrow$, $LE \uparrow \uparrow$



→ Size $\uparrow$, $LE \downarrow$

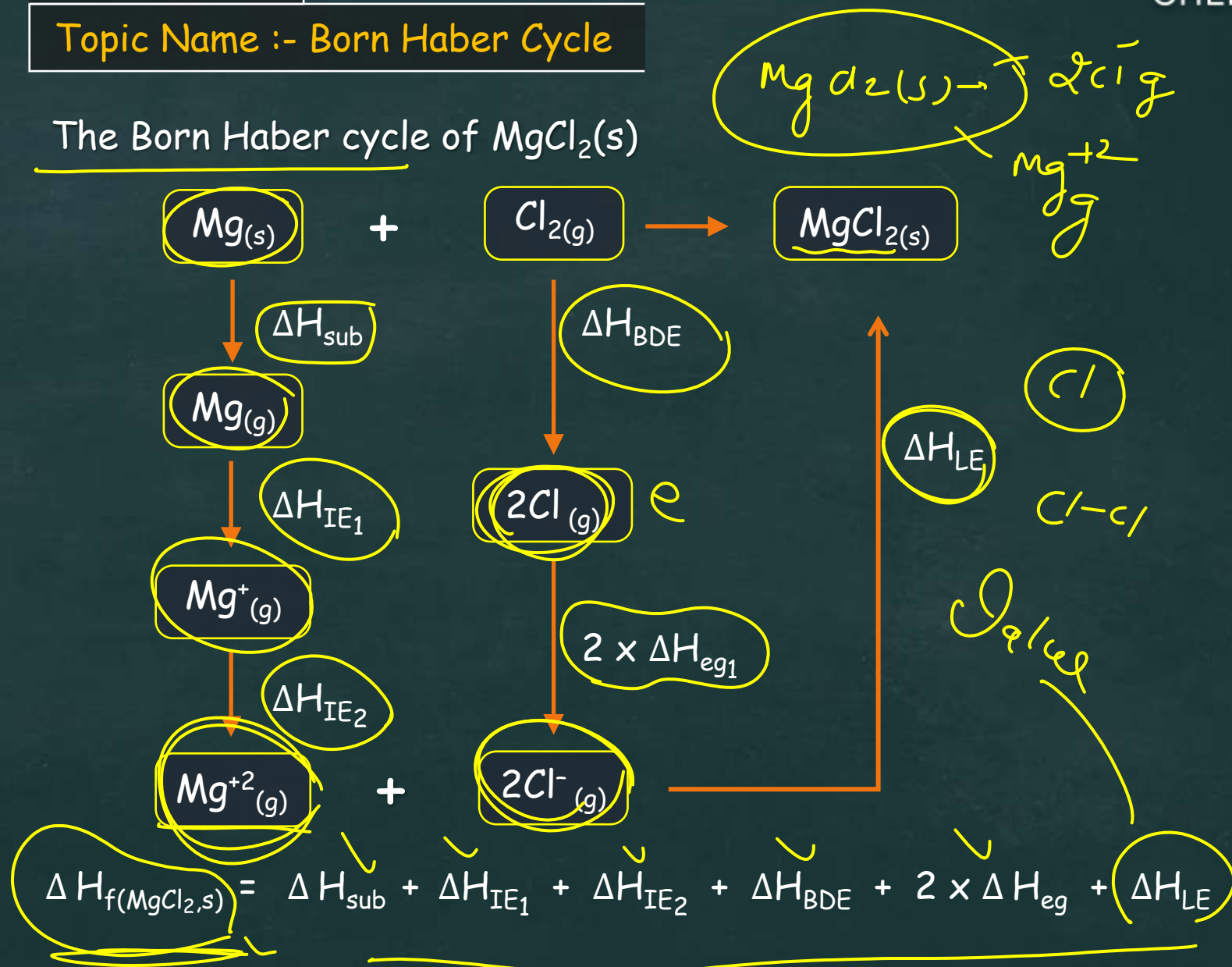


$LE \propto \frac{\text{charge}}{r_{\text{radius}}}$

Concept :- 13

Topic Name :- Born Haber Cycle

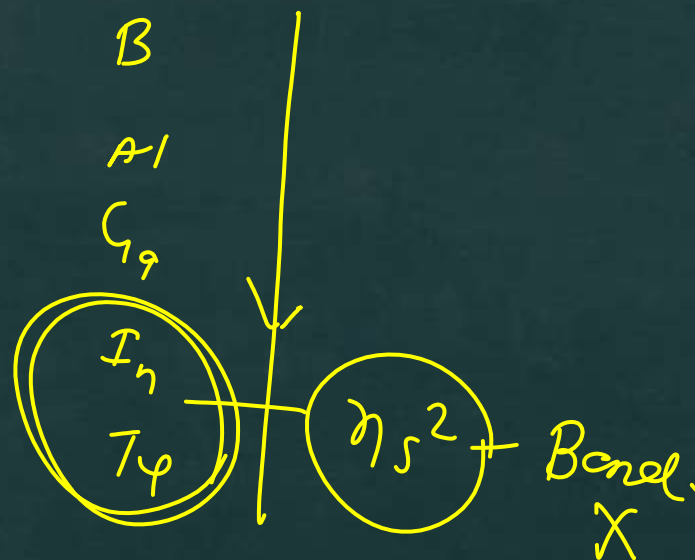
The Born Haber cycle of $\text{MgCl}_2(\text{s})$



Concept :- 14

Topic Name :- Inert pair effect

• This effect is caused due to inertness of outermost s-electrons (ns^2) of heavier p-block elements.



Concept :- 14

Topic Name :- Inert pair effect

	Group-13	Group-14	Group-15
	ns^2np^1 +3	ns^2np^2 +4	ns^2np^3 +5
(group oxidation state)			
	B +3	C +4	N +5
	Al +3	Si +4	P +5
	Ga +3 > +1	Ge +4 > +2	As +5 > +3
	In +3 > +1	Sn +4 > +2	Sb +5 > +3
	Tl +3 < +1	Pb +4 < +2	Bi +5 < +3

Handwritten notes:

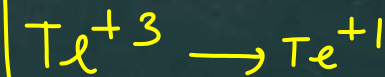
- Higher O.S. (pointing to B, C, N)
- ns² - inert (pointing to Ga, Ge, As)
- Lower O.S. - stable (pointing to Tl, Pb, Bi)

For the last element group oxidation state is highly oxidising in nature.

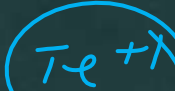
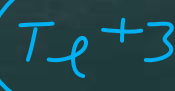
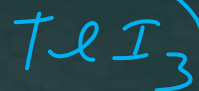
Concept :- 14

Topic Name :- Inert pair effect

Problems on inert pair effect

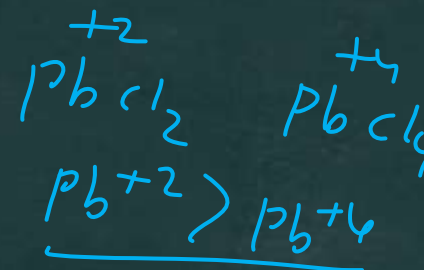


Example

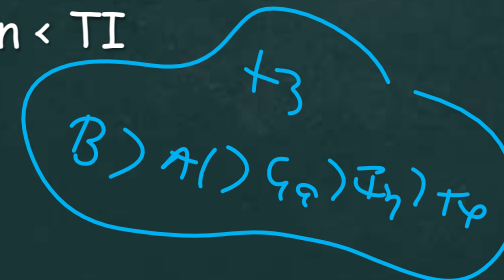
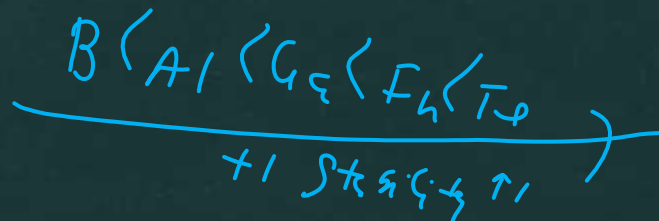


(1) Thallium (III) iodide does not exist.

(2) PbCl₂ is more stable than PbCl₄.



(3) The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence : Al < Ga < In < Tl



Concept :- 15

Topic Name :- Lewis Octet Rule

Atoms can combine either by transfer of valence electrons from one atom to another (gaining or losing) or by sharing of valence electrons in order to have an octet in their valence shells.

Limitations of Lewis Octet rule

▪ Hypervalent : PCl_5 , IF_7

▪ Hypovalent : BF_3 , BeCl_2 (Stoner)

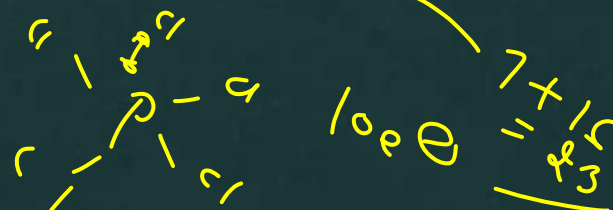
▪ Odd electron species : NO_2 , ClO_2

H
He 7 deplet

Be, B, C, N, O, F
Octet

Si, P, S, Cl

Octet Extend



Concept :- 16

Topic Name :- Type of Bond

Chemical
Bond

Strong Bond
($E_{rel} > 42 \text{ kJ/mol}$)

Ionic
Bond

Covalent
Bond

Metallic
Bond

Co-ordinate
Bond

Covalent + Ionic

Weak Bond
($E_{rel} < 42 \text{ kJ/mol}$)

$E_R < 42$

Hydrogen Bond
($8 < E_{rel} < 42 \text{ kJ/mol}$)

$8 < E_R < 42$

Venp

Weak Force
($E_{rel} < 8 \text{ kJ/mol}$)

$E_r < 8 \text{ kJ/mol}$

Inter-
molecular

Intra-
molecular

Vander
Waal's Forces

Ion-Dipole

Ion-Induced
Dipole

Dipole-Dipole

Dipole-Induced
Dipole

Instantaneous
Dipole- Induced
Dipole

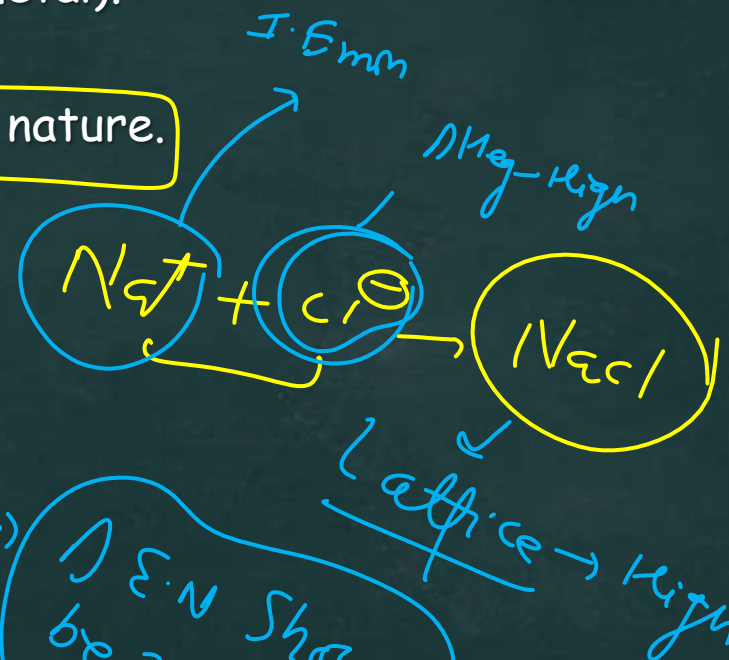
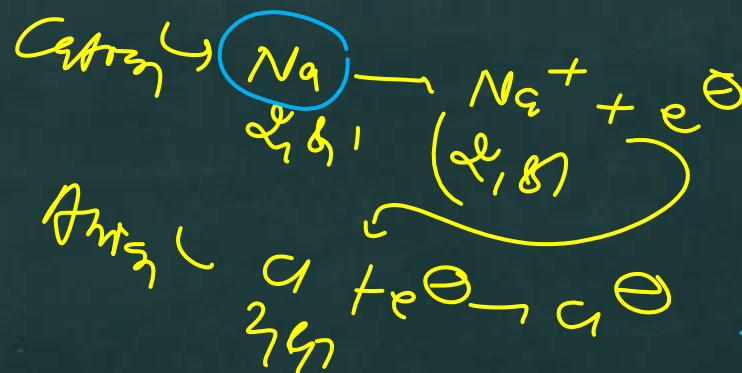
Concept :- 17

Topic Name :- Ionic Bond

Answer
Cation

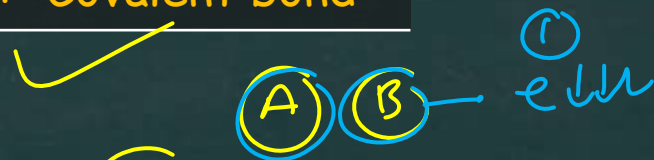
It is the strong electrostatic force of attraction between two opposite ions as a result of complete transfer of electron(s) from one element (less electronegative or metal) to another element (more electronegative or non-metal).

Ionic bond is non-directional in nature.

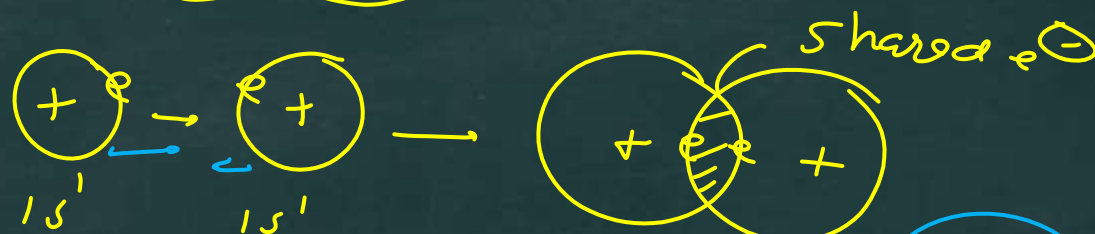


Concept :- 18

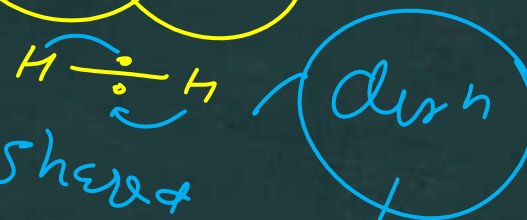
Topic Name :- Covalent bond



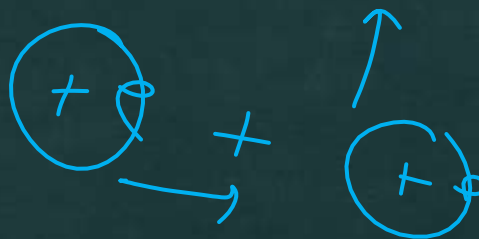
Bond formed by mutual sharing of electrons between two atoms.



Equal no of e^- shared



Isomerism

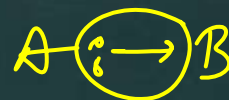


Concept :- 19

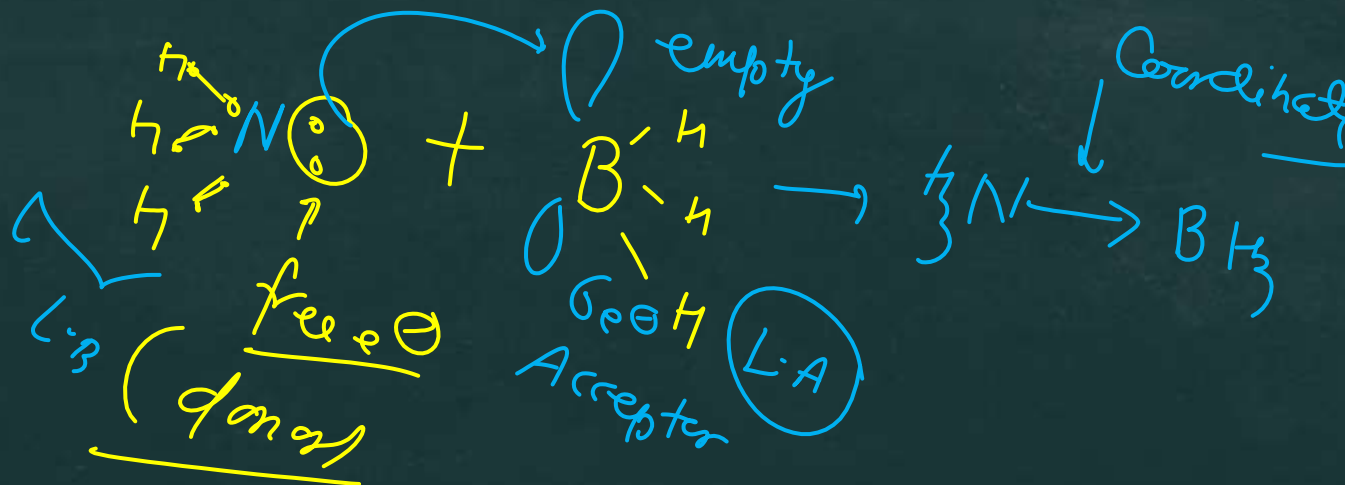
Topic Name :- Co-ordinate Bond / Dative Bond

• It is an interaction between ^{Poor} Lewis acid and ^{Rich} Lewis base.

• It is a special kind of covalent bond. $\rightarrow$ Unequal sharing of e^-



• The bond formed between two atoms in which contribution of an electron pair is made by one of them while sharing is done by both.



Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

- According to valence bond theory, as many unpaired electron(s) is/are present either in their ground state or in the excited state, the atoms will form the same number of covalent bonds.

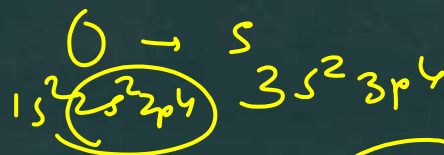
unpaired e^- → Covalent Bond

- If lone pair is present on the central atom it can create a co-ordinate bond that depends on circumstances.

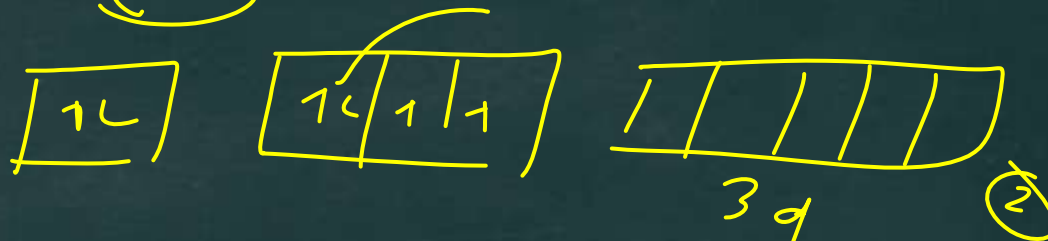
Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

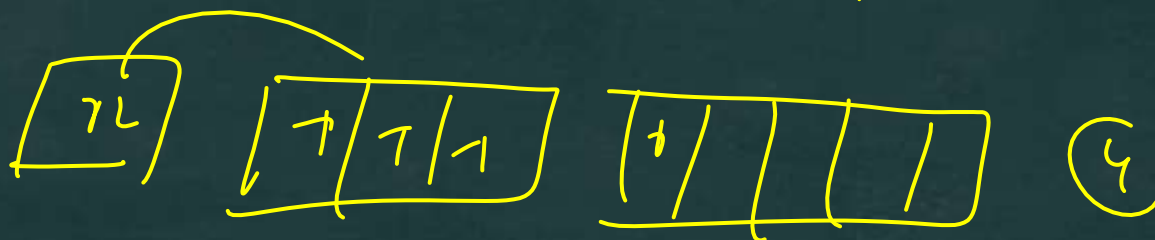
Example



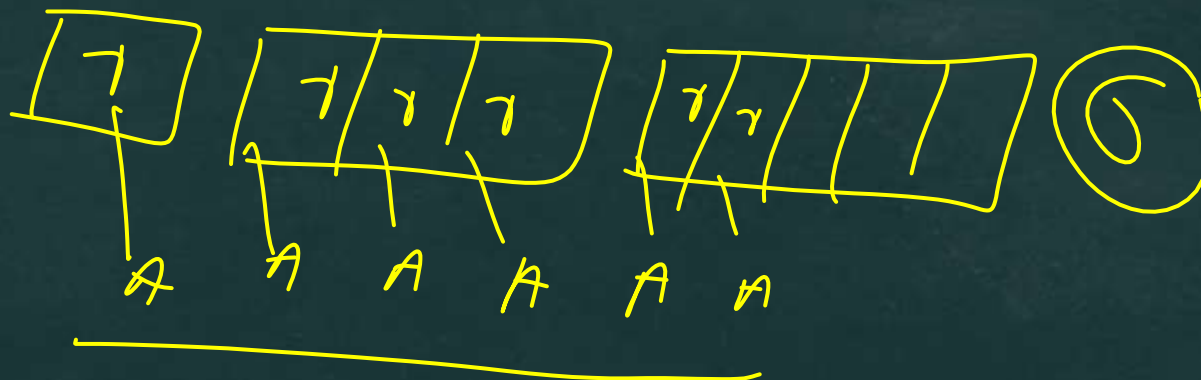
S:



S*:

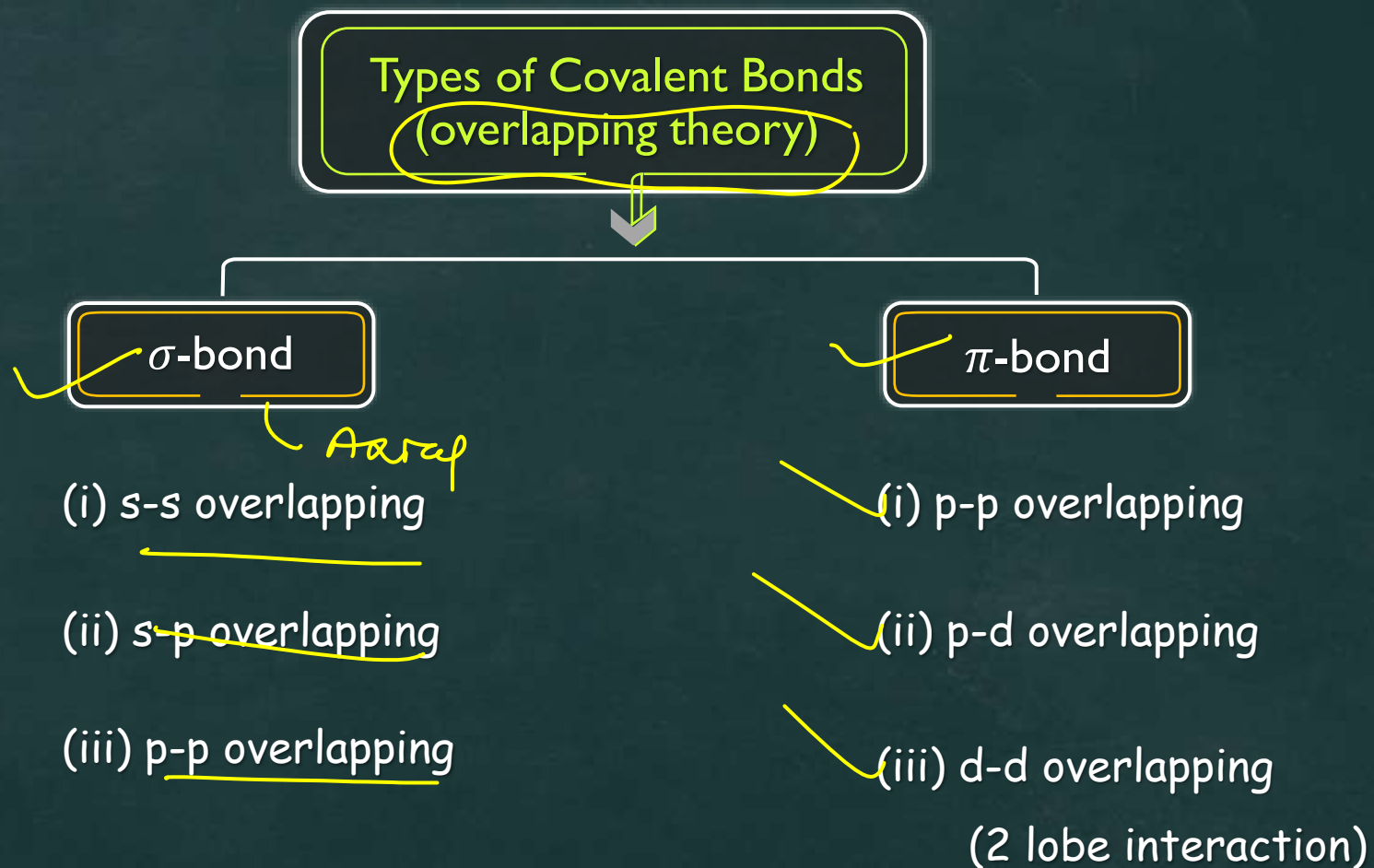


S**:



Concept :- 20

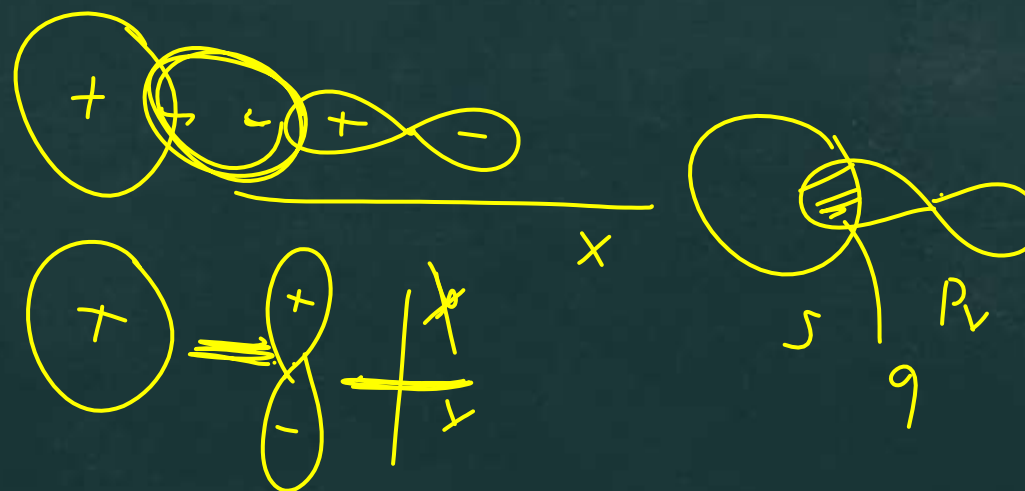
Topic Name :- Valence Bond Theory (VBT)



Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

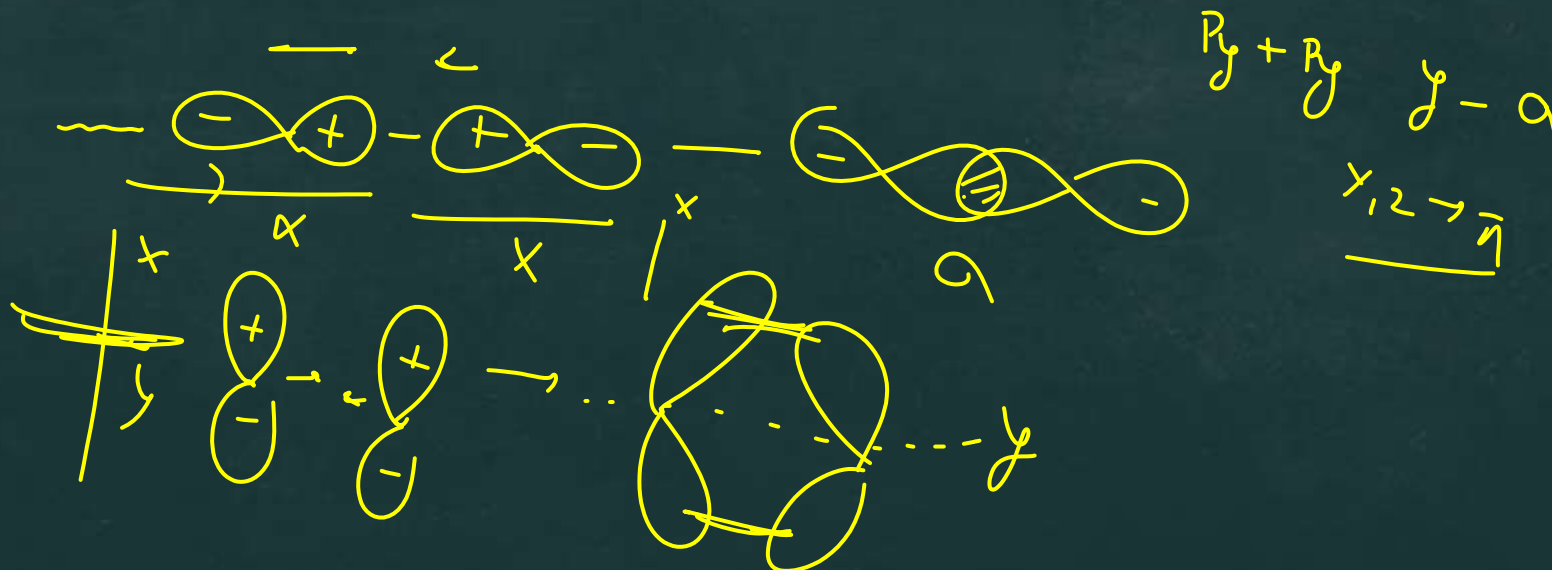
Atomic Orbitals	Internuclear axis	Types of Covalent Bond
$s + p_y \rightarrow y \text{ axis}$ $s + p_x$	x - axis	σ
	y - axis	x
	z - axis	x



Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

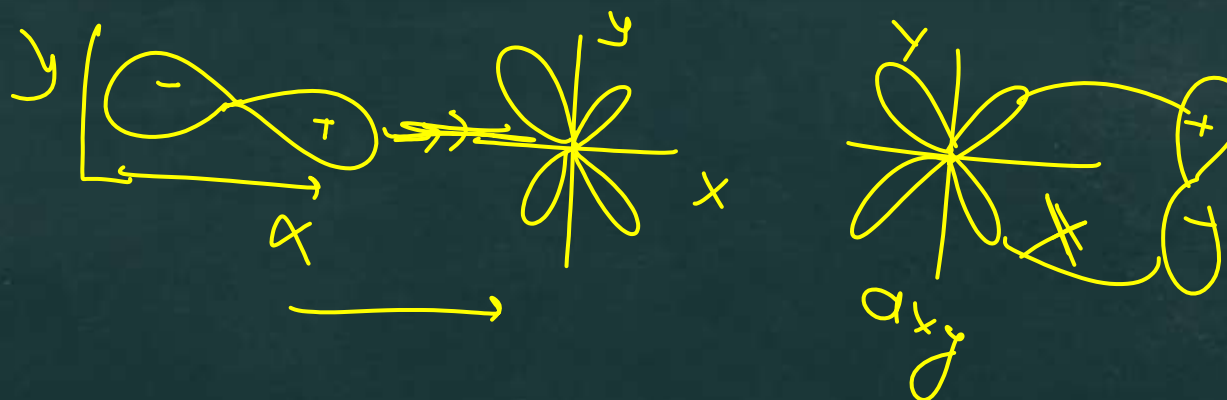
Atomic Orbitals	Internuclear axis	Types of Covalent Bond
$p_x + p_x$ y, z	x - axis	σ
	y - axis	π
	z - axis	π



Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

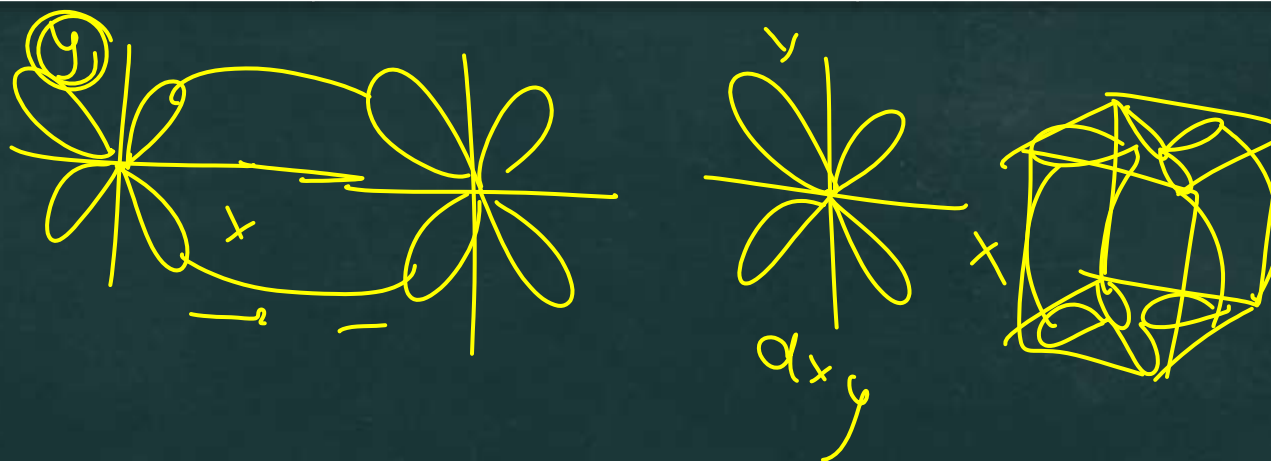
Atomic Orbitals	Internuclear axis	Types of Covalent Bond
$p_x + d_{xy}$	x - axis	σ
	y - axis	π
	z - axis	σ



Concept :- 20

Topic Name :- Valence Bond Theory (VBT)

Atomic Orbitals	Internuclear axis	Types of Covalent Bond
$d_{xy} + d_{xy}$	x - axis	π
	y - axis	π
	z - axis	δ



Concept :- 21

Topic Name :- Strength of Covalent Bond

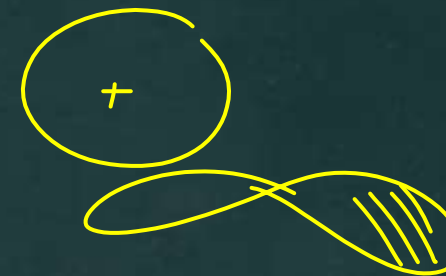
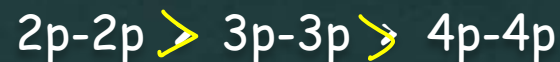
Overlapping area

Case-I : When value of n (shell number) is different then

$$\text{Strength of bond} \propto \frac{1}{\text{Shell no.}(n)} \uparrow$$

$\downarrow$
 δ long $\rightarrow n \uparrow, \delta \downarrow$

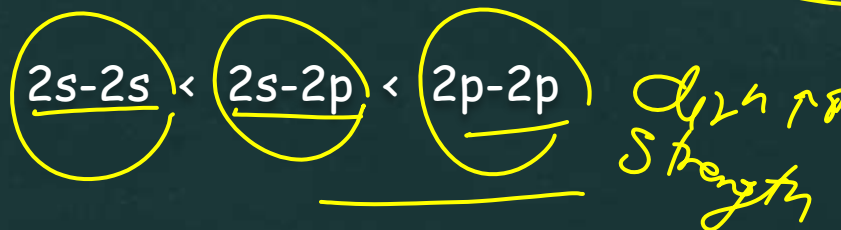
Example



Case-II : When value of n (shell number) is same then

Strength of bond $\propto$ Directional nature of orbitals

Example



Concept :- 21

Topic Name :- Strength of Covalent Bond

Example

✓  σ Bond strength order : $1s-1s > 2s-2s > 3s-3s$ $n \uparrow$, strength $\uparrow$

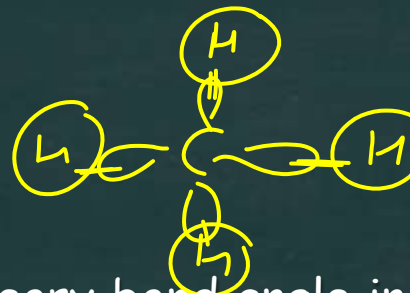
$\longrightarrow n \uparrow$, strength $\downarrow$



✓  Order of π - bond strength

$2p_{\pi}-2p_{\pi} > 2p_{\pi}-3d_{\pi} > 2p_{\pi}-3p_{\pi} > 3p_{\pi}-3p_{\pi}$
 $2 > 23 > 23 > 3$

Drawbacks of VBT



- According to this theory bond angle in CH_4 should be 90° , as these are formed by p-p overlapping, but actually it has four $109^\circ 28'$ angles.

5-6 min

CH_4 All 4 C-H

- O_2 gas is diamagnetic according to VBT but it is paramagnetic

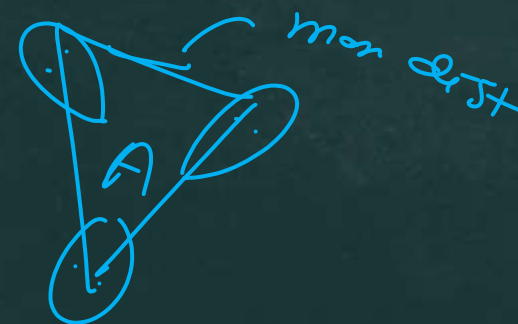
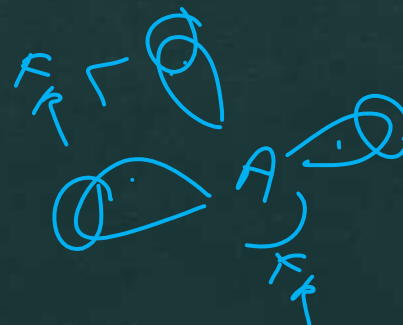


Concept :- 22

Topic Name :- VSEPR

The main postulates of VSEPR theory are as follows :

- The shape of a molecule depends upon the number of valence shell electron pairs (bonded or nonbonded) around the central atom.
- Pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged. ✱ ✱
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.



Concept :- 23

Topic Name :- Hybridisation

Calculation of Hybridisation

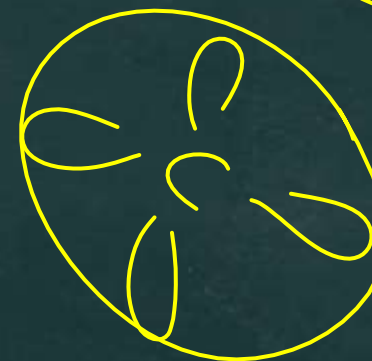
There are various types of hybridisation involving s, p and d orbitals.

The different types of hybridisation are as under :-

don't
take part in
Resonance

AO + AO

→ (HO) + (HO)



Equal
Shape
Size
Energy

$n=2$ sp Linear

$n=3$ sp²

$n=4$ sp³

$n=5$ sp³d

$n=6$ sp³d²

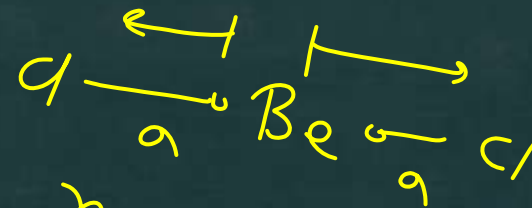
$n=7$ sp³d³

$n = \text{no. of } s \text{ orbital} + \text{no. of } p \text{ orbital} + \text{no. of } d \text{ orbital} + \text{no. of } f \text{ orbital} + \text{no. of } -ve \text{ charge}$

ALLEN
DIGITAL
CHEMISTRY

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp	2	0	BeCl ₂ HgCl ₂	$\mu = 0$	3	∞

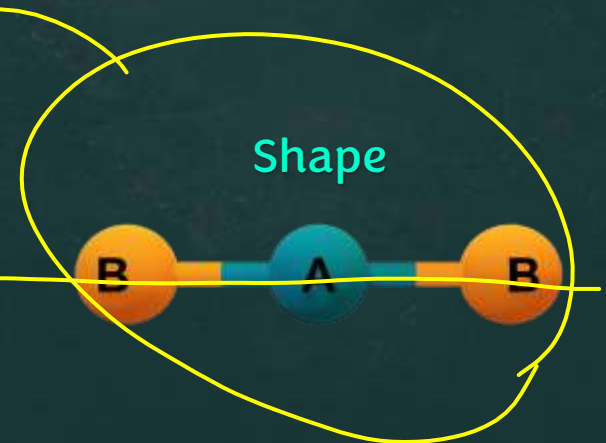


$$\eta = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 0 + 0$$

5p7

$$\ell_{\text{het}} = 0$$

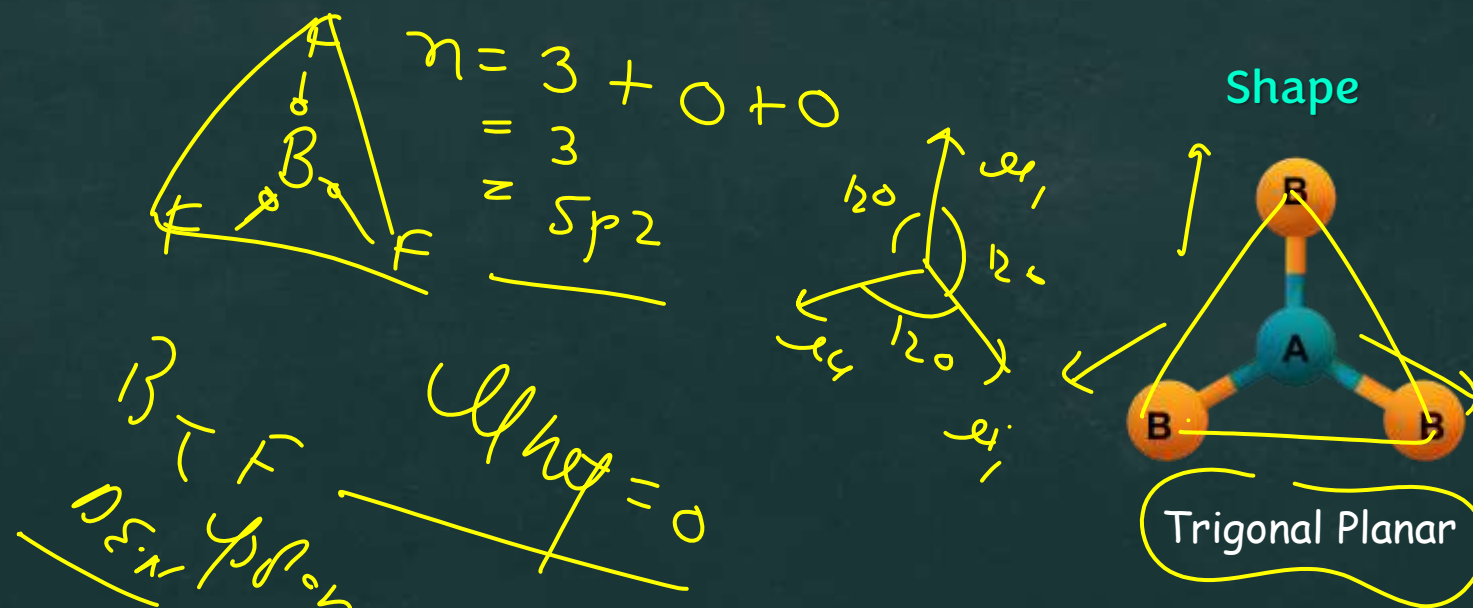
Wheat



Concept :- 23

Topic Name :- Hybridisation

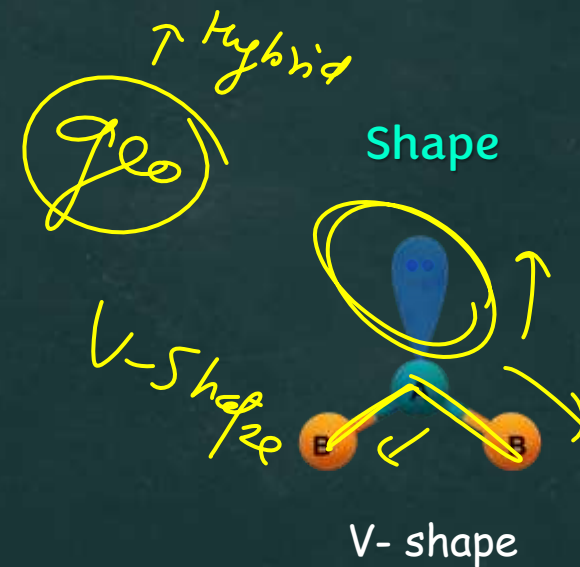
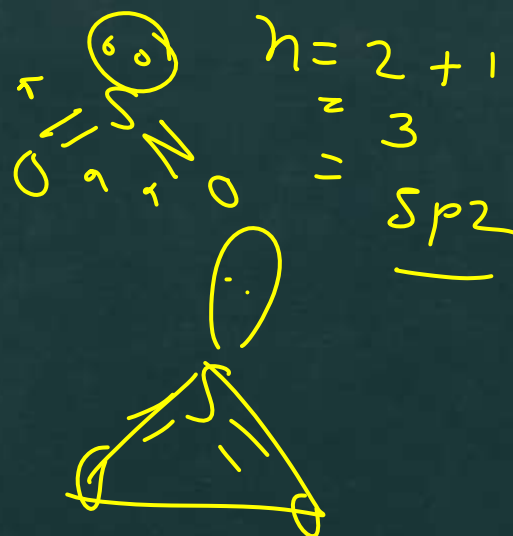
Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^2	<u>3</u>	<u>0</u>	<u>BF_3</u>	$\mu = 0$	<u>4</u>	1



Concept :- 23

Topic Name :- Hybridisation

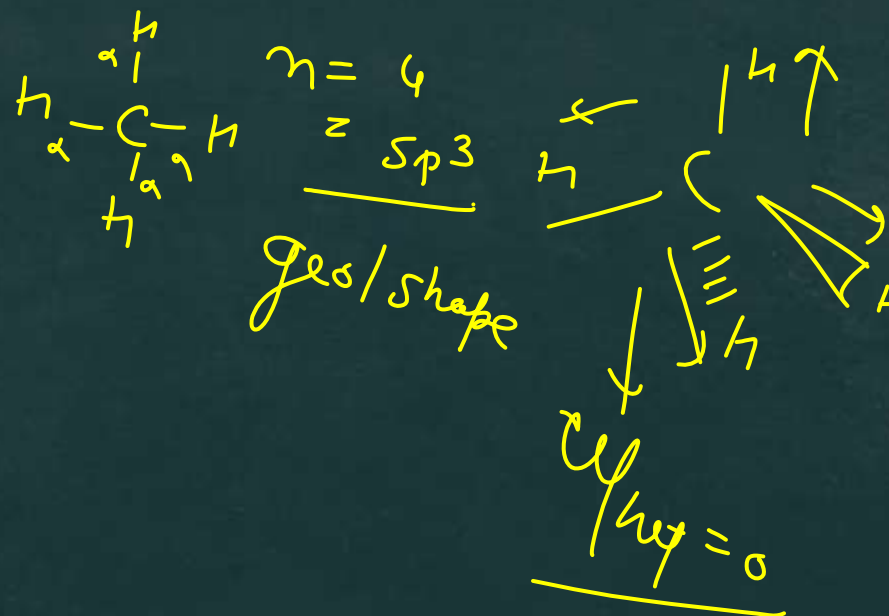
Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^2	2	1	O_3 SO_2	$\mu \neq 0$	3	1



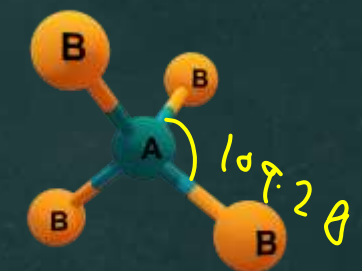
Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3 $(s+p_x+p_y+p_z)$	4	0	CH_4 NH_4^+	$\mu = 0$	3	10



Shape

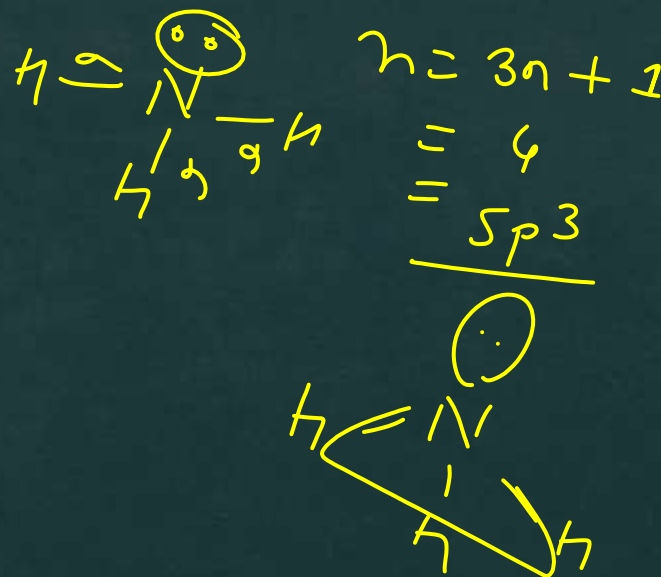


Tetrahedral

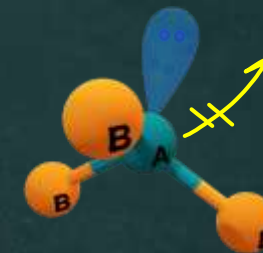
Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3 ($s+p_x+p_y+p_z$)	3	1	NH_3	$\mu \neq 0$	3	4



Shape

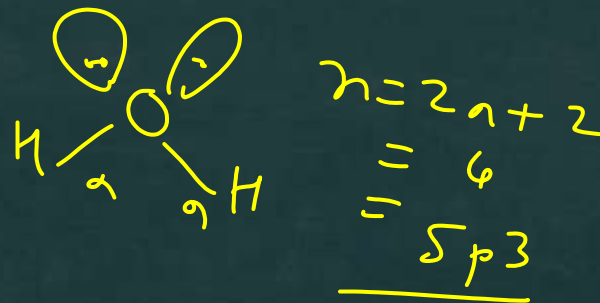


Trigonal Pyramidal

Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3 ($s+p_x+p_y+p_z$)	2	2	H_2O	$\mu \neq 0$	3	1

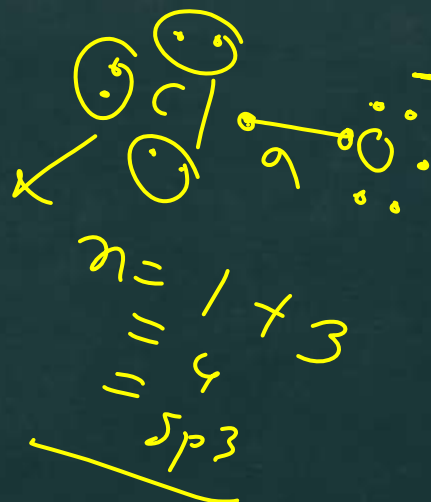


V-shape or bent or Angular

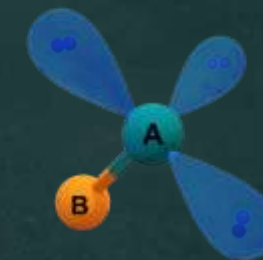
Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	No. of atoms in a plane
sp^3 <u>$(s+p_x+p_y+p_z)$</u>	1	3	ClO^-	2



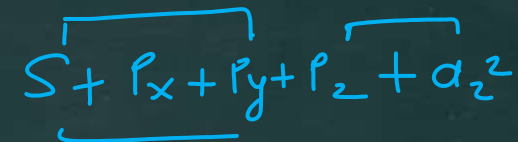
Shape



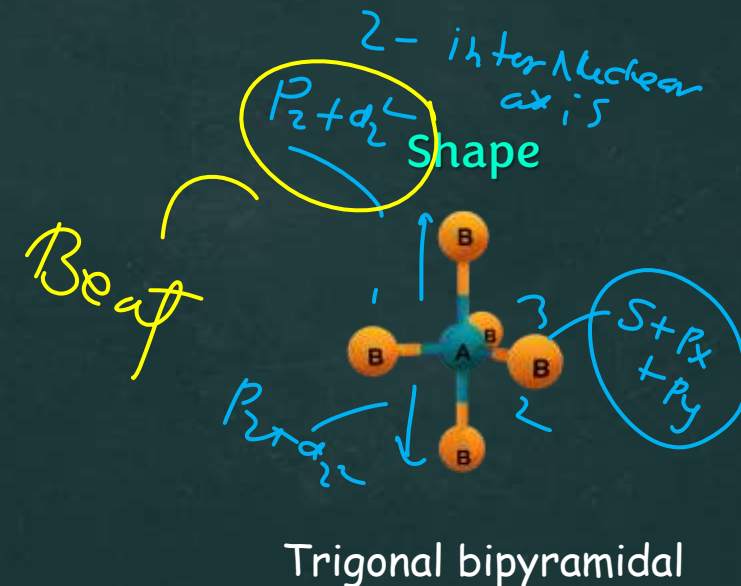
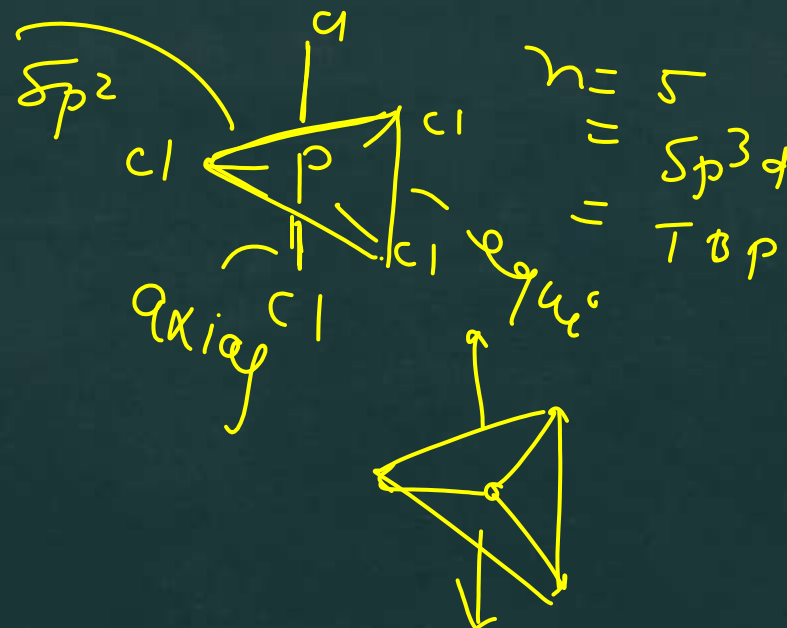
Linear

Concept :- 23

Topic Name :- Hybridisation



Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d ($s + p_x + p_y + p_z + d_{z^2}$)	5	0	PCl_5	$\mu = 0$	4	4



Concept :- 23

Topic Name :- Hybridisation

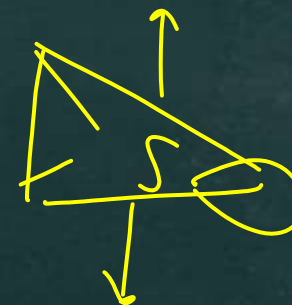
Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d ($s+p_x+p_y+p_z+d_{z^2}$)	4	1	SF_4	$\mu \neq 0$	3	10



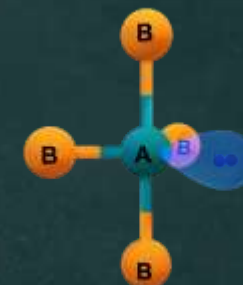
$$n = 4 + 1$$

$$= 5$$

$$sp^3d$$



Shape

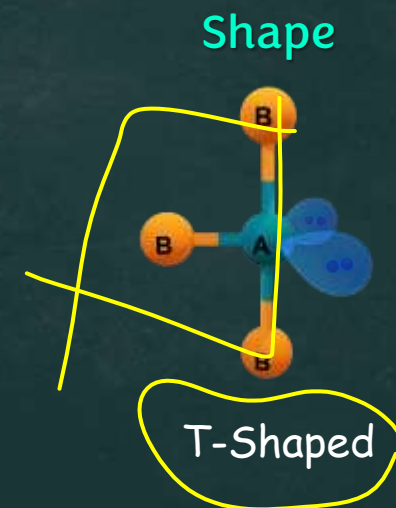
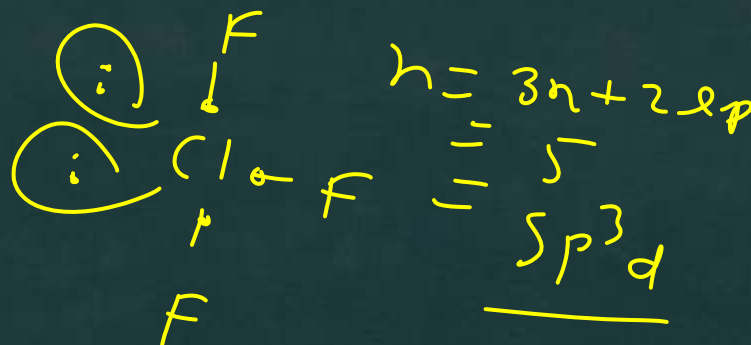


See-saw

Concept :- 23

Topic Name :- Hybridisation

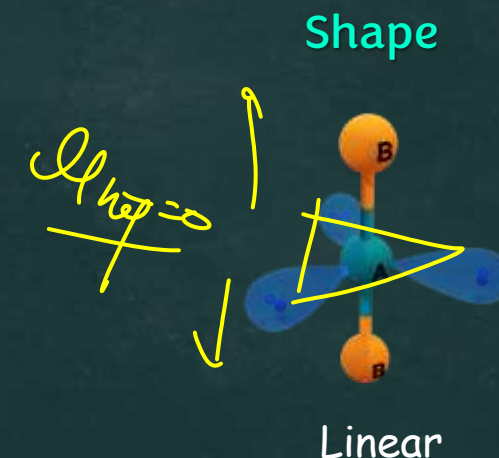
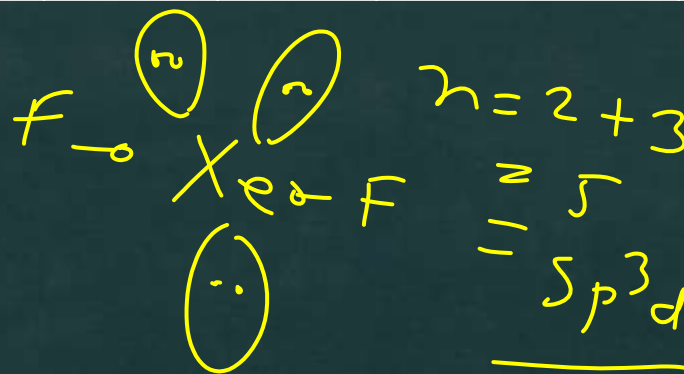
Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d ($s+p_x+p_y+p_z+d_{z^2}$)	3	2	ClF_3	$\mu \neq 0$	4	1



Concept :- 23

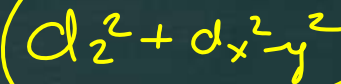
Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane
sp^3d ($s+p_x+p_y+p_z+d_{z^2}$)	2	3	XeF_2 I_3^-	$\mu = 0$	3

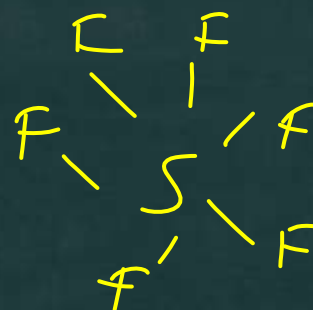


Concept :- 23

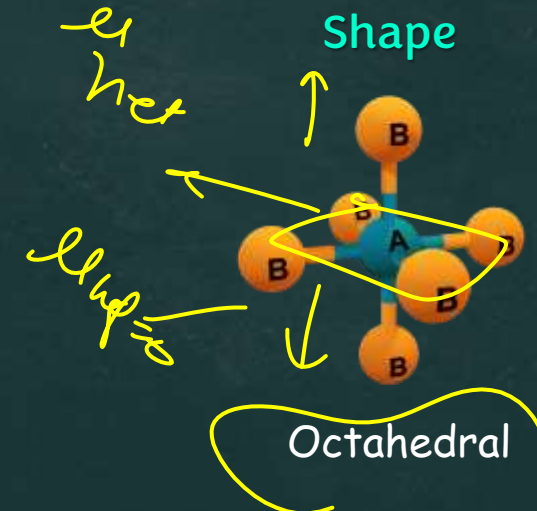
Topic Name :- Hybridisation



Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d^2 ($s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2}$)	6	0	SF_6	$\mu = 0$	5	3



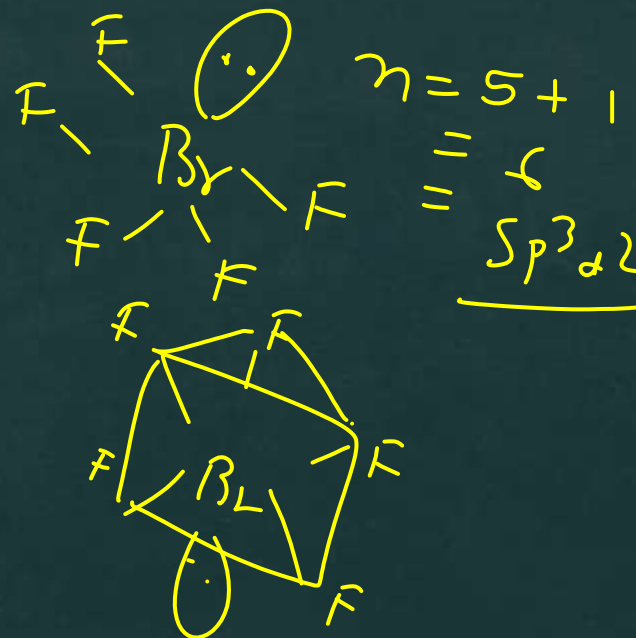
no. of atoms = 6
no. of bonds = 6
 sp^3d^2



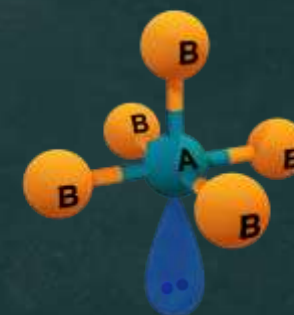
Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d^2 ($s+p_x+p_y+p_z+d_{z^2}+d_{x^2-y^2}$)	5	1	BrF_5	$\mu \neq 0$	4	3



Shape

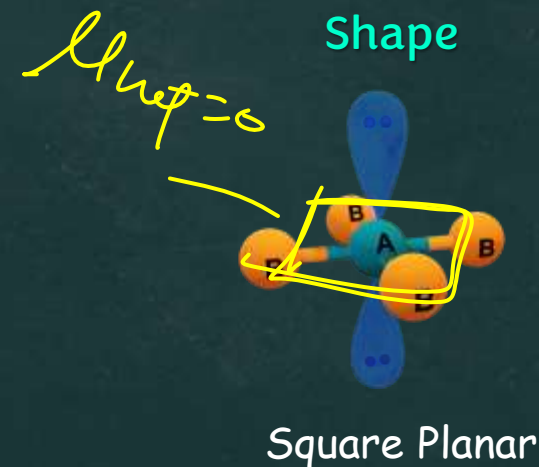
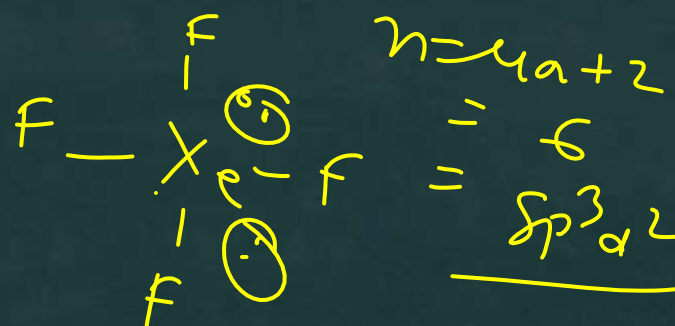


Square pyramidal

Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
<u>sp^3d^2</u> ($s+p_x+p_y+p_z+d_{z^2}+d_{x^2-y^2}$)	4	2	XeF_4	$\mu = 0$	5	1

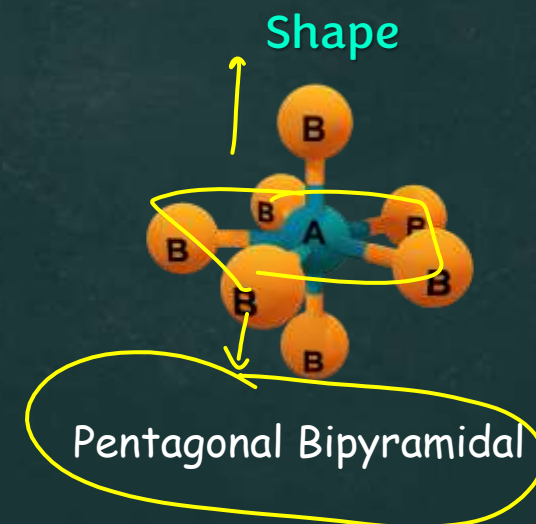
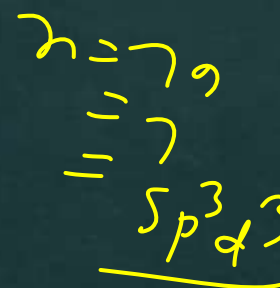


Concept :- 23

Topic Name :- Hybridisation

$$d_{z^2} + d_{x^2-y^2} + d_{xy}$$

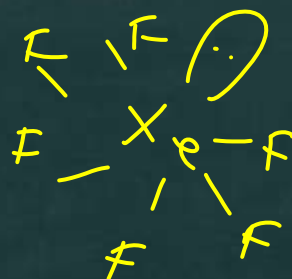
Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment	No. of atoms in a plane	No. of such plane
sp^3d^3 $(s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2} + d_{xy})$	7	0	IF ₇	$\mu_{net} = 0$	6	1



Concept :- 23

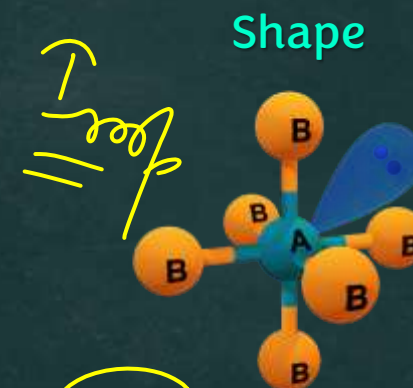
Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment
sp^3d^3 $(s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2} + d_{xy})$	6	1	XeF_6	$\mu \neq 0$



$$n = 6 + 1 = 7$$

$$\underline{sp^3d^3}$$

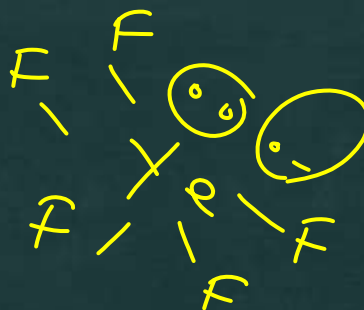


Distorted Octahedron

Concept :- 23

Topic Name :- Hybridisation

Hybridisation	Bond Pair	Lone Pair	Examples	Dipole moment
sp^3d^3 $(s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2} + d_{xy})$	5	2	XeF_5^-	$\mu \neq 0$ $\mu_{net} = 0$

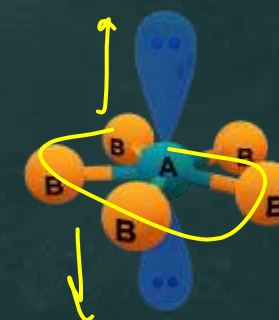


$$n = 5 + 2$$

$$= 7$$

$$= sp^3d^3$$

Shape



Pentagonal Planar

Example

HW

sp

$\text{BeH}_2, \text{BeCl}_2, \text{CO}_2, \text{NO}_2^+, \text{C}_2\text{H}_2, \text{HCN}, \text{HgCl}_2,$
 $\text{N}_2\text{O}, \text{N}_3^-$

sp^2

$\text{BF}_3, \overset{\oplus}{\text{C}}\text{H}_3, \text{CH}_2=\text{CH}_2, \text{Graphite}, \text{NO}_3^-, \text{HNO}_2,$
 $\text{SO}_2, \text{SO}_3, \text{HCO}_3^-, \text{SnCl}_2, \text{SnI}_2, \text{AlCl}_3$

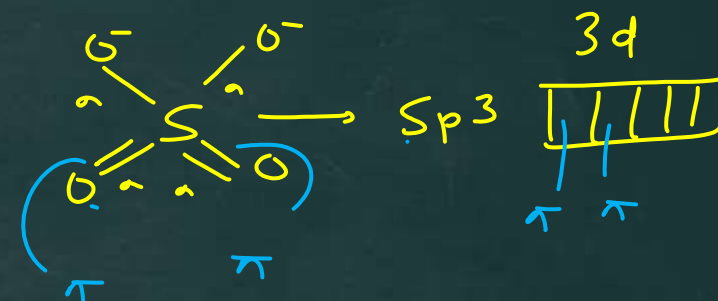
sp^3

$\text{CH}_4, \overset{\oplus}{\text{P}}\text{Cl}_4, \overset{\oplus}{\text{N}}\text{H}_4, \overset{\ominus}{\text{B}}\text{F}_4, \text{BeF}_4^{-2}, \text{MgF}_4^{-2}, \text{NF}_3,$
 $\text{N}(\text{CH}_3)_3, \overset{\ominus}{\text{C}}\text{H}_3, \text{H}_3\text{O}^+, \text{XeO}_3, \overset{\ominus}{\text{N}}\text{H}_2, \text{Cl}_2\text{O}, \text{SCl}_2,$
 $\text{Diamond}, \text{SiO}_2, \text{SiC}$

Concept :- 24

Topic Name :- $p_{\pi}-p_{\pi}$ & $p_{\pi}-d_{\pi}$ Bonds

CO_2	$2p_{\pi}-2p_{\pi}$	2 bond
SO_2	$2p_{\pi}-3p_{\pi}$ $2p_{\pi}-3d_{\pi}$	1 bond 1 bond
SO_4^{2-}	$2p_{\pi}-3d_{\pi}$	2 bond
NO_2^+	$2p_{\pi}-2p_{\pi}$	2 bond
NO_3^-	$2p_{\pi}-2p_{\pi}$	2 bond
POCl_3	$2p_{\pi}-3d_{\pi}$	1 bond
SO_3	$2p_{\pi}-3p_{\pi}$ $2p_{\pi}-3d_{\pi}$	1 bond 2 bond
ClO_4^-	$2p_{\pi}-3d_{\pi}$	3 bond
XeO_3	$2p_{\pi}-5d_{\pi}$	2 bond



$2p_{\pi}-3d_{\pi}$

$2p_{\pi}-2p_{\pi}$
 $2p_{\pi}-2p_{\pi}$



 Example

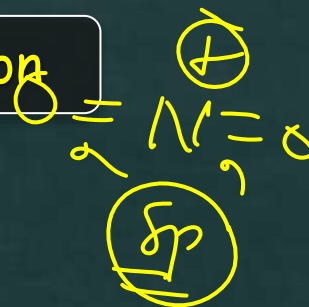
The hybridisations of the atomic orbitals of nitrogen in $\text{NO}_2^\ominus$, $\text{NO}_2^\oplus$ and $\text{NH}_4^\oplus$ respectively are. [JEE (Main) 2021]

sp^3

- (1) sp^3 , sp^2 and sp
(3) sp^3 , sp and sp^2

- (2) sp , sp^2 and sp^3
(4) sp^2 , sp and sp^3

Solution



Ans. (4)



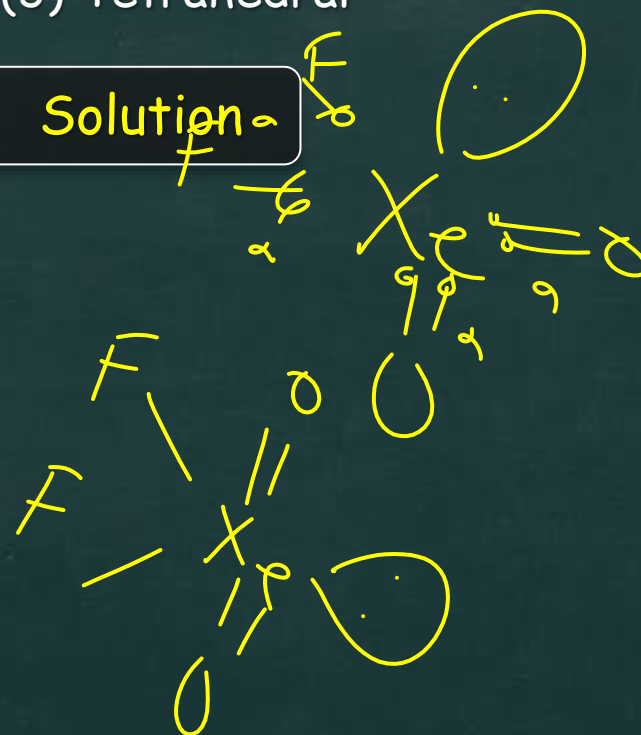
Example

The shape of XeO_2F_2 molecule is :

[JEE (Advanced) 2012]

- (1) Trigonal bipyramidal
- (2) Square planar
- (3) Tetrahedral
- ✓ (4) See-saw

Solution



$$n = \text{un} + \text{I.e.p}$$

$$= 5 + 1$$

$$= 6$$

$$\text{sp}^3\text{d}$$

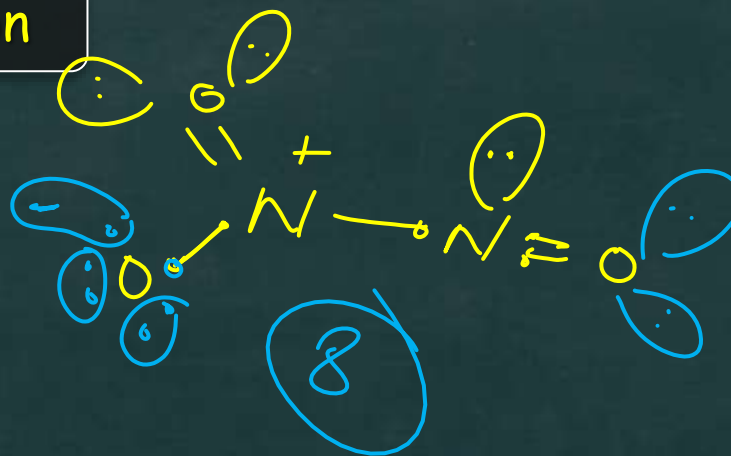
Ans. (4)

 Example

The total number of lone pairs of electrons in N_2O_3 is :

[JEE (Advanced) 2015]

Solution



Ans. (8)



Example

AB_3 is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer)

[JEE Mains-2021]

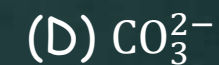
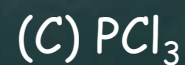
Solution



Ans. (2)

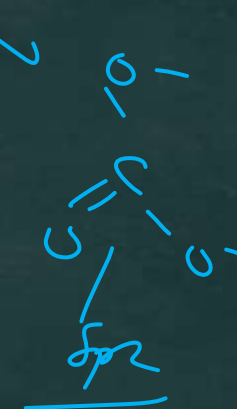
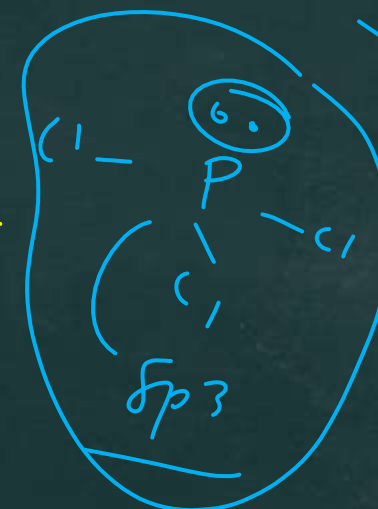
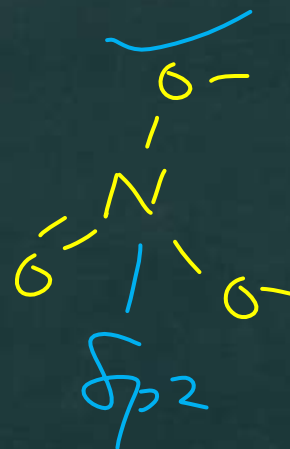
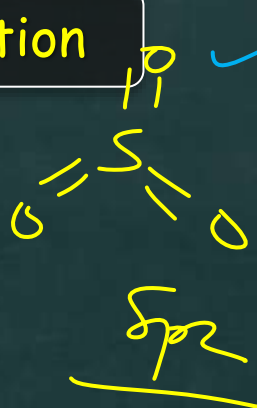
 Example

The number of species having non-pyramidal shape among the following is _____.



[JEE Mains-2021]

Solution



Ans. (3)

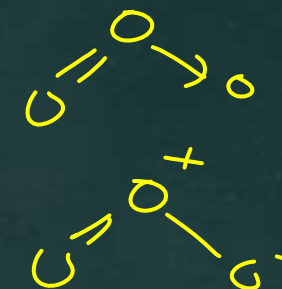


Example

[JEE Mains-2022]

Match List - I with List - II.

List - I (Compound)		List - II (Shape)	
(A)	BrF_5	(i)	bent
(B)	$[\text{CrF}_6]^{3-}$	(II)	square pyramidal
(C)	O_3	(III)	trigonal bipyramidal
(D)	PCl_5	(IV)	octahedral



Choose the correct answer from the options given below :

- (1) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (2) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (3) (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
- (4) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

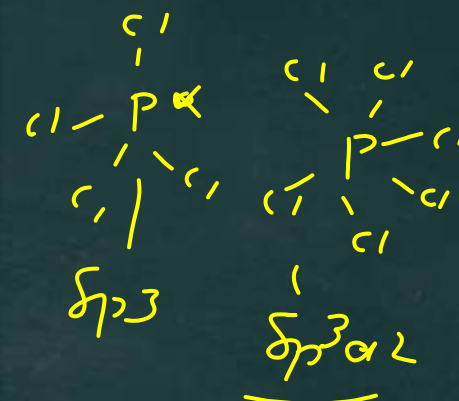
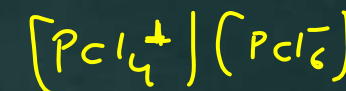
Solution

Ans. (3)

Concept :- 25

Topic Name :- Hybridisation in Ionic Solid

Species	Cationic part	Anionic part
PCl_5	$\text{PCl}_4^+ (\text{sp}^3)$ ✓	$\text{PCl}_6^- (\text{sp}^3\text{d}^2)$
PBr_5	$\text{PBr}_4^+ (\text{sp}^3)$ ✓	Br^-
XeF_6	$\text{XeF}_5^+ (\text{sp}^3\text{d}^2)$ ✓	F^-
N_2O_5	$\text{NO}_2^+ (\text{sp})$	$\text{NO}_3^- (\text{sp}^2)$
$\text{N}_2\text{O}_3(\text{s})$	NO^+	$\text{NO}_2^- (\text{sp}^2)$
$\text{N}_2\text{O}_4(\text{s})$	NO^+	$\text{NO}_3^- (\text{sp}^2)$
$\text{I}_2\text{Cl}_6 (\text{liquid})$	$\text{ICl}_2^+ (\text{sp}^3)$	$\text{ICl}_4^- (\text{sp}^3\text{d}^2)$
$\text{I}_2(\text{liquid})$	$\text{I}_3^+ (\text{sp}^3)$	$\text{I}_3^- (\text{sp}^3\text{d})$ ✓
Cl_2O_6	$\text{ClO}_2^+ (\text{sp}^2)$	$\text{ClO}_4^- (\text{sp}^3)$



sp³ or 2



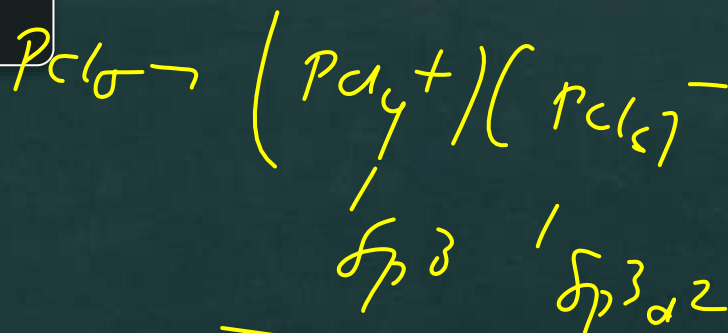
Example

The structure of PCl_5 in the solid state is :

[JEE (Main) 2020]

- (1) Square pyramidal
- (2) Tetrahedral $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$
- (3) Square planar $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$
- (4) Trigonal bipyramidal

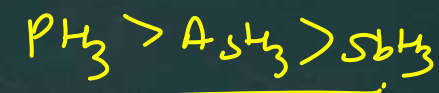
Solution



Ans. (2)

Concept :- 26

Topic Name :- Drago's Rule

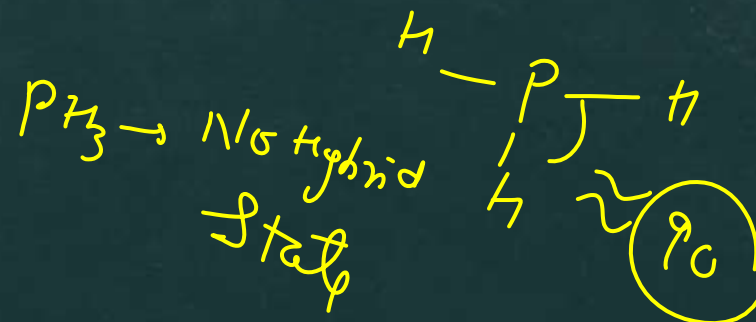


Species	Bond angle
✓ PH_3	94°
✓ AsH_3	92.5°
✓ SbH_3	91°

Species	Bond angle
H_2S	92°
H_2Se	91°
H_2Te	90.5°

pure p-orbital bond

Bond angle observed is almost 90° which indicates that no or negligible hybridisation takes place and bond is formed by almost pure 'p' atomic orbitals.



 Example

The compound that has the largest H-M-H bond angle (M = N, O, S, C), is:

(1) H_2O

(2) CH_4

(3) NH_3

(4) H_2S

[JEE (Main) 2020]

Solution

sp^3

109.28°

sp^3

$< 109.28^\circ$

sp^3

Ans. (2)

Concept :- 27

Topic Name :- Dipole Moment

• It is the product of charge accumulated on an ~~element~~ *in molecule* and internuclear distance (bond length).

$$\mu_{th} = q \times d \quad \text{unit : debye}$$

where μ_{th} = theoretical dipole moment

q = charge accumulated on either A or B (not with Sign)

d = bond length of A - B

$$(1D = 10^{-18} \text{ esu.cm})$$

$$(1D = 3.33 \times 10^{-30} \text{ C.m})$$

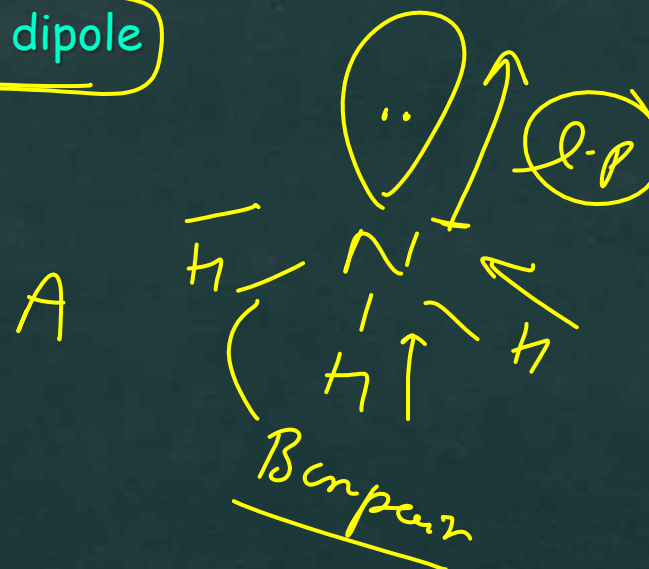
Concept :- 27

Topic Name :- Dipole Moment

• In a molecules two type of dipole moment present

(1) Bond pair dipole

(2) Lone-pair dipole



Concept :- 27

Topic Name :- Dipole Moment

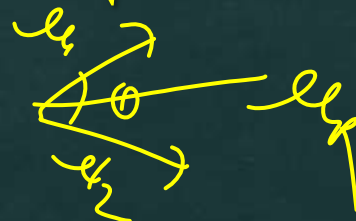
Application of dipole moment

Calculation of % ionic character in covalent compound

$$\% \text{ IC} = \frac{\mu_{ob}}{\mu_{th}} \times 100$$

$$\% \text{ IC} = \frac{\mu_{ob}}{q \times d} \times 100$$

$$\mu_{net} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$$



HCl NaCl

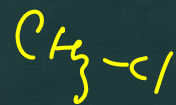
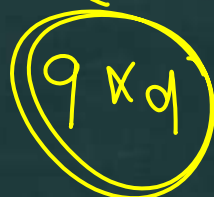
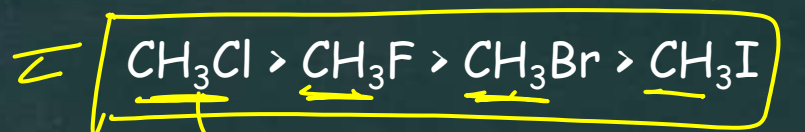
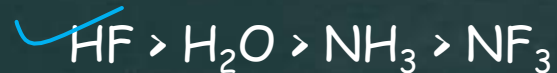
$$\mu_{th} = q \times d / \text{esu-cm}$$

$$\mu_{obs} = \sum \mu_{cm} \text{ given}$$

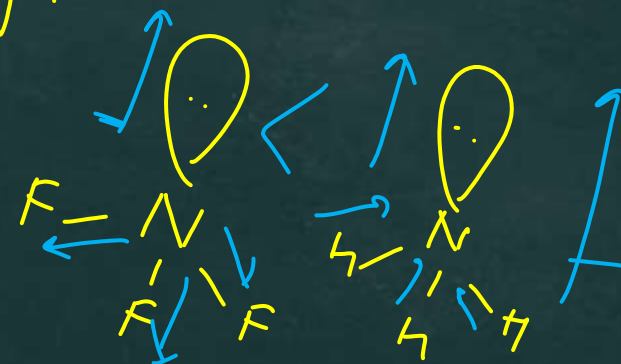
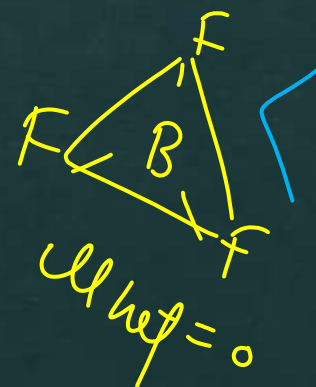
Concept :- 27

Topic Name :- Dipole Moment

Some important orders of dipole moments (μ)



↑ μ ↓



Concept :- 28

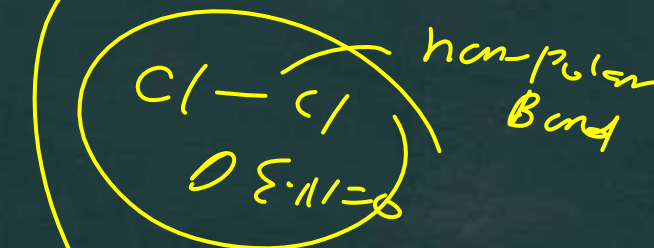
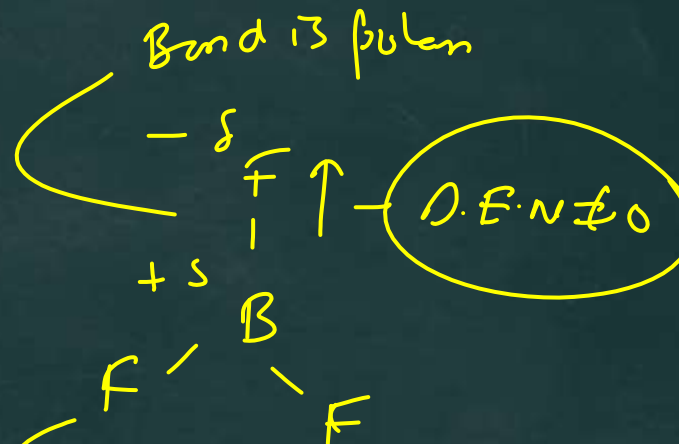
Topic Name :- Polarity of species

Polarity of bond

- (i) Polar Bond : $\Delta EN \neq 0$
- (ii) Non-Polar Bond : $\Delta EN = 0$

Polarity of molecule

- (i) Polar Molecule : $\mu \neq 0$
- (ii) Non-Polar Molecule : $\mu = 0$



$\mu = 0$
 non-polar molecule

Concept :- 29

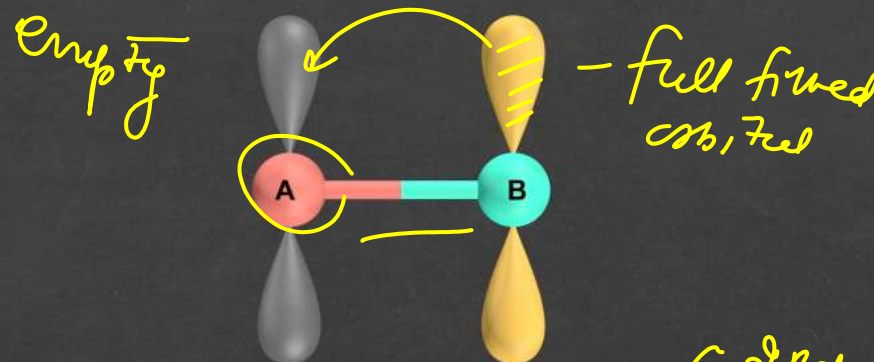
Topic Name :- Back Bonding

Introduction

2-2 ✓ 2-4 ✗
2-3 ✓ 2-5 ✗✗

• Lone pair sometimes may be transferred from filled orbital of an atom to unfilled (vacant) orbital of adjacent atom in a covalent molecule or ion.

• This phenomena is known as back bonding.



Back Bond
π-Bond

• It is a type of intramolecular (within the molecule) Lewis acid - Lewis base interaction.

Concept :- 29

Topic Name :- Back Bonding

Conditions for Back Bonding

- There should be a covalent bond between two atoms.
- One of them atom ^{Extra e^-} must have lone pair (generally, from IInd period).
- Another atom must have vacant orbital (generally, from IInd & IIIrd period).

Concept :- 29

Topic Name :- Back Bonding

Types of Back Bond

✓ (1) $\overset{A}{2p_{\pi}} - \overset{B}{2p_{\pi}}$ Back Bond

✓ (2) $2p_{\pi} - 3p_{\pi}$ Back Bond

(3) $3p_{\pi} - 3p_{\pi}$ non-effective

✓ (4) $2p_{\pi} - 3d_{\pi}$ Back Bond

(5) $2p_{\pi} - 4d_{\pi}$ non-effective

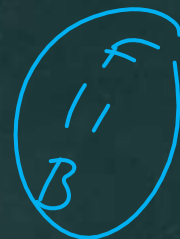
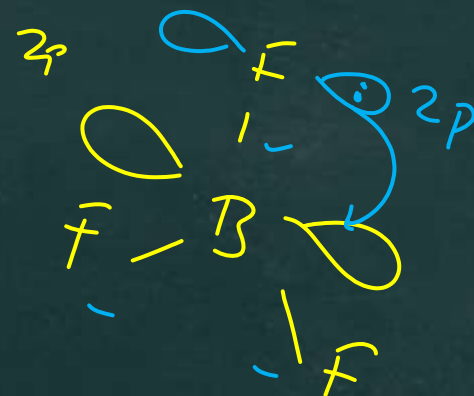
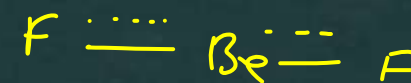
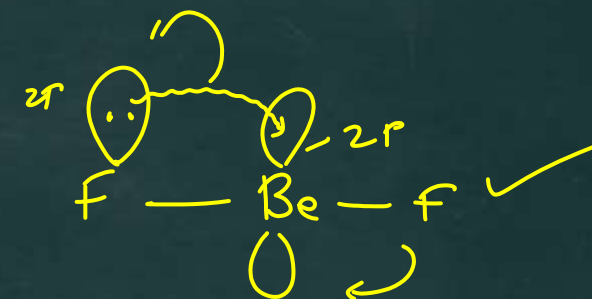
(6) $3p_{\pi} - 3d_{\pi}$ non-effective

Concept :- 29

Topic Name :- Back Bonding

Example	Type of Back Bonding
BeF ₂ <u>2p_n - 2p_n</u> ✓	
<u>BeCl₂</u> ✓	✓
BF ₃ <u>2p_n - 2p_n</u> ✓	✓
<u>AlCl₃</u> <u>3p - 3p</u>	✗

(2p)
(3p)



Concept :- 29

Topic Name :- Back Bonding

Example	Type of Back Bonding
$B(OCH_3)_3$	✓
$N(SiH_3)_3$	$2p\pi-3d\pi$ ✓
$P(SiH_3)_3$	✗
OCl_2	Non-effective back bonding



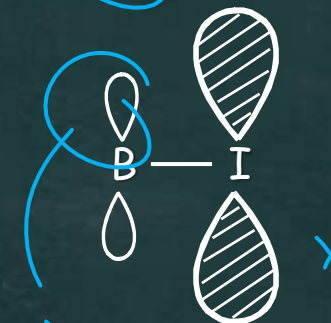
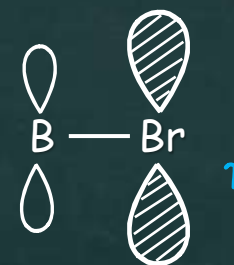
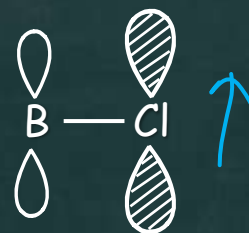
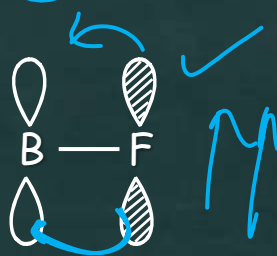
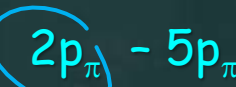
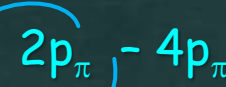
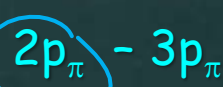
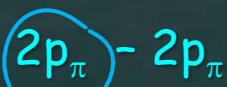
 Example

Compare extent of Back Bonding and Lewis acidic nature in the following:



Solution

Extent of BB ↓



Extent of Back bonding :



Lewis acidic nature :



 Example

Why bond angle in Cl_2O is greater than H_2O ? Discuss.

Solution

On the basis of electronegativity



Expected angle should be $\text{H}_2\text{O} > \text{Cl}_2\text{O}$



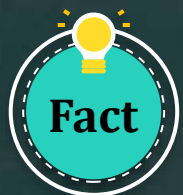
Example

Why bond angle in Cl_2O is greater than H_2O ? Discuss.

Solution

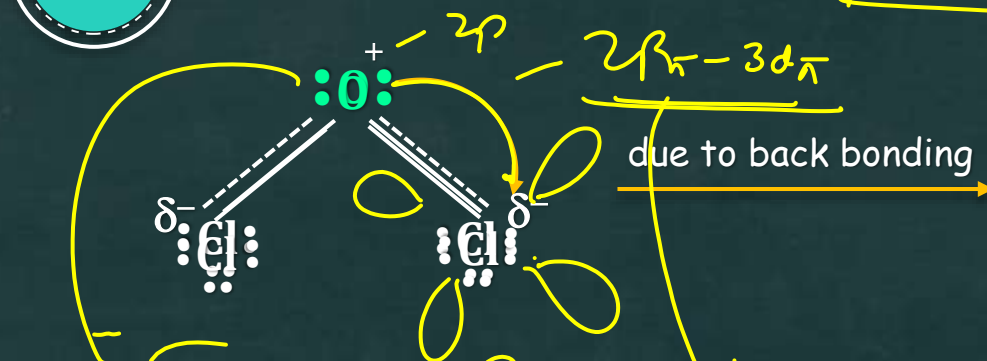
$2p_\pi - 3d_\pi$

111°



Fact

Experimental angle order is $\text{Cl}_2\text{O} > \text{H}_2\text{O}$



111°

Resonance hybrid

Nature of Back bonding = $2p_\pi - 3d_\pi$

Bond angle increases due to following two reasons :

Steric crowding (more prominent)

Back bonding (non-effective)

Back bonding is not effective

 Example

Discuss the reason of **Lewis acidic nature** order in the following ?



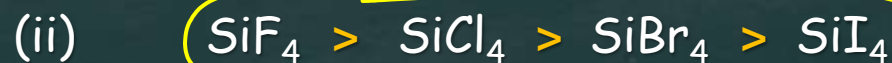
Solution

Extent of B-B ↓
empty orbital presence ↑
L.A ↑↑

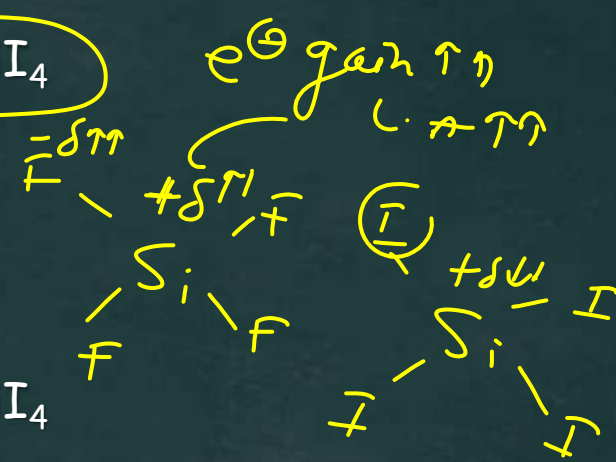
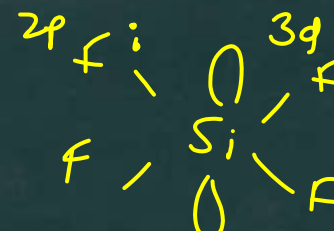


 Example

Discuss the reason of Lewis acidic nature order in the following ?



Solution

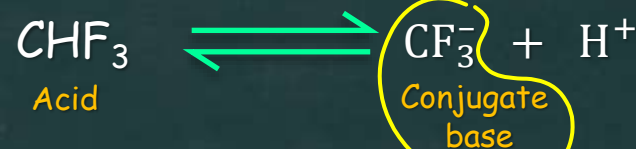


In ' SiF_4 ' partial positive charge on ' Si ' is more prominent

 Example

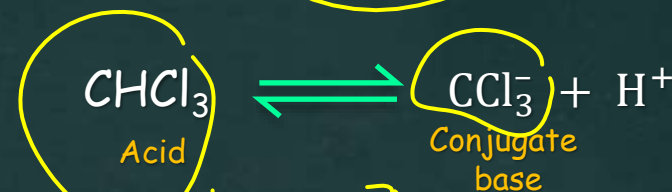
Why CHCl_3 is more acidic than CHF_3 ? Discuss.

Solution

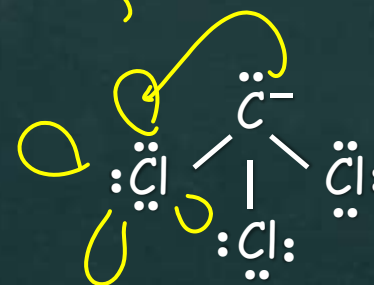
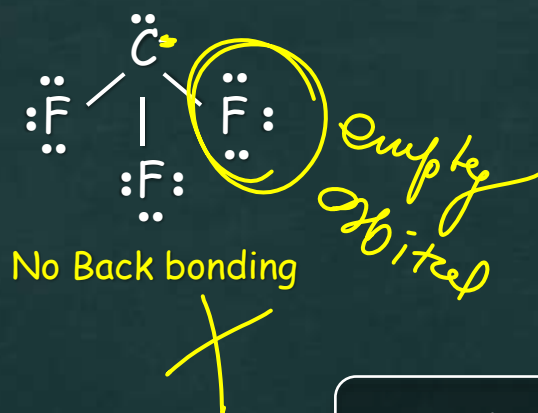


Stable

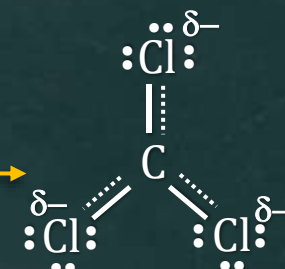
B.B



Stable ↑



due to back bonding



$\text{CHCl}_3 > \text{CHF}_3$ (Acidic nature)

Concept :- 30

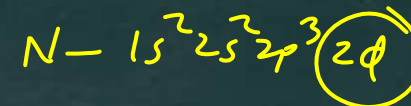
Topic Name :- Molecule does not exist

Lack of 2d orbitals

Molecule exist



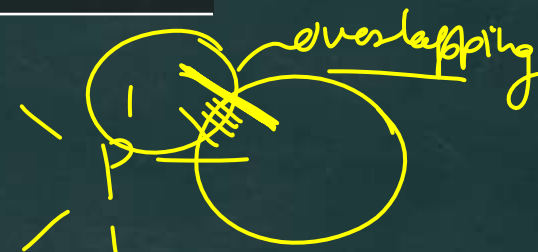
Molecule does not exist



Concept :- 30

Topic Name :- Molecule does not exist

Steric crowding



Molecule exist	Molecule does not exist
$\checkmark$ PF_5 $\checkmark$ SF_4 $\checkmark$ SF_6 $\checkmark$ IF_7	$\text{PI}_5(\text{g})$ $\text{SCl}_4, \text{SBr}_4, \text{SI}_4$ $\text{SCl}_6, \text{SBr}_6, \text{SI}_6$ $\text{ICl}_7, \text{IBr}_7$

Concept :- 30

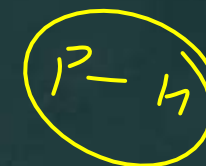
Topic Name :- Molecule does not exist

Valence Bond Theory (V.B.T)

Molecule exist



Molecule does not exist



$\text{P}-3\text{d}$

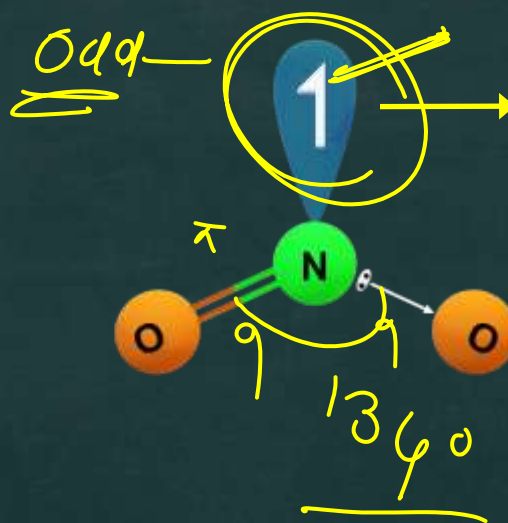
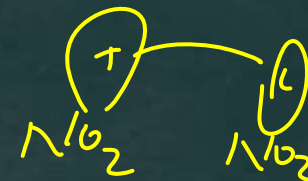
Energy Gap

Concept :- 31

Topic Name :- ~~Odd~~ Electron Molecules

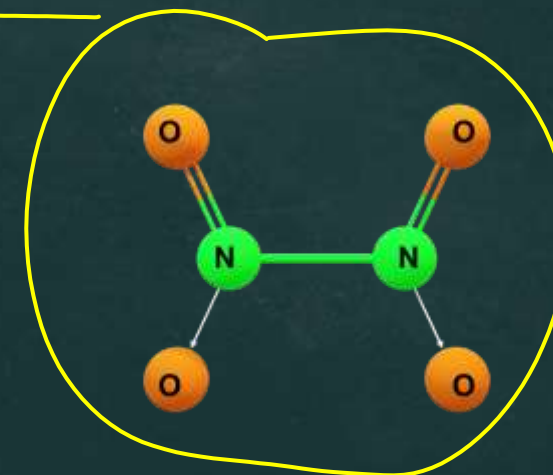
$$\text{NO}_2 = 7 + 8 + 8 = 23$$

- $\theta = 134^\circ$
- Hybridisation : ~~Approx.~~ sp^2
- Hybrid Orbitals = 3



Unpaired electron in one of the sp^2 hybrid orbital

sp^2 (odd)

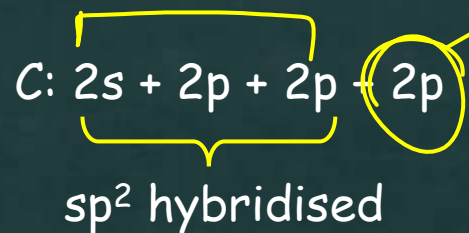


Concept :- 31

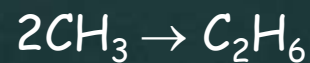
Topic Name :- Odd Electron Molecules



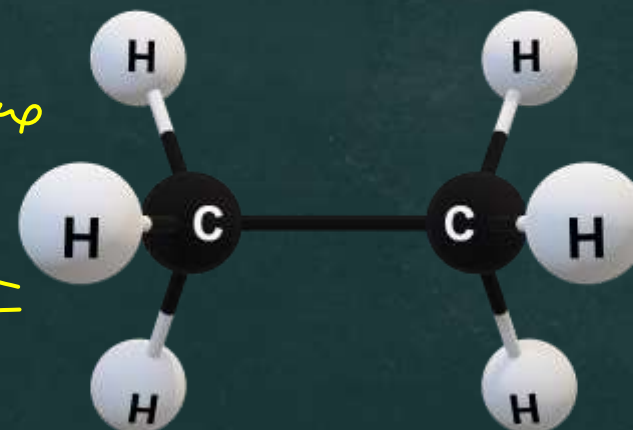
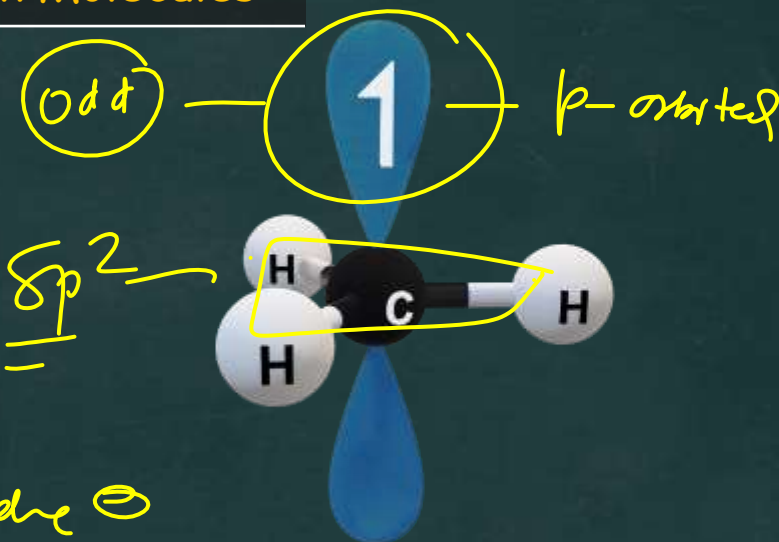
- $\theta = 120^\circ$
- Hybridisation : sp^2
- Hybrid Orbitals = 3



odd e⁻



Σ bond
W₆₈₇₂



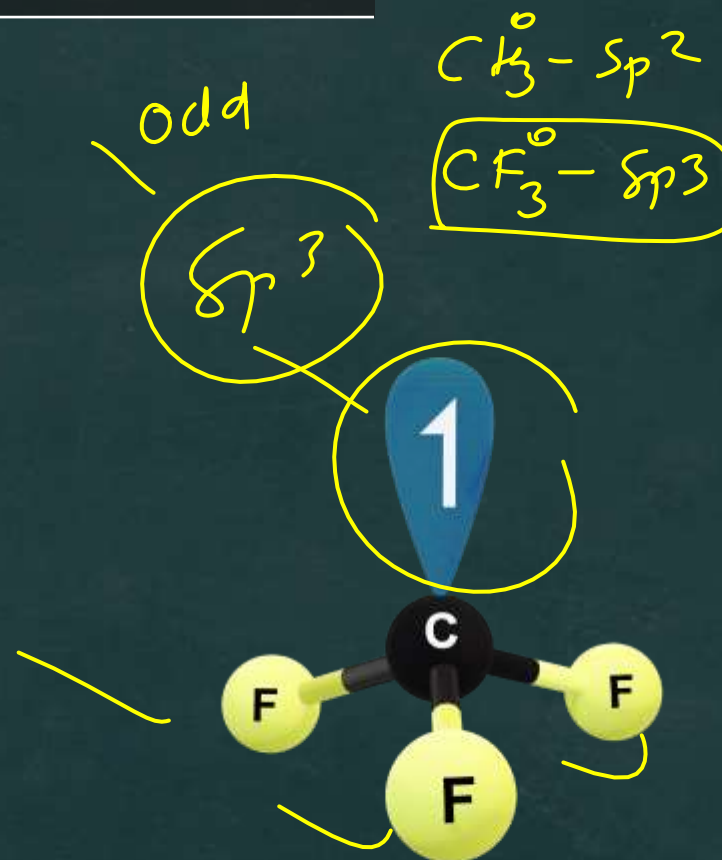
Concept :- 31

Topic Name :- Odd Electron Molecules

$\dot{\text{C}}\text{F}_3$

- $\theta \approx 109^\circ 28'$
- Hybridisation : sp^3
- Hybrid Orbitals = 4

C: $2s + 2p + 2p + 2p$
 $\underbrace{\hspace{1.5cm}}_{sp^3 \text{ hybridised}}$

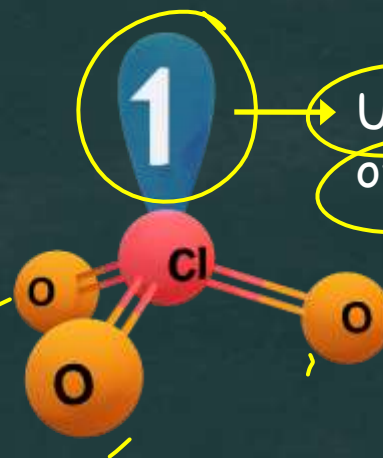


Concept :- 31

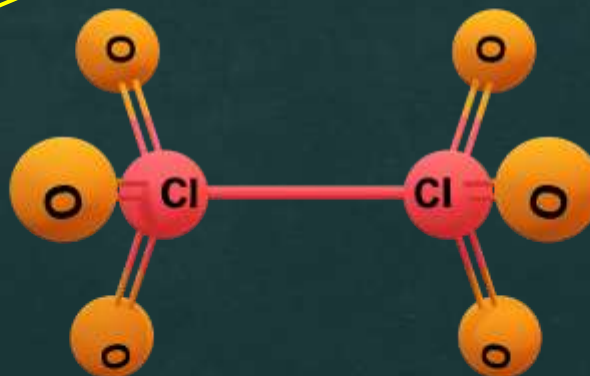
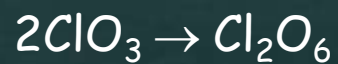
Topic Name :- Odd Electron Molecules



- $\theta = 109^\circ 28'$
- Hybridisation : sp^3
- Hybrid Orbitals = 4



Unpaired electron present in one of the sp^3 hybridised orbital



Concept :- 31

Topic Name :- Odd Electron Molecules

ClO_2

Unpaired e^-

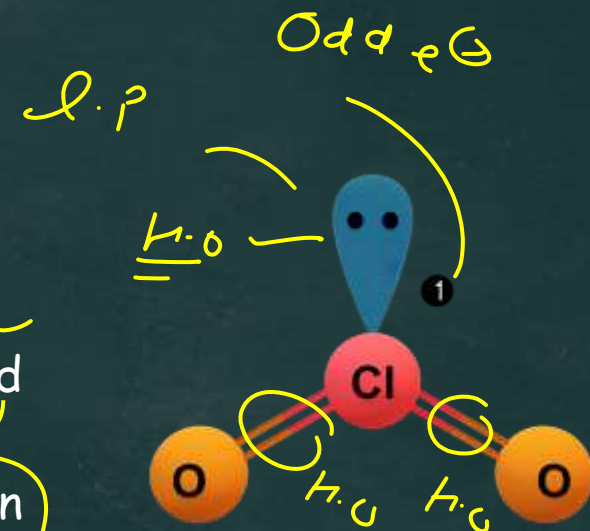
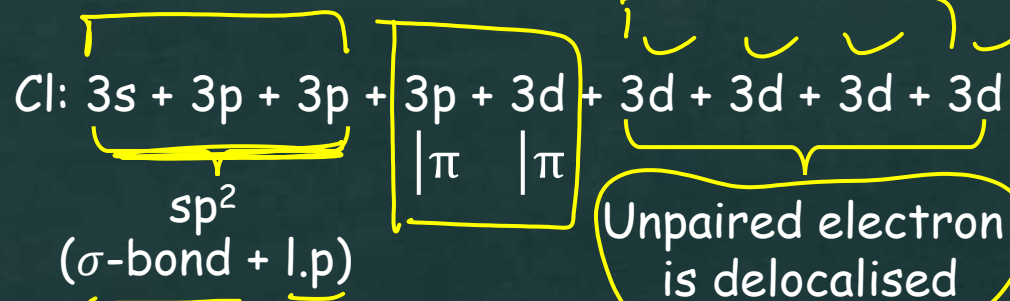
Stable

↳ 4 d orbitals

▪ $\theta = 117^\circ$ |||

▪ Hybridisation : Approx. sp^2

▪ Hybrid Orbitals = 3



Do not form dimer

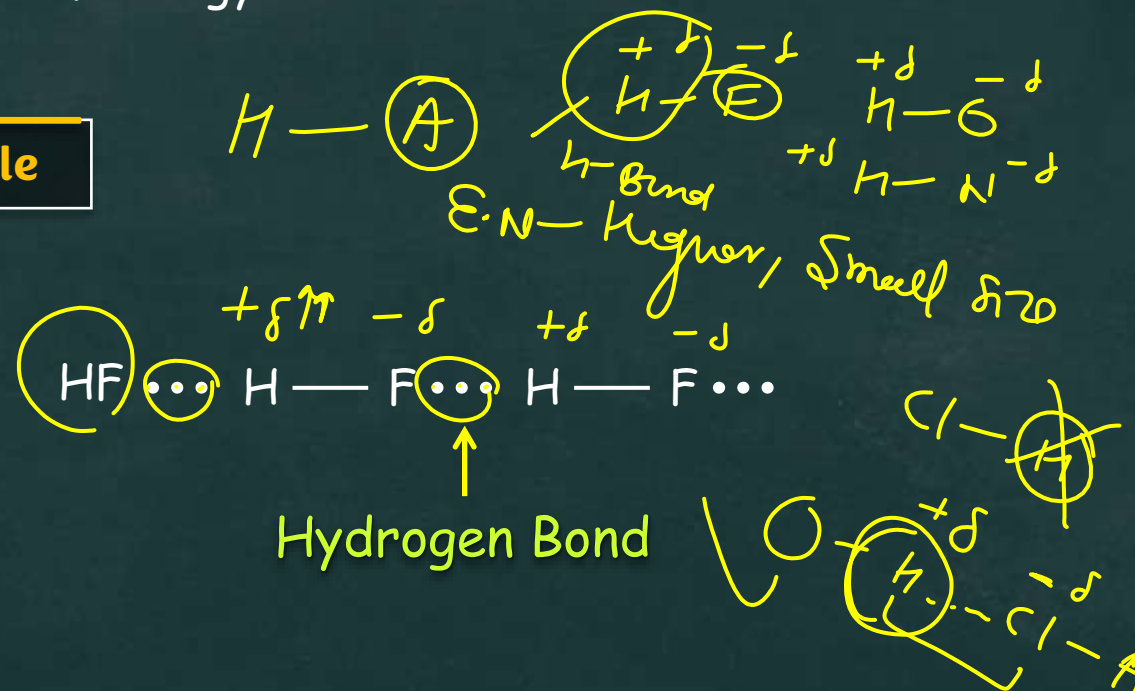
Concept :- 32

Topic Name :- Hydrogen Bonding

8 < H < 42 kJ/mole

It is a weak electrostatic force of attraction between hydrogen and more electronegative element which is smaller in size like F, N, O (sometimes Chlorine also) and have at least one lone pair and the amount of energy released is in between 8 to 42 KJ/mole.

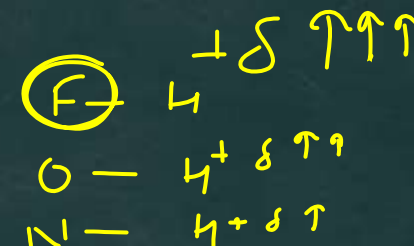
Example



Concept :- 32

Topic Name :- Hydrogen Bonding

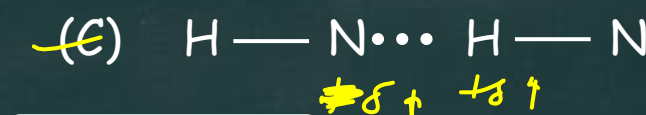
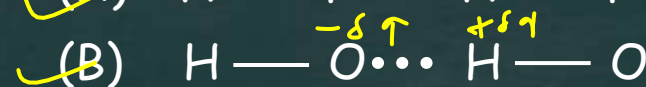
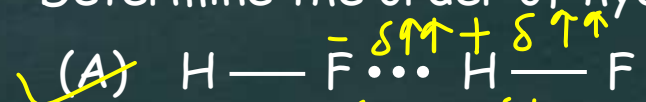
Strength of hydrogen bonding



H-bond strength $\propto$ Charge accumulated on hydrogen atom

 Example

Determine the order of hydrogen bond strength



Solution

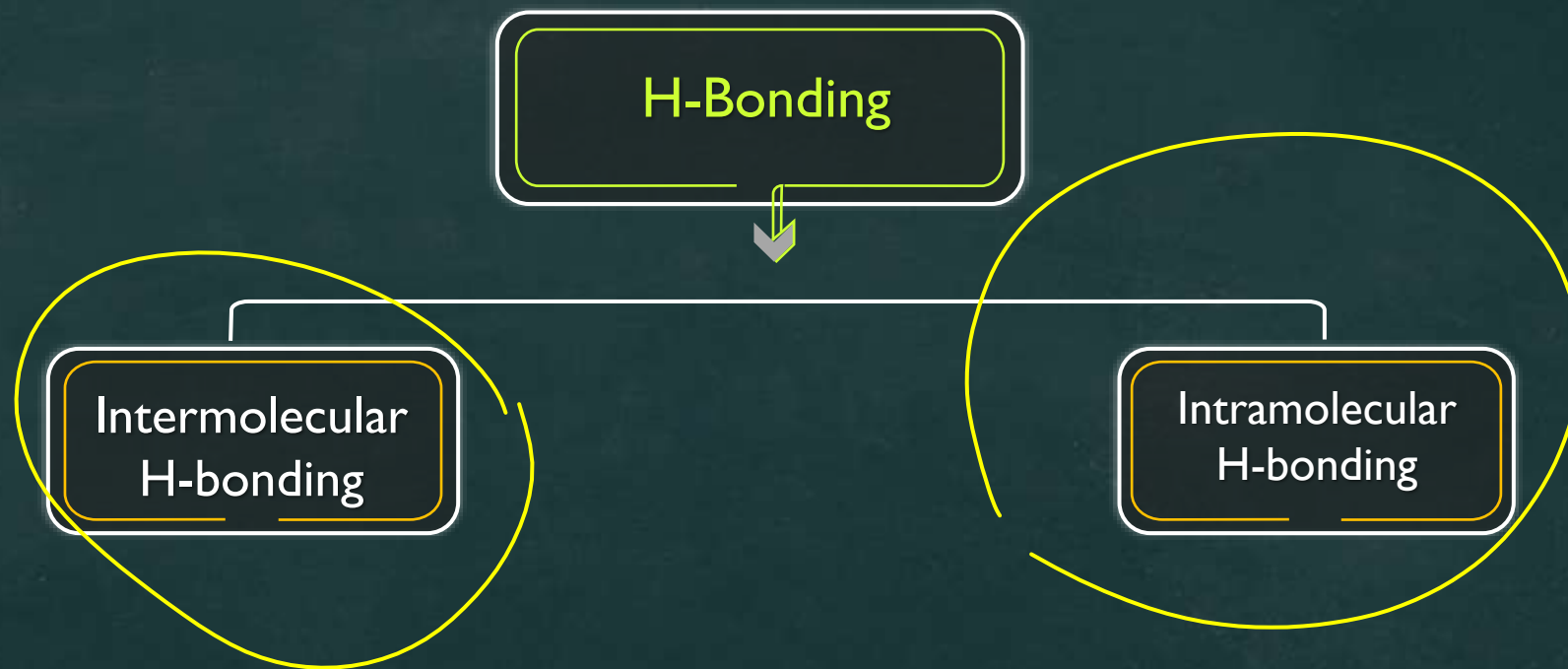
$A > B > C$

Ans. $A > B > C$

Concept :- 32

Topic Name :- Hydrogen Bonding

Types of hydrogen bonding



Concept :- 32

Topic Name :- Hydrogen Bonding

Intermolecular H-bonding

- Between same or different molecule

- Association

- Viscosity : More

- Heat of vaporization : More

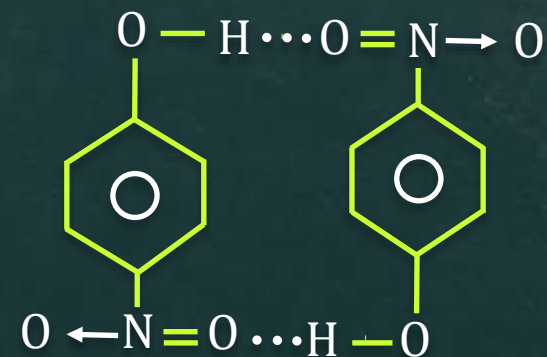
- Boiling point : More



Extra Amount ↑↑

Example

para-nitrophenol



Concept :- 32

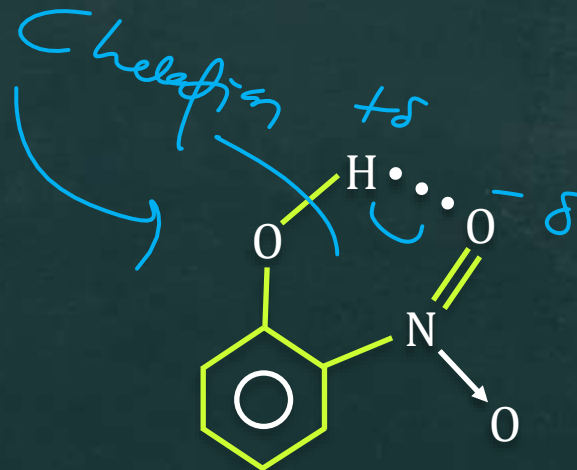
Topic Name :- Hydrogen Bonding

Intramolecular H-bonding

- Within molecule
- Ring- ■ Chelation
- Viscosity : Less
- Heat of vaporization : Less
- Boiling point : Less

Example

ortho-nitrophenol



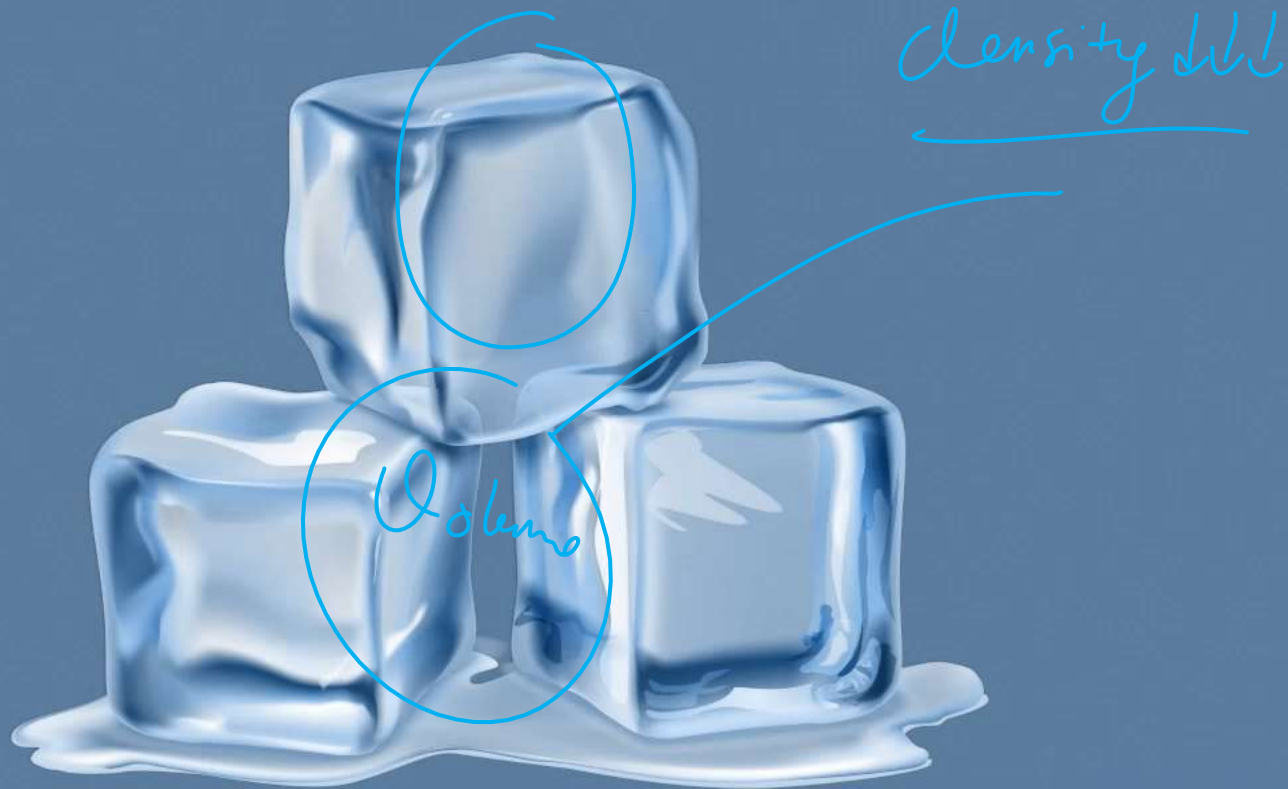
Concept :- 32

Topic Name :- Hydrogen Bonding

Hydrogen Bonding in Ice

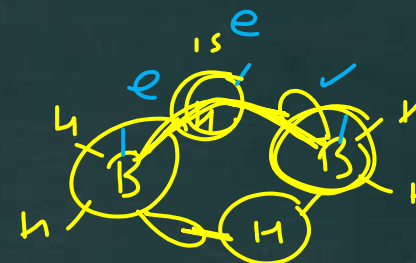
Ice has Cage like structure, so its volume increases and density decreases.

Cage → density ↓↓↓



Concept :- 33

Topic Name :- Bridge Bonding



Compound	Bridge	Hybrid	Planar/ Non-planar	Structure
B_2H_6	$3c-2e^-$	sp^3-s-sp^3	Non-planar	
Al_2H_6	$3c-2e^-$	sp^3-s-sp^3	Non-planar	
Be_2H_4	$3c-2e^-$	sp^3-s-sp^3	Planar	

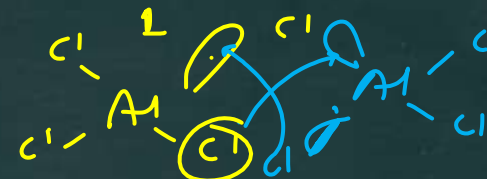
Concept :- 33

Topic Name :- Bridge Bonding

Compound	Bridge	Hybrid	Planar/ Non-planar	Structure
$(\text{BeH}_2)_n$	$3c-2e^-$	sp^3-s-sp^3	Non-planar <i>2c-2e to 3c-2e</i>	
$\text{Al}_2(\text{CH}_3)_6$	$3c-2e^-$	$sp^3-sp^3-sp^3$	Non-planar	

Concept :- 33

Topic Name :- Bridge Bonding

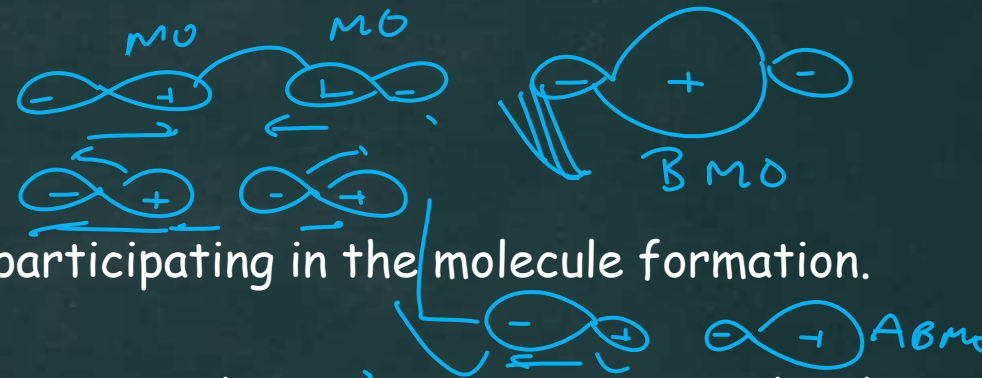


Compound	Bridge	Hybrid	Planar/ Non-planar	Structure
<u>Al_2Cl_6</u>	<u>$3\text{c}-4\text{e}^-$</u>	<u>$\text{sp}^3-\text{sp}^3-\text{sp}^3$</u>	Non-planar	
<u>Al_2Br_6</u>	$3\text{c}-4\text{e}^-$	$\text{sp}^3-\text{sp}^3-\text{sp}^3$	Non-planar	
<u>Be_2Cl_4</u>	$3\text{c}-4\text{e}^-$	$\text{sp}^2-\text{sp}^3-\text{sp}^2$	Planar	

Concept :- 34

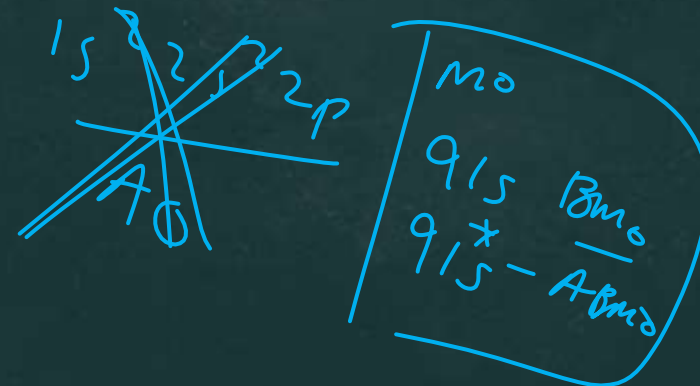
Topic Name :- Molecular Orbital Theory

According to MOT



- All atomic orbitals are participating in the molecule formation.
- As many number of atomic orbitals are participating in molecule formation, same number of molecular orbitals will be formed.
- Formed molecular orbitals are having varying energy and filling of electron in molecular orbitals follows these three rules or principles:

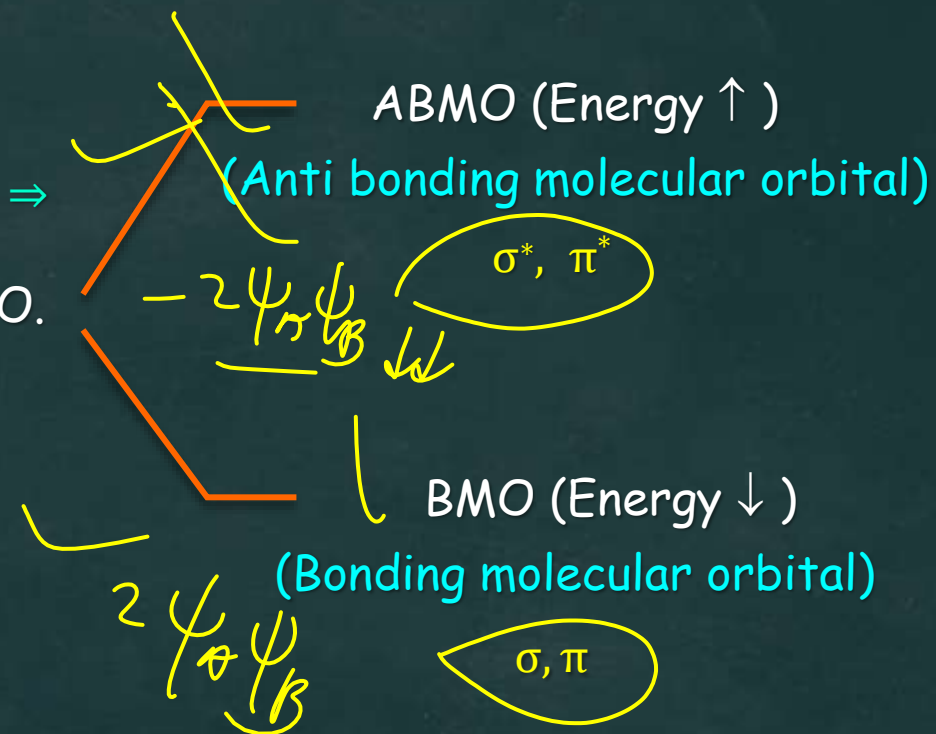
1. Pauli's Exclusion Principle
2. Aufbau Principle
3. Hund's rule



Concept :- 34

Topic Name :- Molecular Orbital Theory

Molecular Orbital Formation $\Rightarrow$

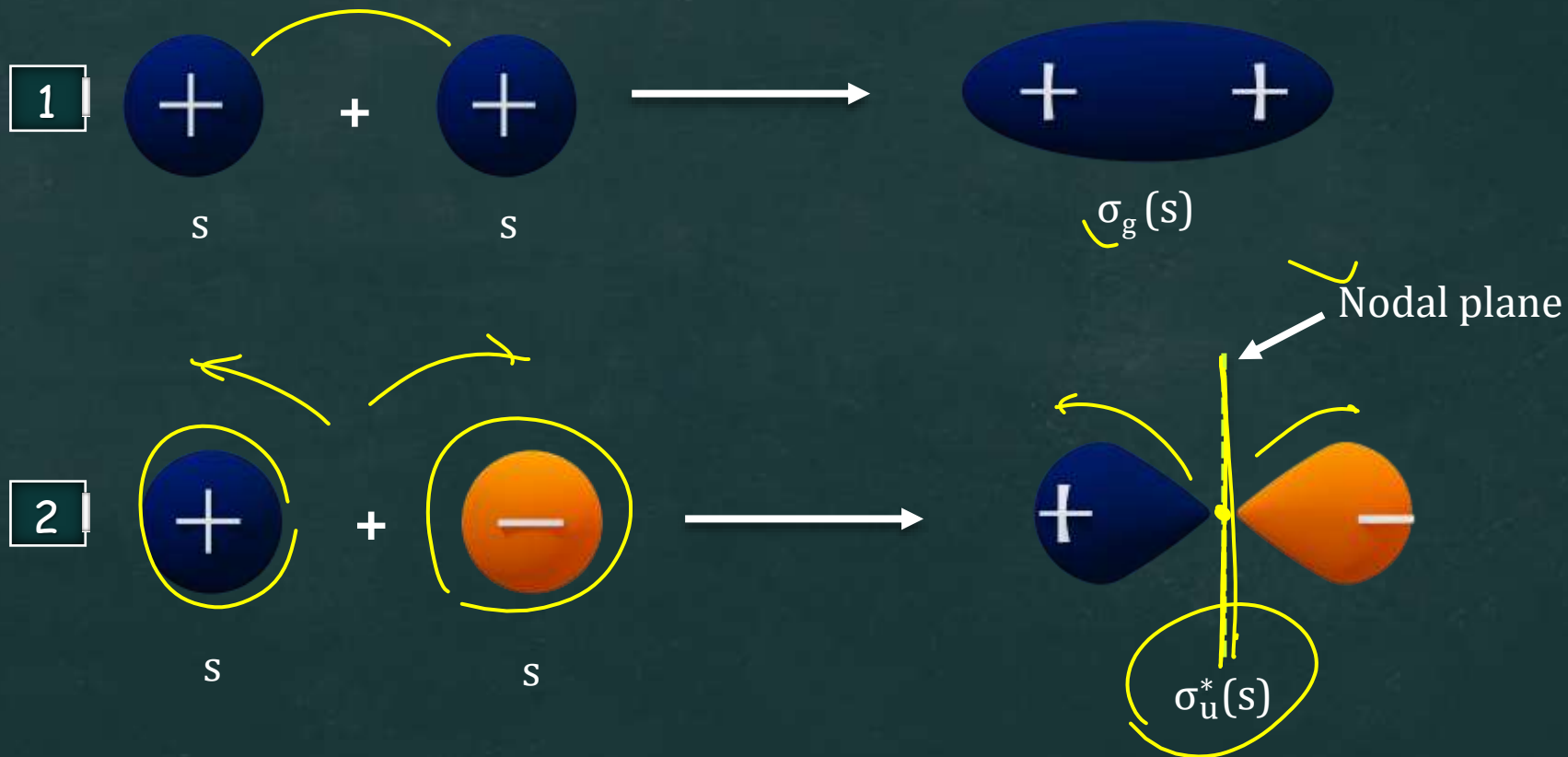


Concept :- 34

Topic Name :- Molecular Orbital Theory

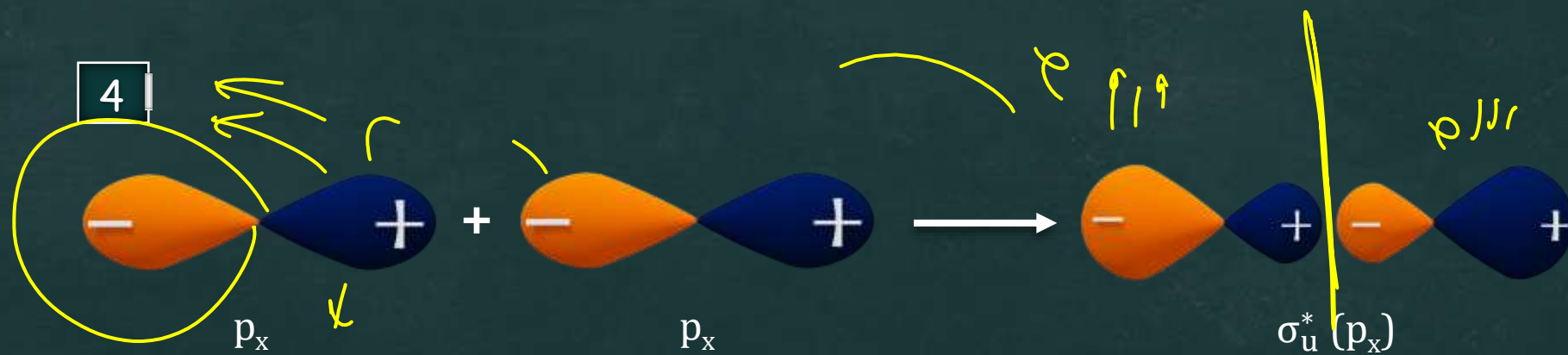
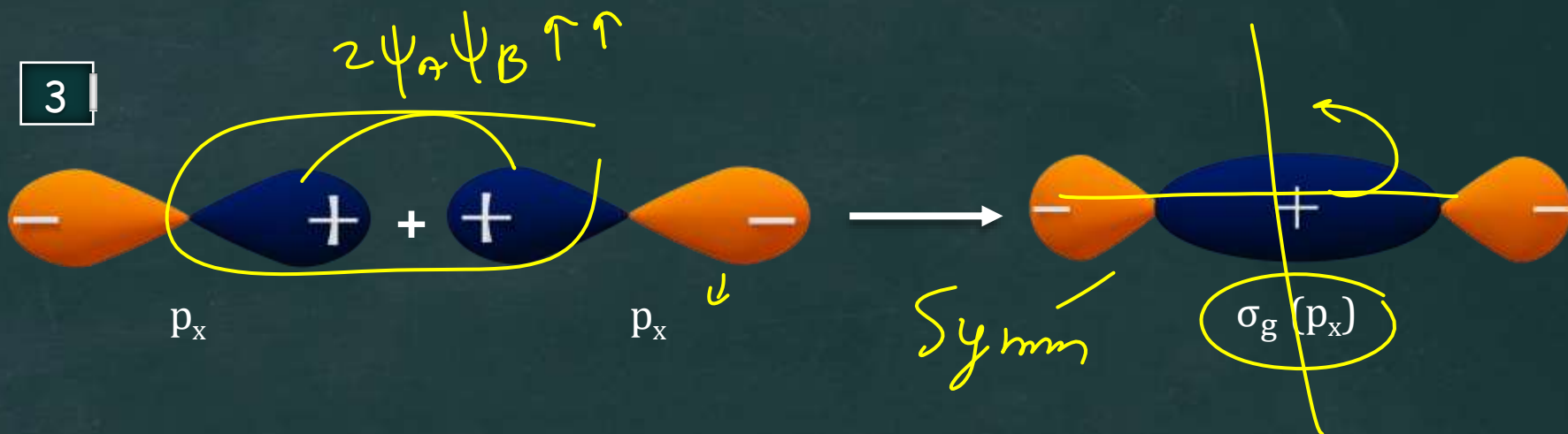
Linear Combination of Atomic Orbitals

σ molecular orbital : It is formed by following ways :



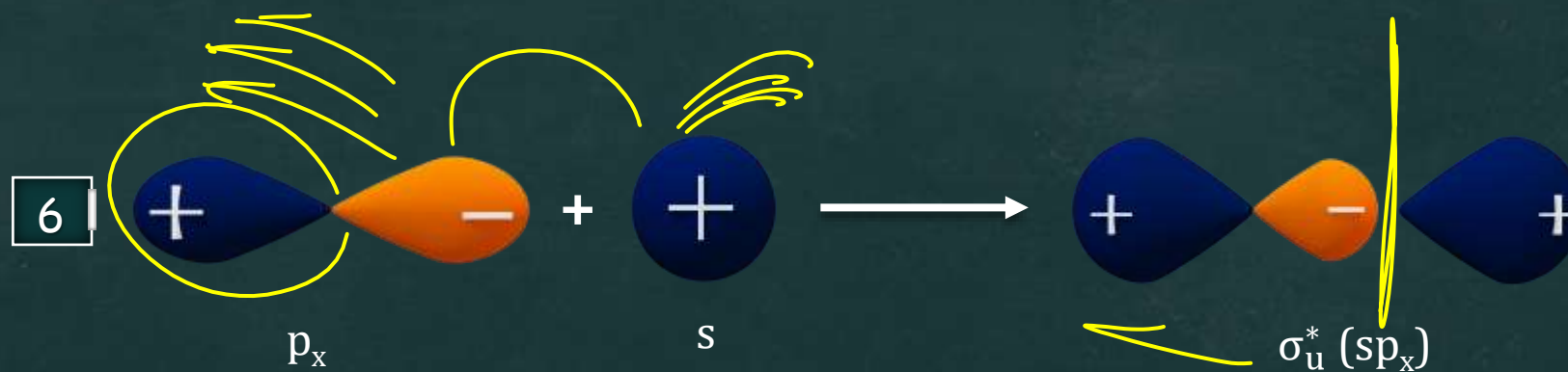
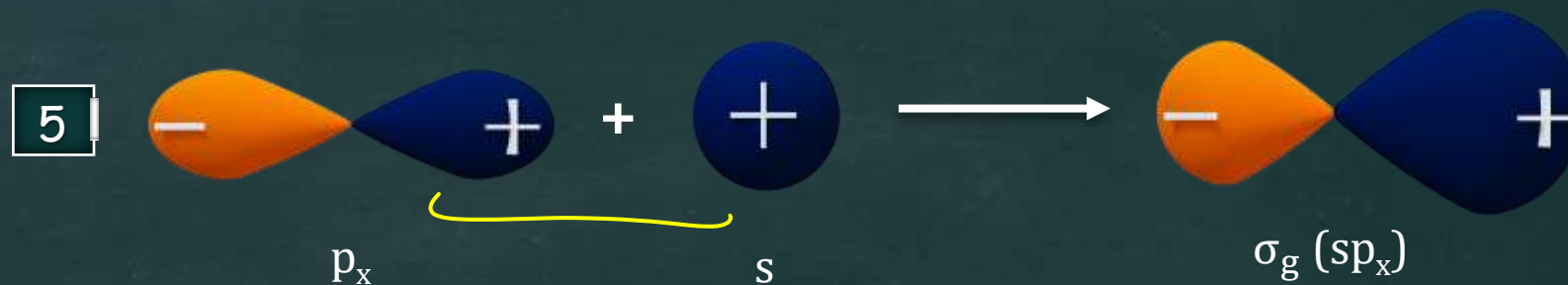
Concept :- 34

Topic Name :- Molecular Orbital Theory



Concept :- 34

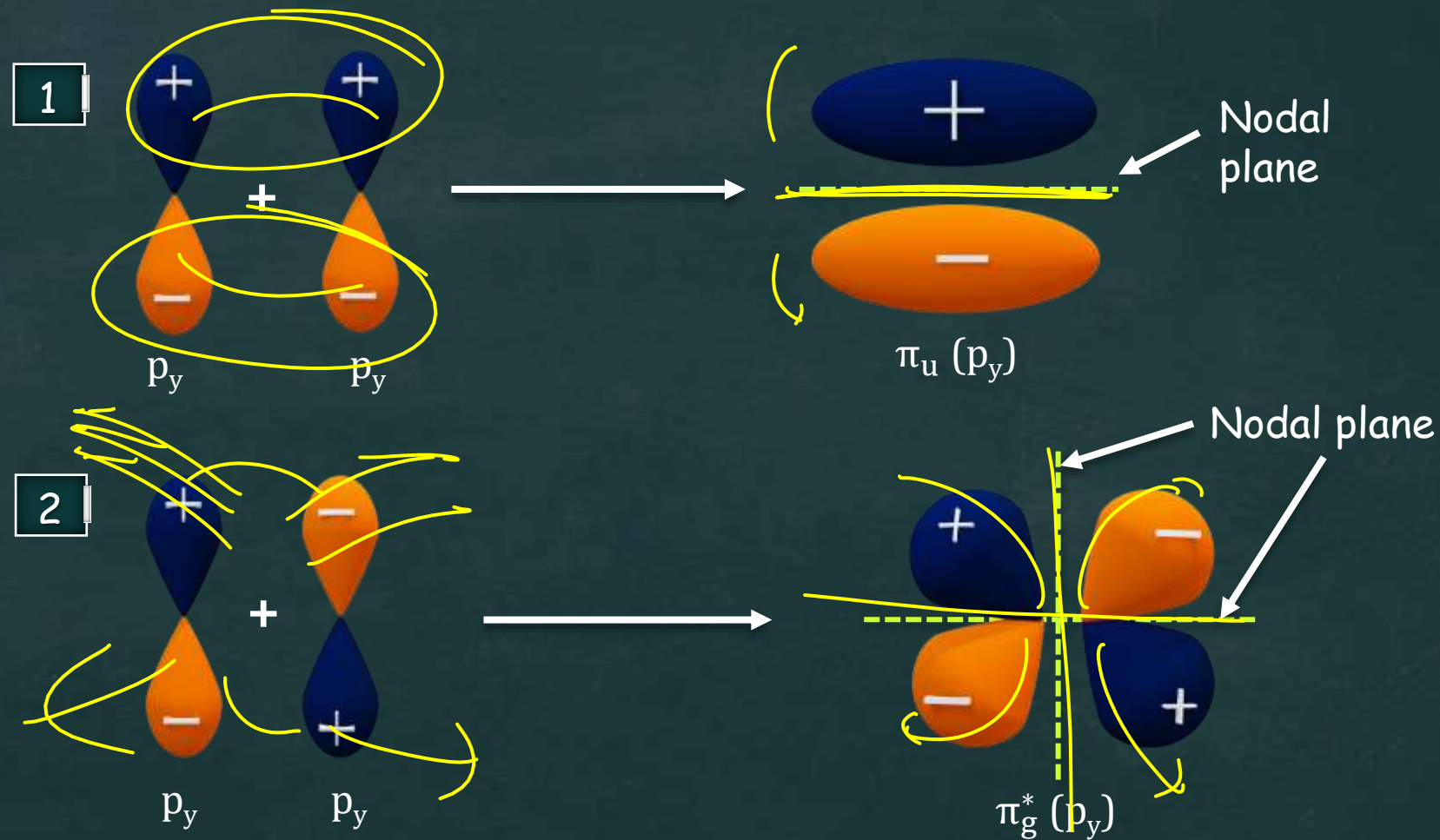
Topic Name :- Molecular Orbital Theory



Concept :- 34

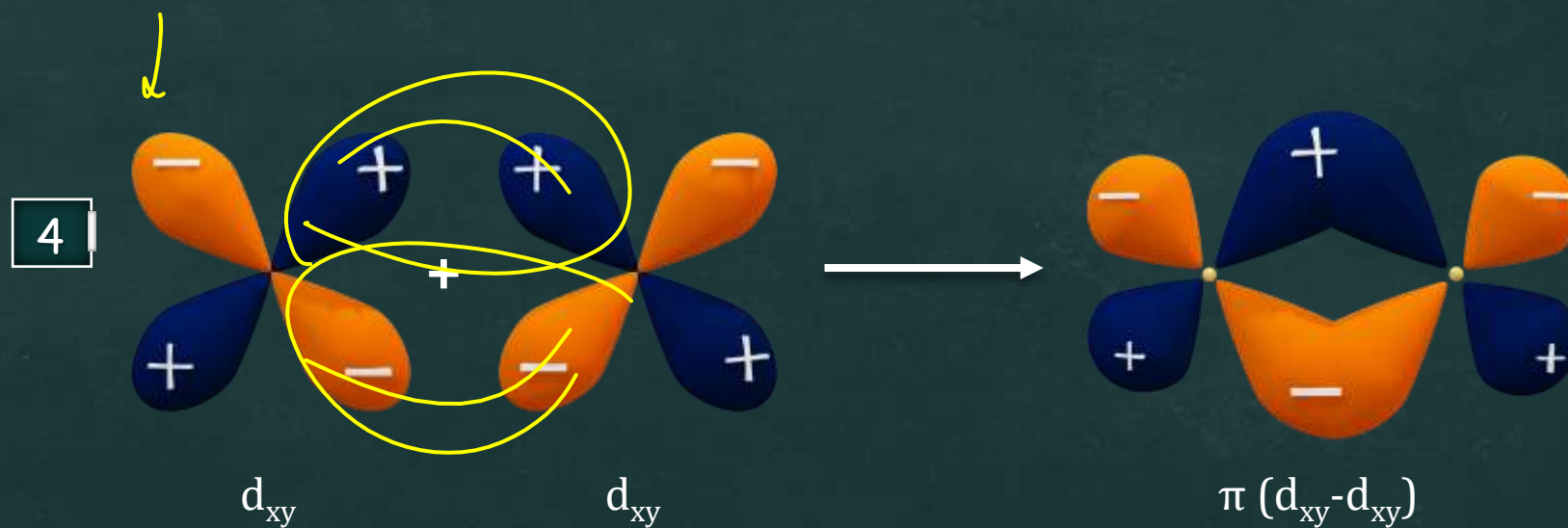
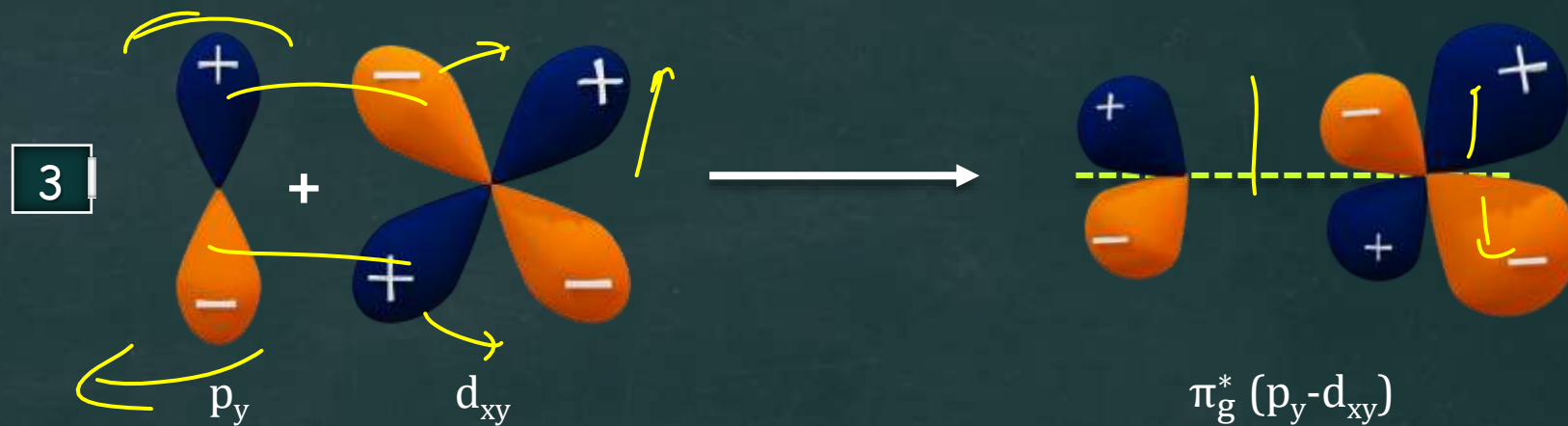
Topic Name :- Molecular Orbital Theory

π molecular orbital : It is formed by following ways :



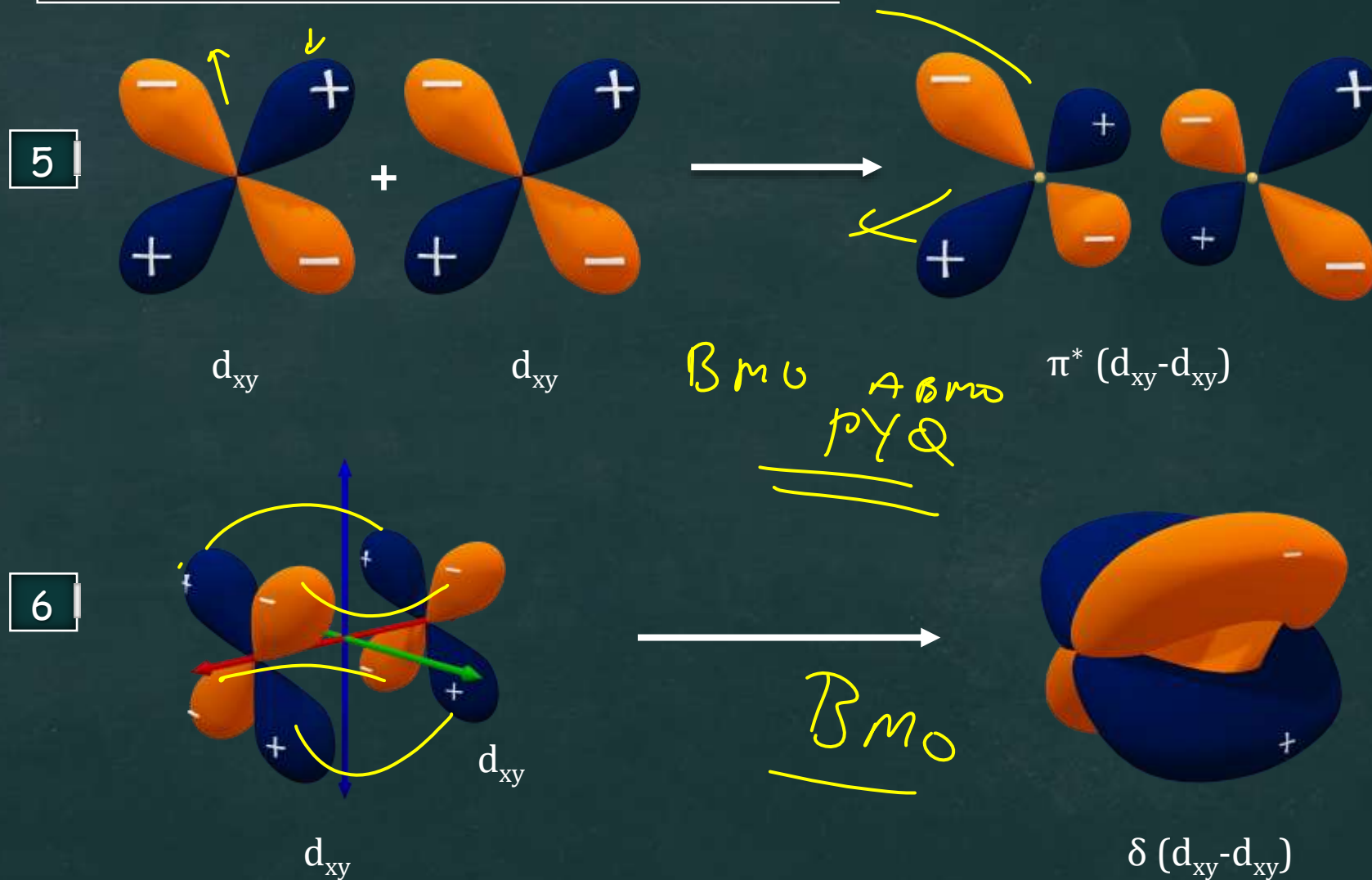
Concept :- 34

Topic Name :- Molecular Orbital Theory



Concept :- 34

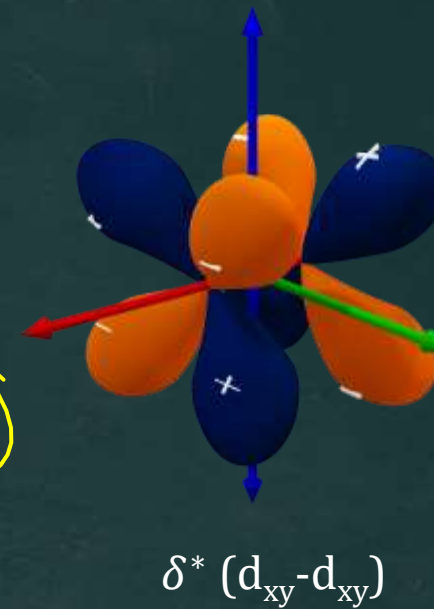
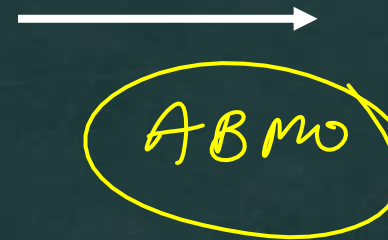
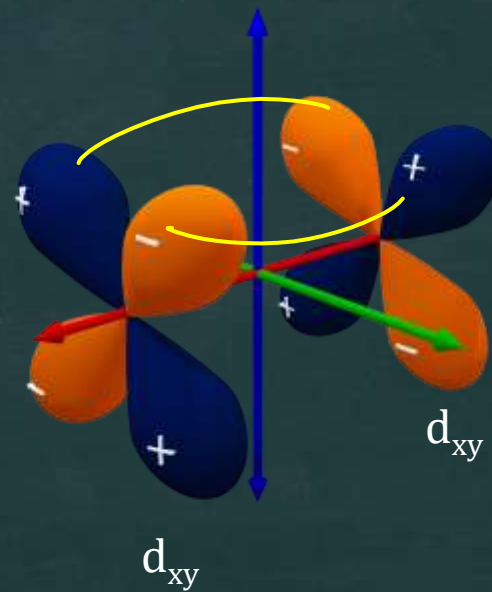
Topic Name :- Molecular Orbital Theory



Concept :- 34

Topic Name :- Molecular Orbital Theory

7



Concept :- 34

Topic Name :- Molecular Orbital Theory

H_2^+ , H_2^- , He_2^+
 H_2, He_2 and their charge species. (sp-mixing not present)

$$\sigma_{1s} < \sigma_{1s}^*$$

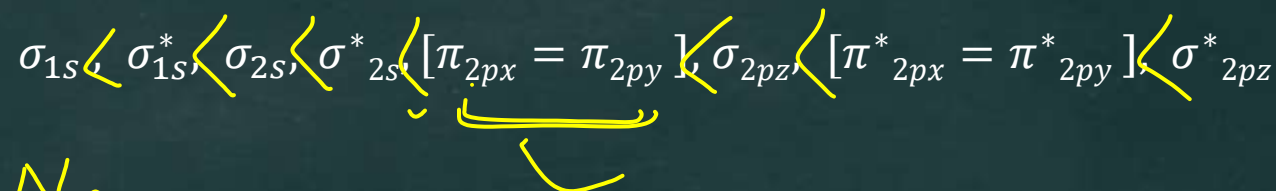
$$H_2 \Rightarrow 2e \quad \sigma_{1s}^2 \quad \sigma_{1s}^{*0} = \frac{1}{2}(2-0)$$

$$B.O. = \frac{1}{2}(N_{BMO} - N_{ABMO}) = 1$$

Concept :- 34

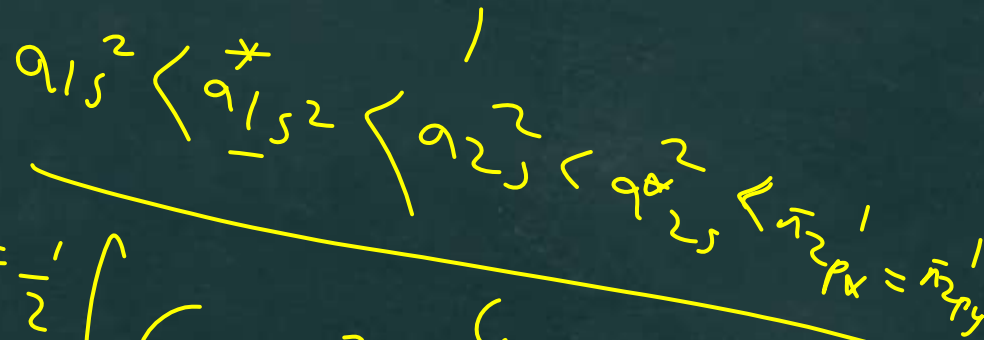
Topic Name :- Molecular Orbital Theory

• $\text{Li}_2, \text{Be}_2, \text{B}_2, \text{C}_2, \text{N}_2$ and their charged species (sp -mixing present)



$\text{N}_2 \rightarrow$

$\text{B}_2 = 10$



$$B.O = \frac{1}{2} (6 - 4) = 1$$

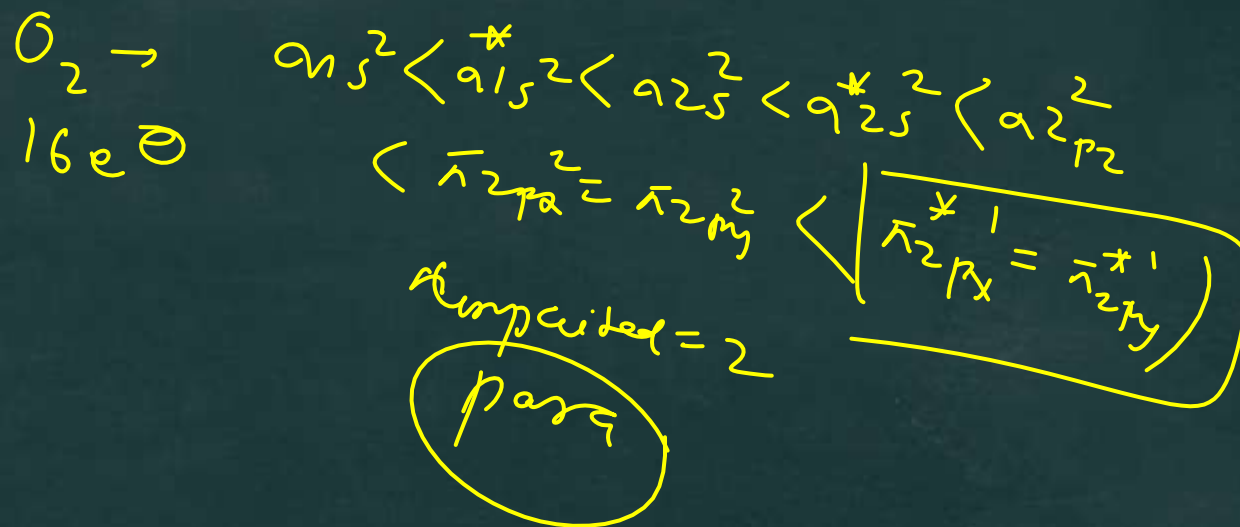
Unpaired = 2
paramagnetic

Concept :- 34

Topic Name :- Molecular Orbital Theory

• O_2 , F_2 and Ne_2 and their charged species (O_2^+ , O_2^- , O_2^{2-}) (sp-mixing not present)

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < [\pi_{2px} = \pi_{2py}] < [\pi_{2px}^* = \pi_{2py}^*] < \sigma_{2pz}^*$$



Concept :- 34

Topic Name :- Molecular Orbital Theory

total no. of e⁻ 10, 16
even → Dia
odd → para

Species (molecular or ions)	Total number of e ⁻ s	Bond Order	Number of unpaired electrons (n)	Remark or magnetic behavior
H ₂ ⁺	1	0.5 ✓	1	Para magnetic
H ₂ , He ₂ ²⁺	2 -	1	0	Diamagnetic
H ₂ ⁻ , He ₂ ⁺	3	0.5	1	Para magnetic
He ₂	4 .	0	0	Diamagnetic
Li ₂ ⁺ , He ₂ ⁻	5	0.5	1	Para magnetic
Li ₂ , He ₂ ²⁻ , Be ₂ ²⁺	6	1	0	Diamagnetic
Be ₂ ⁺ , Li ₂ ⁻	7	0.5	1	Para magnetic
Be ₂ , Li ₂ ²⁻	8	0	0	Diamagnetic
Be ₂ ⁻ , B ₂ ⁺	9	0.5	1	Para magnetic
B ₂ , Be ₂ ²⁻ , HF	10	1	2	Para magnetic

Concept :- 34

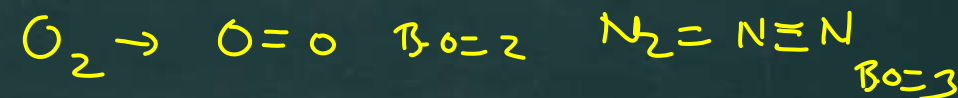
Topic Name :- Molecular Orbital Theory

Species (molecular or ions)	Total number of e^- s	Bond Order	Number of unpaired electrons (n)	Remark or magnetic behavior
B_2^-, C_2^+	11	1.5	1	Para magnetic
$C_2, B_2^{2-}, N_2^{2+}, CN^+$	12	2	0	Diamagnetic
C_2^-, N_2^+	13	2.5	1	Para magnetic
$N_2, CO, NO^+, C_2^{2-},$ CN^-, O_2^{2+}	14	3	0	Diamagnetic
N_2^-, NO, O_2^+	15	2.5	1	Para magnetic
NO^-, O_2	16	2	2	Para magnetic
O_2^-	17	1.5	1	Para magnetic
F_2, O_2^{2-}, HCl	18	1	0	Diamagnetic
F_2^-	19	0.5	1	Para magnetic
Ne_2	20	0	0	Diamagnetic

Concept :- 35

Topic Name :- Bond Order, Bond Strength, Bond Energy, Bond Length

Bond Order



Effective number of bonds between two atoms in a molecules

Bond Length

$$\text{B.O} \propto \text{B.S} \propto \frac{1}{\text{B.L}}$$

The higher the bond order, the stronger the bond between two atoms and hence shorter the bond length.

Bond Strength

As the bond order increases, the atoms in bonding come closer to each other so the bond strength increases.

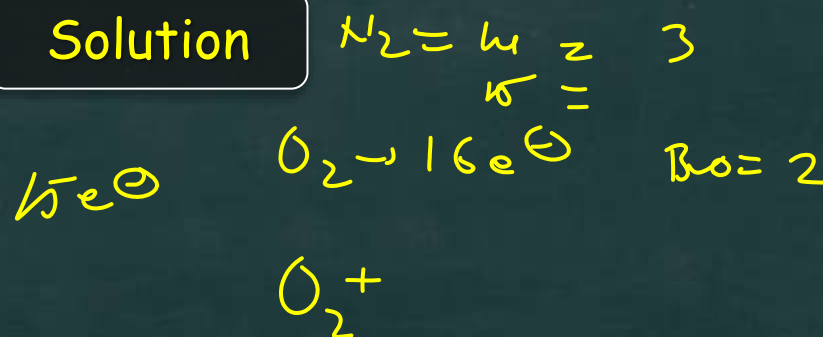
$$\text{Bond order} \propto \text{Bond strength} \propto \text{B.D.E.} \propto \text{B.E.} \propto \frac{1}{\text{BL}}$$



Example

Compare bond order (BO), bond length (BL) and bond strength in O_2 , O_2^+ , O_2^- , O_2^{2+} , O_2^{2-} ?

Solution



$$B.O \propto B.S \propto \frac{1}{B.L}$$

14	-	3
15	-	2.5
16	-	2
17	-	1.5
18	-	1
19	-	0.5

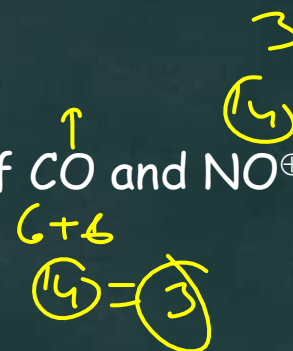
Ans.

- $B.O = O_2^{2+} > O_2^+ > O_2 > O_2^- > O_2^{2-}$
- $B.L = O_2^{2+} < O_2^+ < O_2 < O_2^- < O_2^{2-}$
- $B.E = O_2^{2+} > O_2^+ > O_2 > O_2^- > O_2^{2-}$



Example

The difference between bond orders of CO and NO^+ is $\frac{x}{2}$ where $x = \underline{\hspace{1cm}}$.
(Round off to the Nearest Integer)



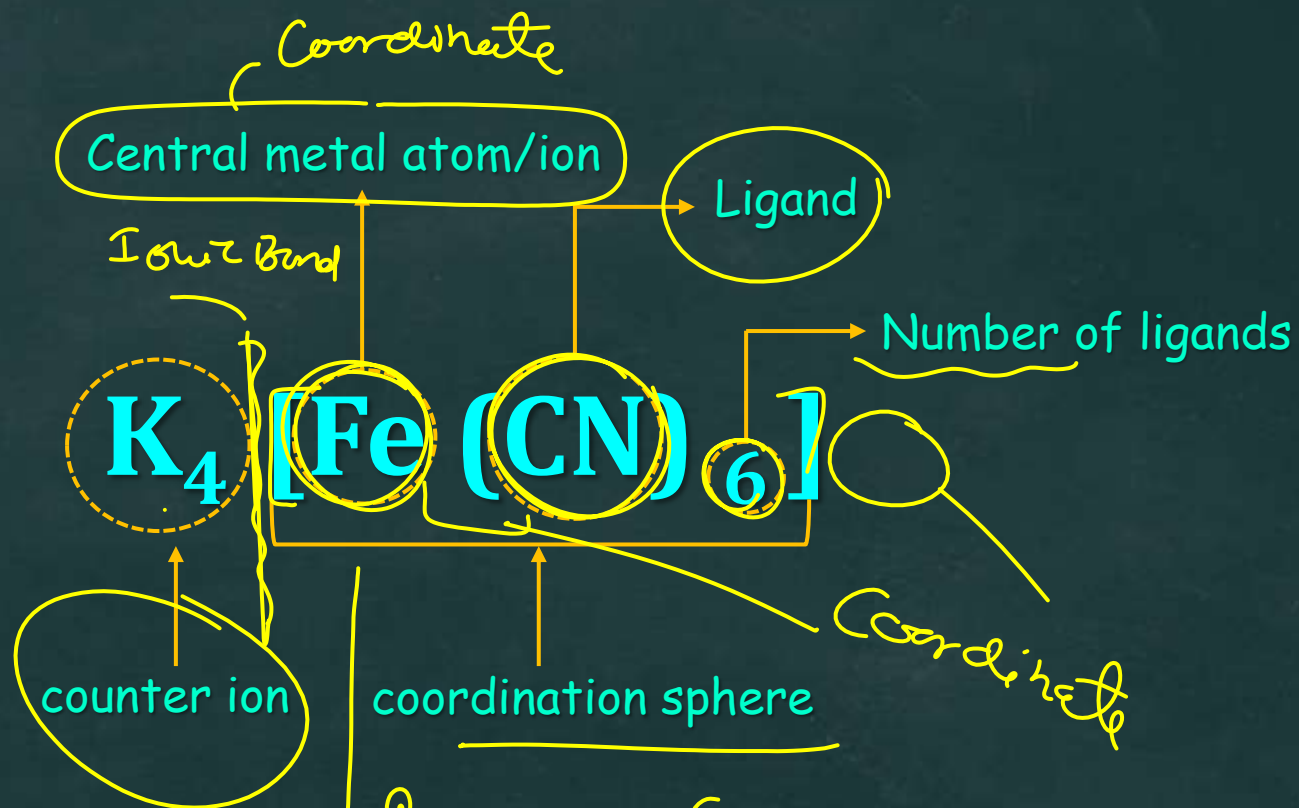
[JEE Mains-2021]

Solution

Ans. (0)

Concept :- 36

Topic Name :- Terminology in Coordination

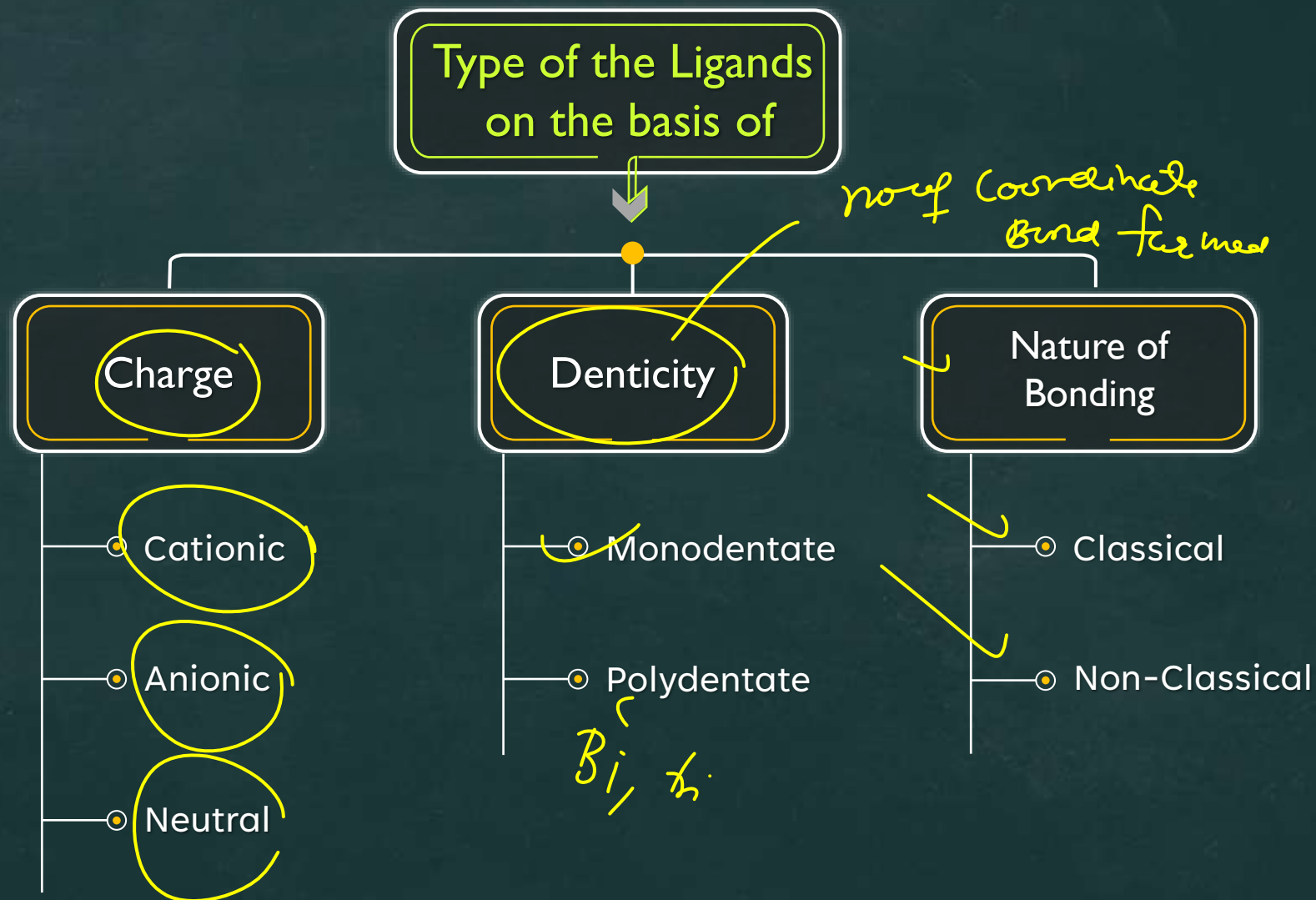


Number of ligands = 6
Coordination number = 6



Concept :- 37

Topic Name :- Type of Ligands



Concept :- 37

Topic Name :- Type of Ligands

Amido $\text{N}_2 = \text{N} \equiv \text{N}$

Ligand Formula	Name of Ligand	Monodentate	Anionic Ligand	Cationic Ligand	Neutral Ligand
NH_3 ✓	ammine / ammonia ✓	✓	✗	✗	✓
NH_2^- ✓	amido ✓	✓	✓	✗	✗
NH^{2-} ✓	imido ✓	✓	✓	✗	✗
N^{3-} ✓	nitrido ✓	✓	✓	✗	✗
N_3^- ✓	azido ✓	✓	✓	✗	✗
F^- ✓	fluoro / fluorido	✓	✓	✗	✗
Cl^- ✓	chloro / chlorido	✓	✓	✗	✗
H_2O ✓	aqua	✓	✗	✗	✓
CO ✓	carbonyl	✓	✗	✗	✓
NO ✓	nitrosyl	✓	✗	✗	✓
PH_3 ✓	phosphine	✓	✗	✗	✓

Concept :- 37

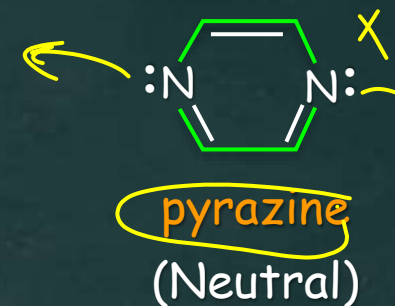
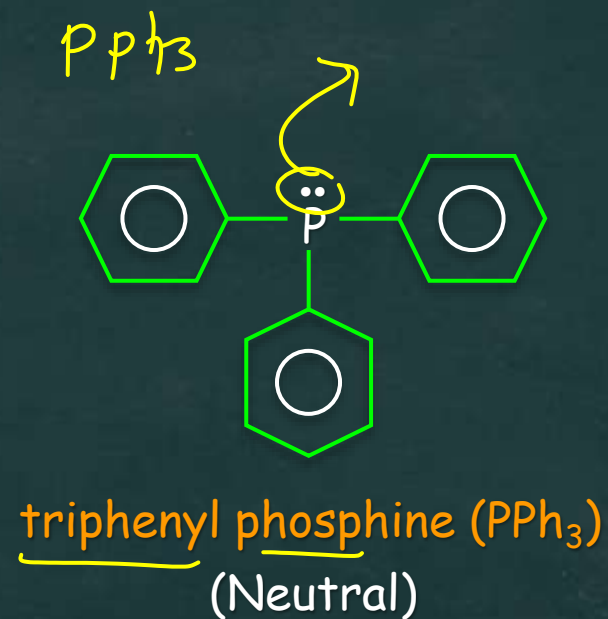
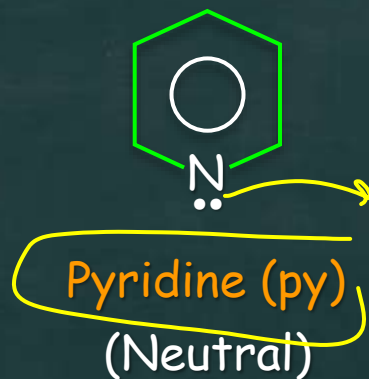
Topic Name :- Type of Ligands

Ligand Formula	Name of Ligand	Monodentate	Anionic Ligand	Cationic Ligand	Neutral Ligand
Br^-	bromo/bromido	✓	✓	✗	✗
I^-	iodo/iodido	✓	✓	✗	✗
H^-	hydrido	✓	✓	✗	✗
O^{2-}	oxo/oxido	✓	✓	✗	✗
O_2^{2-}	peroxo/peroxido	✓	✓	✗	✗
O_2^-	superoxo/superoxido	✓	✓	✗	✗
$\text{CH}_3\text{COO}^\ominus$	acetato	✓	✓	✗	✗
CO_3^{2-}	carbonato	✓	✓	✗	✗
SO_4^{2-}	sulphato	✓	✓	✗	✗
OH^-	hydroxo/hydroxide	✓	✓	✗	✗
$\text{S}_2\text{O}_3^{2-}$	thiosulphato	✓	✓	✗	✗

Concept :- 37

Topic Name :- Type of Ligands

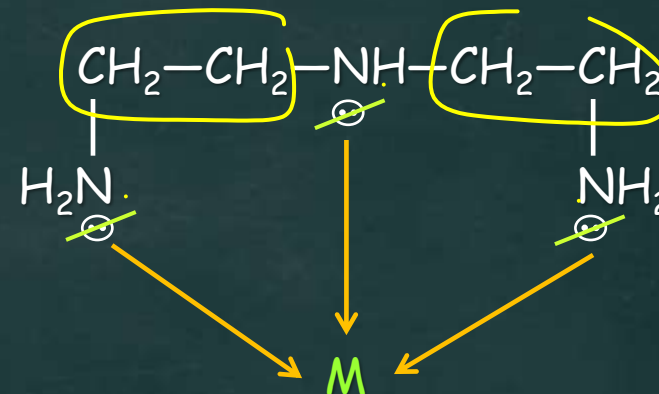
Structure of some monodentate ligands



Concept :- 37

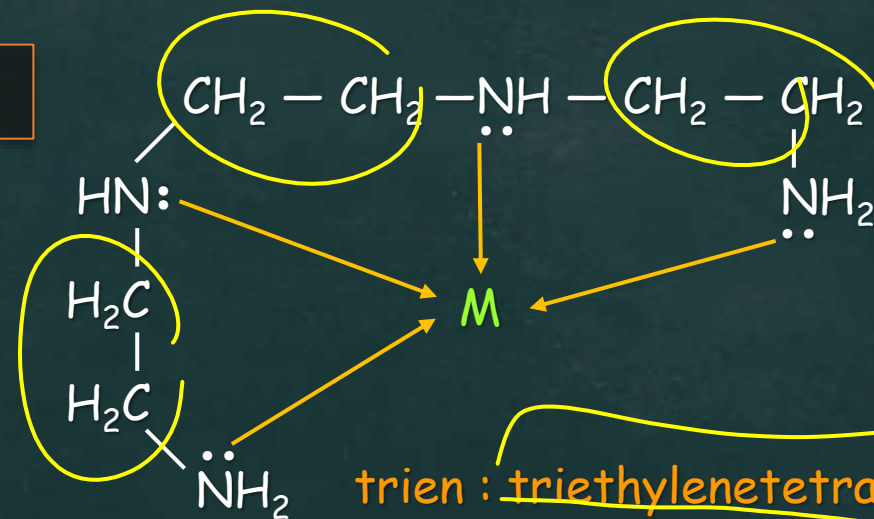
Topic Name :- Type of Ligands

(B) Tridentate ligands



dien Diethylenetriamine

(C) Tetradentate ligands

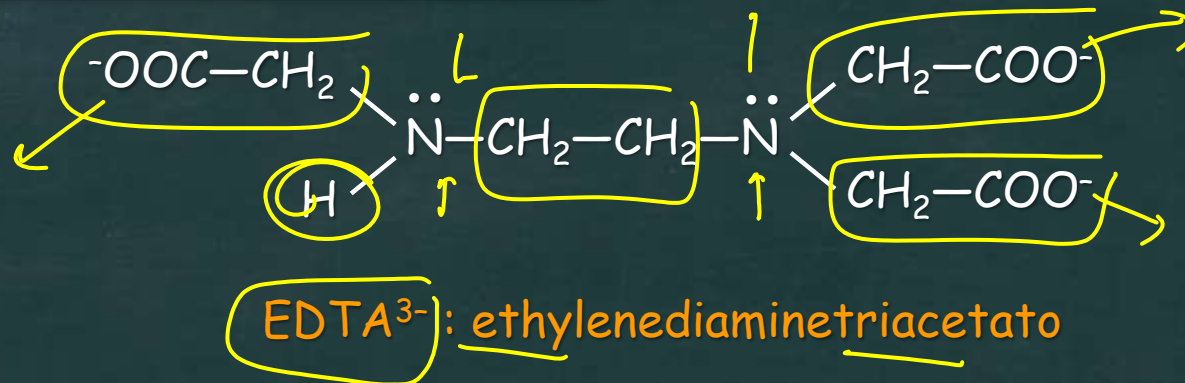


trien : triethylenetetraamine

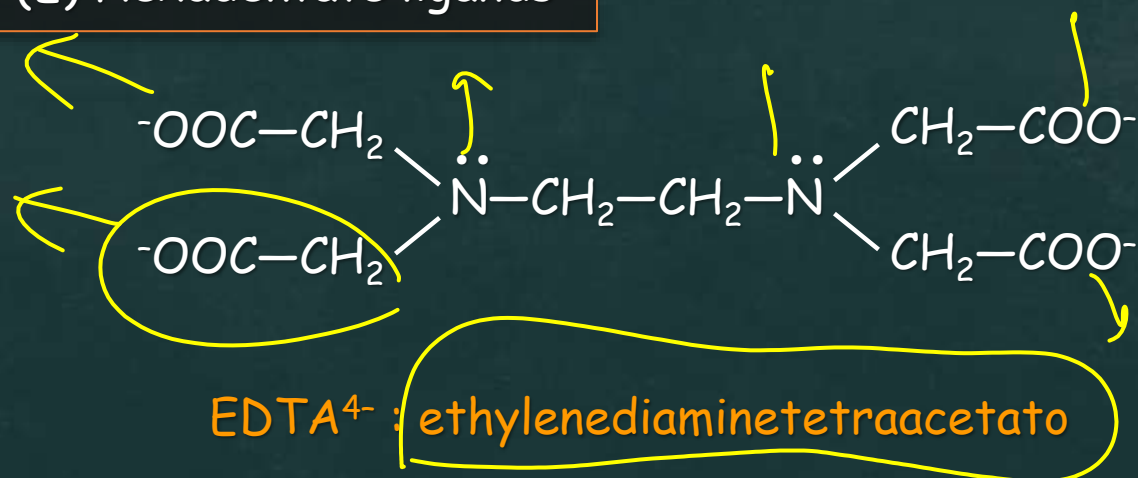
Concept :- 37

Topic Name :- Type of Ligands

(D) Pentadentate ligands



(E) Hexadentate ligands



 Example

The total number of coordination sites in ethylenediaminetetraacetate (EDTA^{4-}) is _____.

[JEE Mains-2020]

Solution

Here,

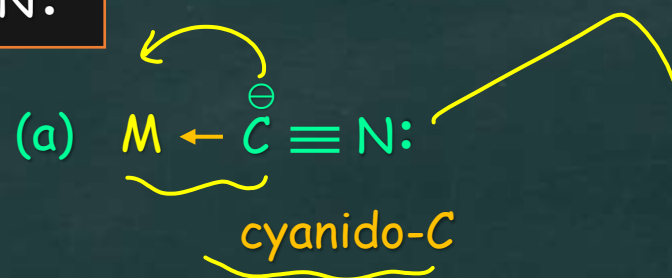
6 $\rightarrow$ Coordination

Ans. (6)

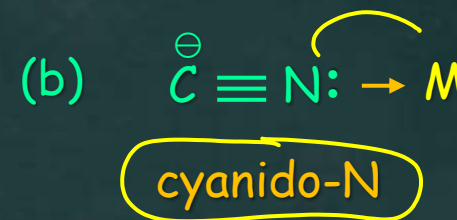
Concept :- 37

Topic Name :- Type of Ligands

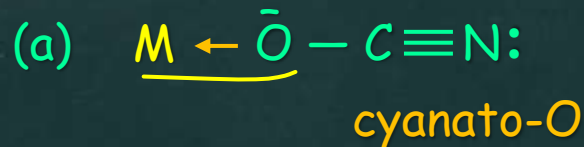
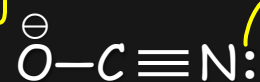
Ambidentate ligands



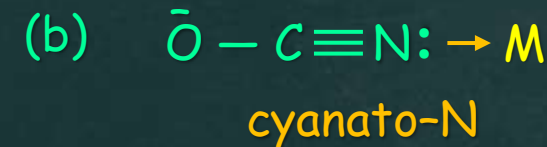
(C-atom coordinated to metal)



(N-atom coordinated to metal)



(O-atom coordinated to metal)



(N-atom coordinated to metal)



Example

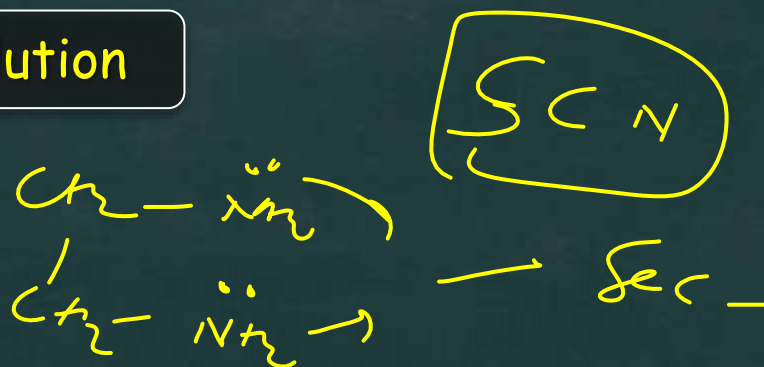
Number of ambidentate ligands in a representative metal complex



[JEE Mains-2023]

[en = ethylenediamine]

Solution



Ans. (4)

Concept :- 37

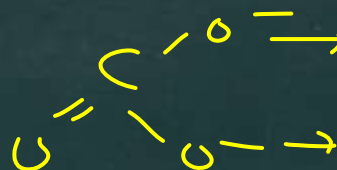
Topic Name :- Type of Ligands

Flexidentate ligands

mono

 Ligands which show flexibility in denticity are called flexidentate ligand.

Example



CO_3^{2-} , SO_4^{2-} , SO_3^{2-} , NO_3^- , N_2 , O_2 , $\ddot{\text{N}}\text{H}_2 - \ddot{\text{N}}\text{H}_2$, PO_4^{3-} , $\text{CH}_3 - \text{COO}^-$ etc.

Bridging ligands

Example

$\text{H}_2\ddot{\text{O}}:$, $\ddot{\text{O}}\text{H}^-$, $:\ddot{\text{N}}\text{H}_2^-$, F^- , Cl^- , Br^- , I^- , O^{2-} , $:\text{SCN}^-:$

Concept :- 37

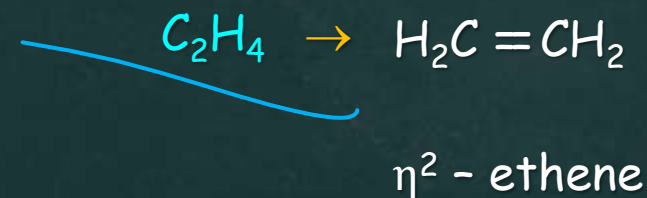
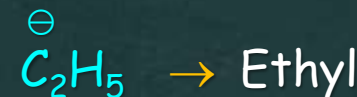
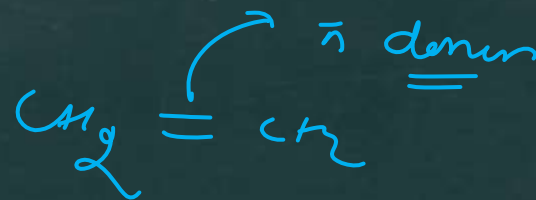
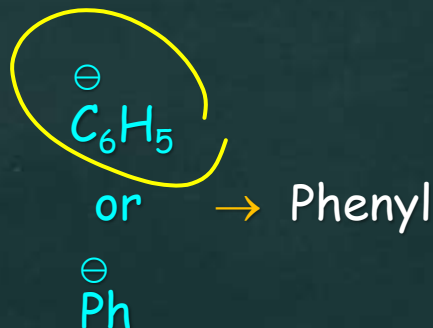
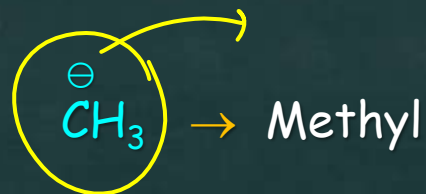
Topic Name :- Type of Ligands

Organic ligands



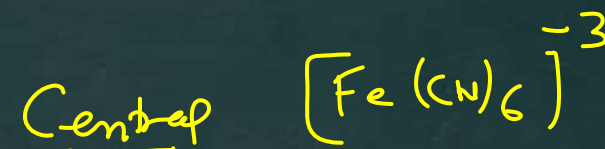
No suffix will be added in the name of organic ligand.

Example

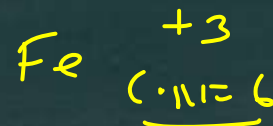


Concept :- 38

Topic Name :- Werner's Theory

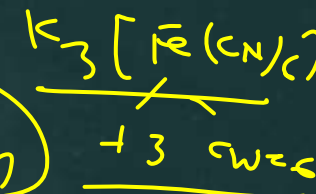


According to Werner coordination compounds metal have two types of valencies.



Primary valency

C.M - Oxidation



C.N - no of coordinate
Bond formed by C.M

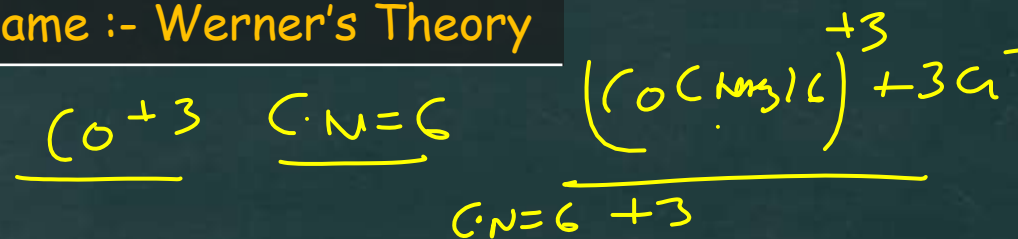
Secondary valency

Corresponds to the oxidation
number of metal

Corresponds to the coordination
number of metal

Concept :- 38

Topic Name :- Werner's Theory



Compounds		Colour	No. of Charge on Complex Ion	No. of Ions		
Old Formula	New Formula			Cation	Anion	Total
<u>$\text{CoCl}_3 \cdot 6\text{NH}_3$</u>	<u>$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$</u>	Yellow	+3	1	3	4
$\text{CoCl}_3 \cdot 5\text{NH}_3$	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	Purple	+2	1	2	3
$\text{CoCl}_3 \cdot 4\text{NH}_3$	Trans- $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$	Green	+1	1	1	2
$\text{CoCl}_3 \cdot 3\text{NH}_3$	$[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$	Blue green	—	—	—	—



Example

The primary and secondary valencies of cobalt respectively in $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ are :

(1) 3 and 5

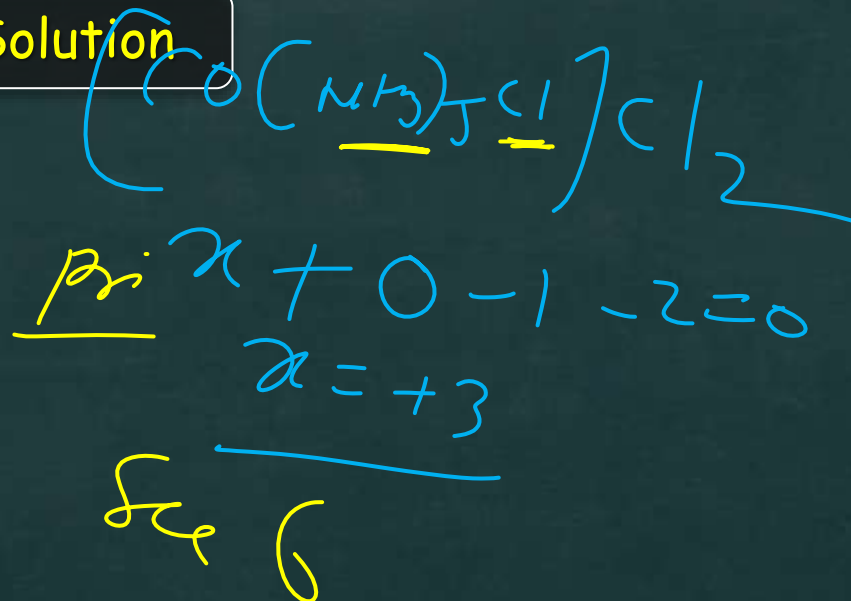
(2) 2 and 6

(3) 2 and 8

~~(4) 3 and 6~~

[JEE Mains-2023]

Solution



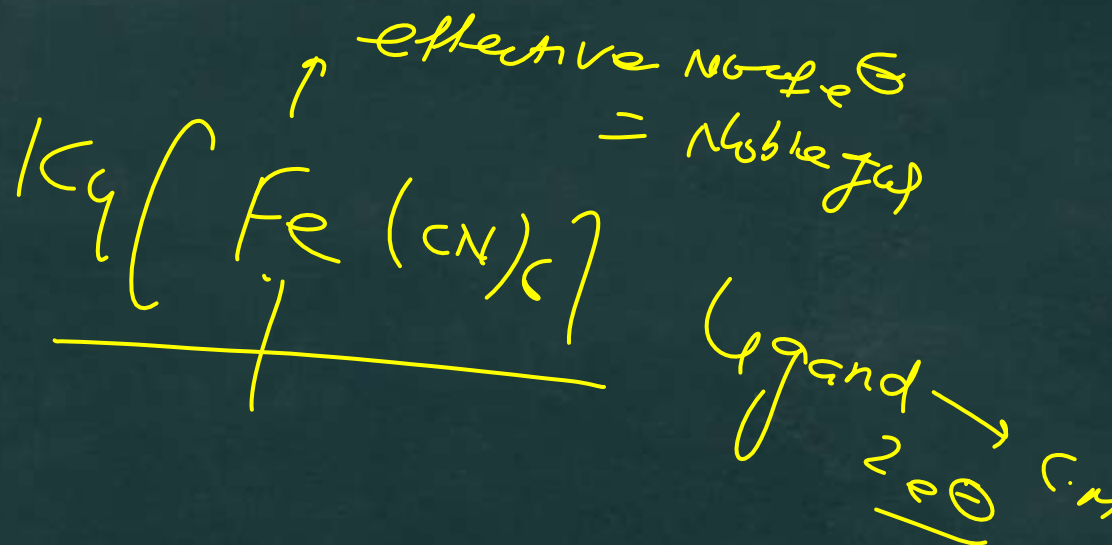
Ans. (4)

Concept :- 39

Topic Name :- Sidgwick theory of EAN

EAN of
central atom =

Atomic number of central atom - (O.N. of central atom)
+ (total number of electrons obtained from various ligands)



Concept :- 39

Topic Name :- Sidgwick theory of EAN

This rule is still well applicable for carbonyl compounds (Metal carbonyls).

Complex Ion	EAN Rule				EAN
	Atomic Number of the Metal	Oxidation State of the Metal	No. of Electrons on Metal Cation / Atom	No. of Electrons Donated by the Ligands	
$\text{Fe}(\text{CO})_5$	26	0	26	$2 \times 5 = 10$	$26 + 10 = 36(\text{Kr})$
$\text{Cr}(\text{CO})_6$	24	0	24	$2 \times 6 = 12$	$24 + 12 = 36(\text{Kr})$
$\text{Ni}(\text{CO})_4$	28	0	28	$2 \times 4 = 8$	$28 + 8 = 36(\text{Kr})$

$$\text{Metal} - \text{O} \times 5 \times 2 = \text{EAN}$$

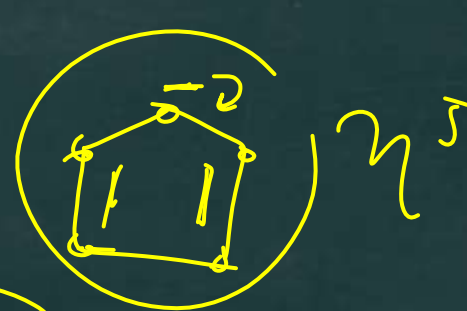
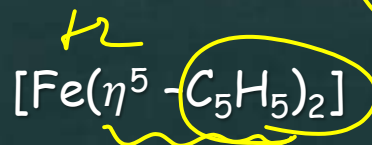
Concept :- 39

Topic Name :- Sidgwick theory of EAN

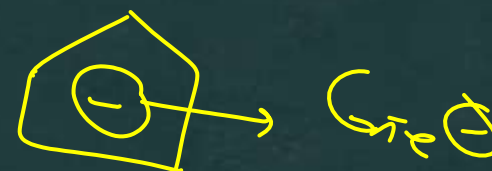
Calculation of E.A.N in π -bonded complexes

All π -electrons should be counted in calculation of EAN in π -bonded complexes

Example



$6\pi e^-$



$$EAN = 26 - 2 + 6 \times 2 = 36 \text{ (follow EAN)}$$

Concept :- 39

Topic Name :- Sidgwick theory of EAN

Note

✓ *For Coordinate*
 $(3e^-)$ $3e^-$ donor.
 Nitrosyl group (NO) is a 3 electron donor.

NO^+ ($2e^-$ donor)	NO ($3e^-$ donor)
<u>Brown ring complex</u> $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})^+]\text{SO}_4$ $\text{EAN} = 26 - 1 + 5 \times 2 + 1 \times 2 = 37$	$[\text{Cr}(\text{NO})_4]$ $\text{EAN} = 24 - 0 + 4 \times 3 = 36$

Concept :- 40

Topic Name :- IUPAC Nomenclature

Non-ionic or Neutral Complexes

Metal	Name
$\text{[Co(NH}_3\text{)}_3\text{(NO}_2\text{)}_3\text{]}$ <i>legone</i> <i>n-o-z-o n+3</i>	<u>triamminetrinitrocobalt(III)</u> <i>Or</i> triamminetrinitrito-N cobalt(III)
<u>Ni(CO)_4</u>	<u>tetracarbonylnickel(0)</u>

Concept :- 40

Topic Name :- IUPAC Nomenclature

Complexes Containing Complex Cations

Metal	Name
$[\text{Co}(\text{en})_3]^{\text{+3}}\text{Cl}_3$	<u>tris</u> (<u>ethylenediamine</u>) <u>cobalt</u> (III) <u>chloride</u>
$[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$	pentaaquanitrosyliron (I) sulphate

(Counter ion)

Concept :- 40

Topic Name :- IUPAC Nomenclature

Coordination Compounds Containing Complex Anions

metal + "ate"

Metal	Name
$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$	iron hexacyanoferrate(II)
$\text{Na}[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)]$	sodium amminebromochloronitrito-N platinate(II)

ate

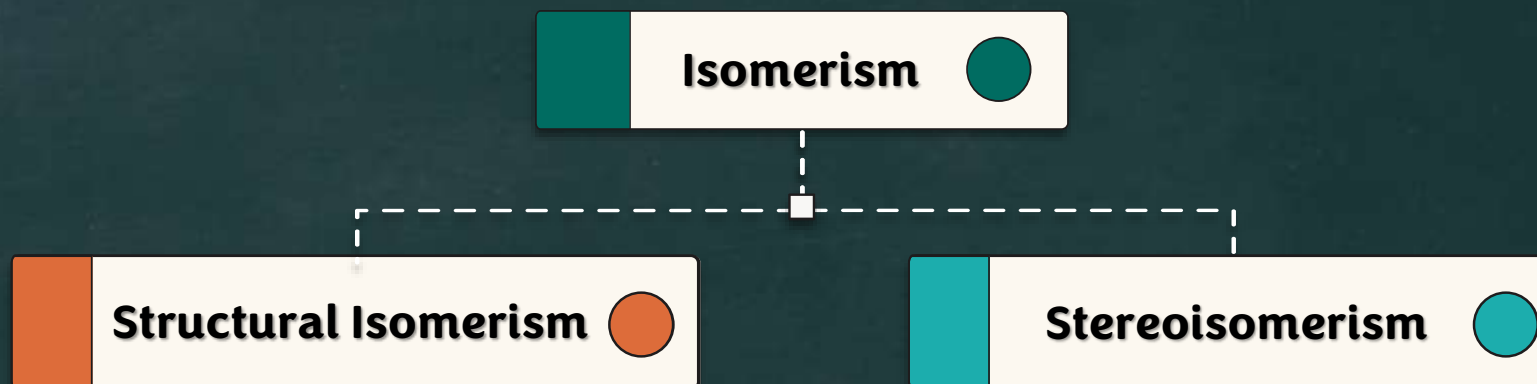
Concept :- 40

Topic Name :- IUPAC Nomenclature

Naming of Coordination compound having Cation and Anion both as Complex Ions

Cation *Anion* *at*

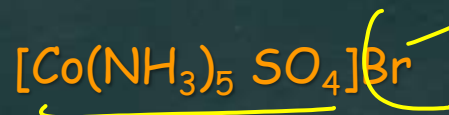
Metal	Name
$[Pt(NH_3)_4][PtCl_4]$	<u>tetraammineplatinum(II)</u> <u>tetrachloroplatinate(II)</u>
$[Pt(NH_3)_4Cl_2][PtCl_4]$	tetraamminedichloroplatinum(IV) tetrachloroplatinate(II)



Concept :- 41

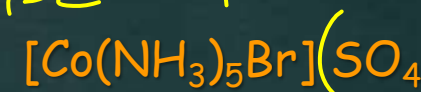
Topic Name :- Structural Isomerism

Ionisation isomerism

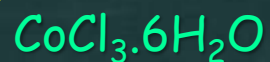


M.F - Same

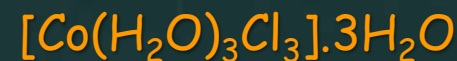
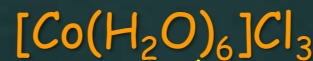
diff-2 ion



Hydrate or solvate isomerism



now 6 H₂O as ligand all diff-2

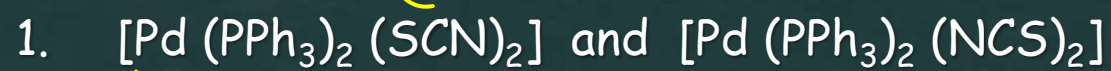


Concept :- 41

Topic Name :- Structural Isomerism

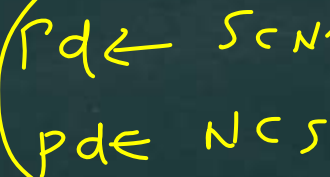
Linkage Isomerism

Example



→ Linkage Isomer ←

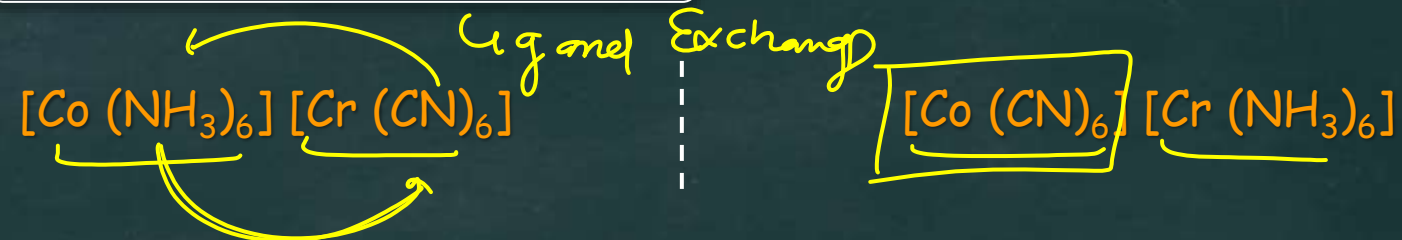
Ammin



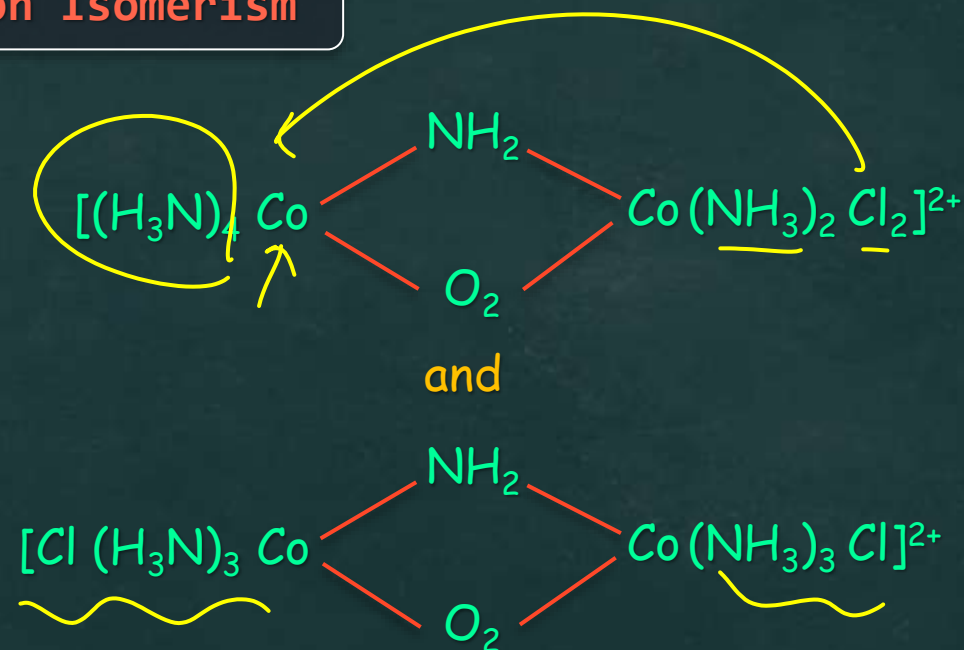
Concept :- 41

Topic Name :- Structural Isomerism

Coordination Isomerism



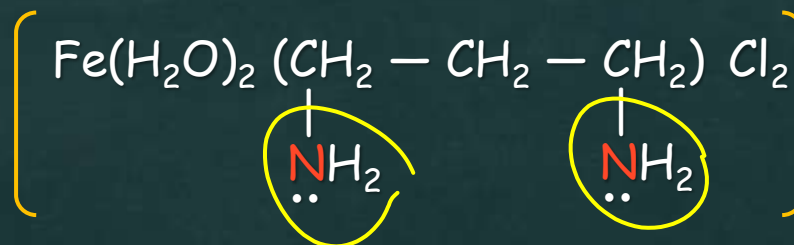
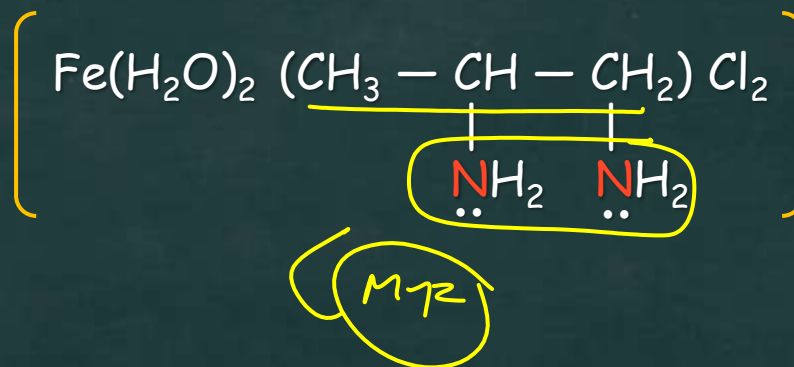
Coordination Position Isomerism



Concept :- 41

Topic Name :- Structural Isomerism

Ligand Isomerism

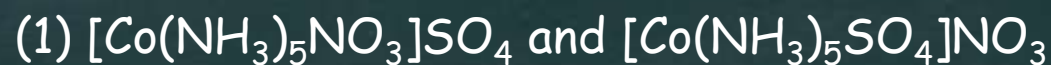




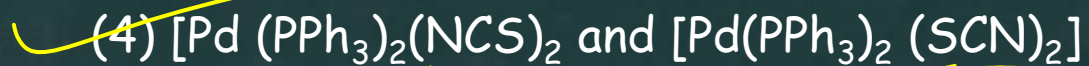
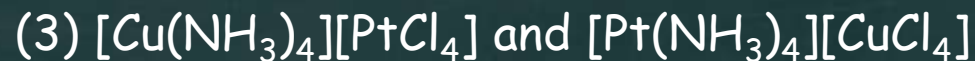
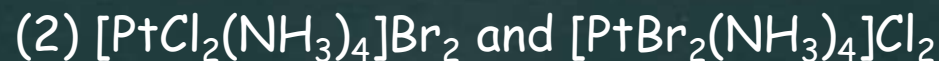
Example

Ambi

Which of the following pairs represents linkage isomers ?



[AIEEE-2009]



Solution

Ans. (4)



Example

Match List-I with List-II :

[JEE Mains-2021]

List-I

- (a) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$
- (b) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
- (c) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (d) $\text{cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$

List-II

- (i) Linkage isomerism
- (ii) Solvate isomerism
- (iii) Co-ordination isomerism
- (iv) Optical isomerism

Choose the correct answer from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
- (3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Solution

Ans. (1)

Concept :- 42

Topic Name :- Stereoisomerism

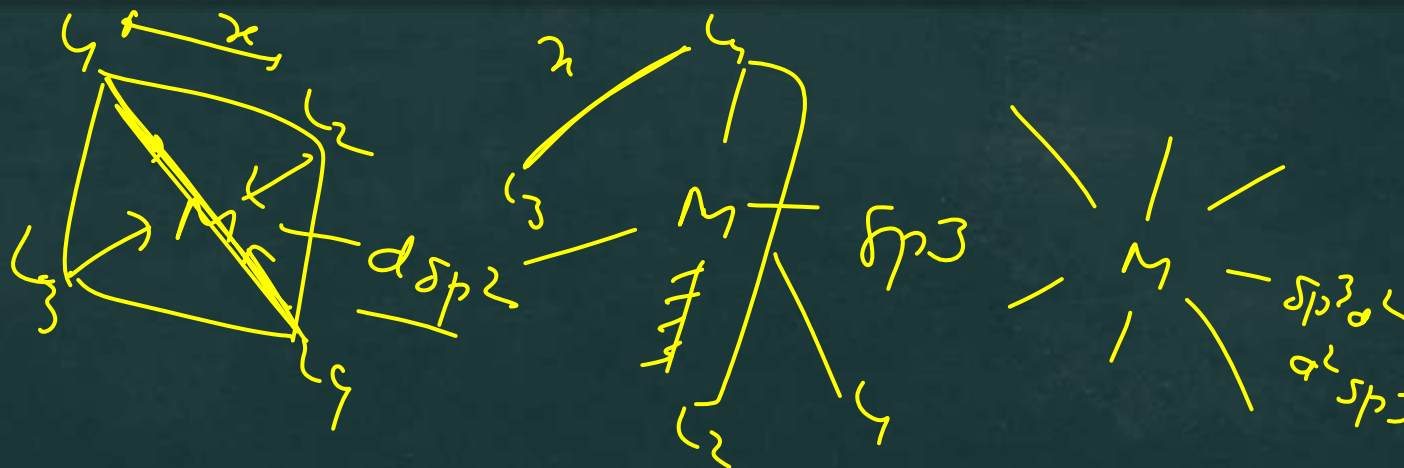


Co-ordination compounds which are having same molecular formula but different orientation of ligands around central atom are known as stereoisomers and this phenomenon is known as stereo-isomerism.

Concept :- 42

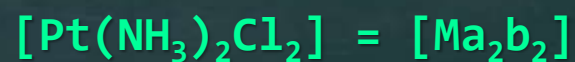
Topic Name :- Stereoisomerism

	CN	G.I.	O.I.
Square Planar	4 (dsp^2)	✓	✗ ✗
Tetrahedral	4 (sp^3)	✗	✓ ✓
Octahedral	6 (sp^3d^2/d^2sp^3)	✓	✓

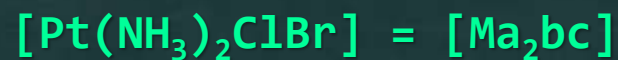
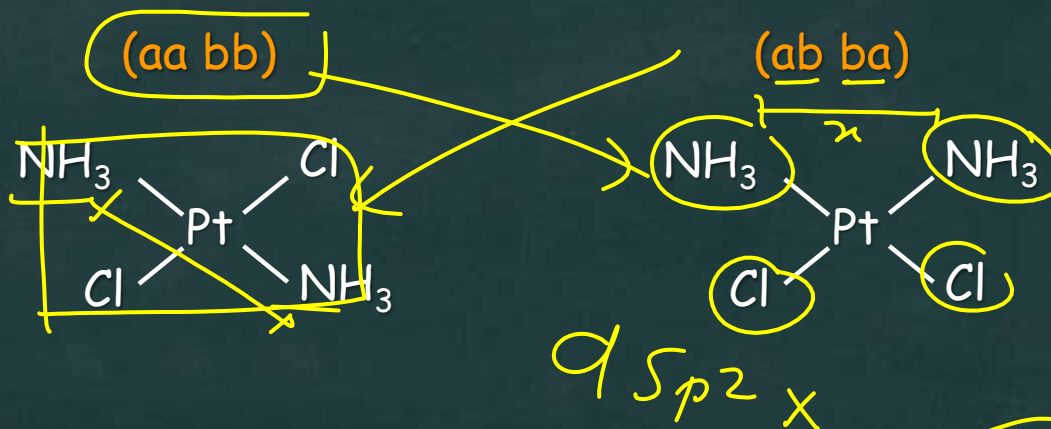


Concept :- 43

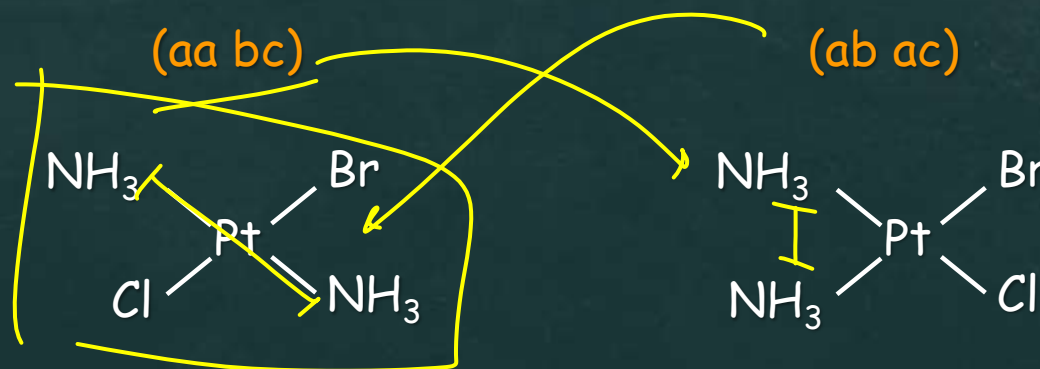
Topic Name :- Stereoisomerism for square planar



a a b b

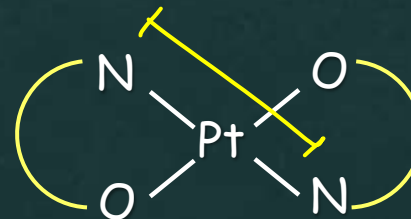
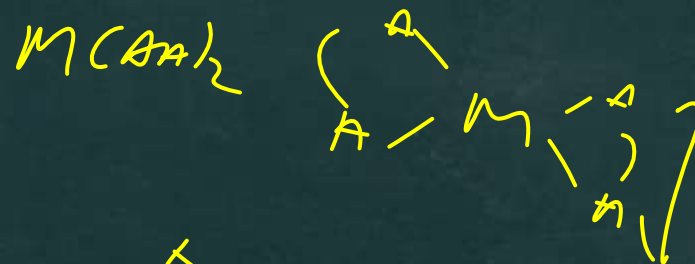
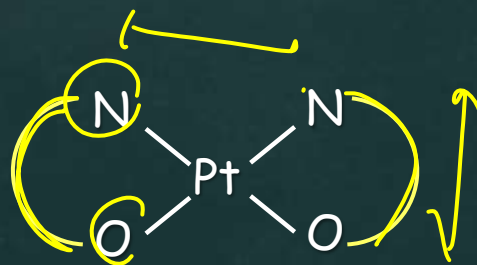
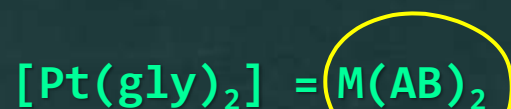
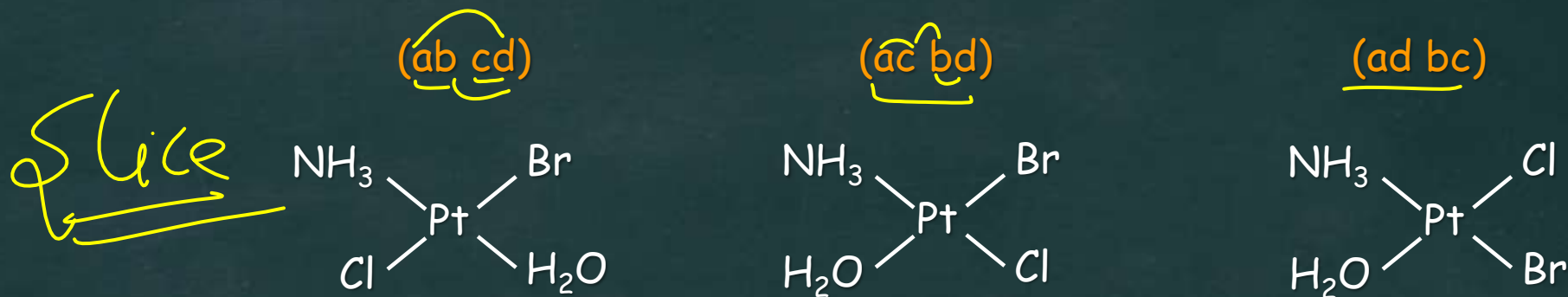
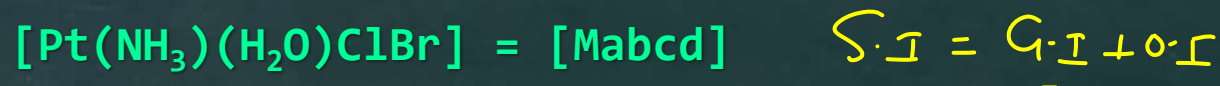


a a b c



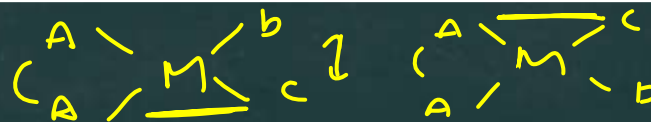
Concept :- 43

Topic Name :- Stereoisomerism for square planar



Concept :- 43

Topic Name :- Stereoisomerism for square planar



Sr. No.	Complexes	Geometrical isomers	Stereo-isomers	Optically Inactive	Optically Active	E.P.
1.	$[Ma_4]$	Not Defined	Not Defined	1	0	0
2.	$[Ma_3b]$	Not Defined	Not Defined	1	0	0
3.	$[Ma_2b_2]$	2	2	2	0	0
4.	$[Ma_2bc]$	2	2	2	0	0
5.	$[Mabcd]$	3	3	3	0	0
6.	$[M(AA)_2]$	Not Defined	Not Defined	1	0	0
7.	$[M(AA)b_2]$	Not Defined	Not Defined	1	0	0
8.	$[M(AA)bc]$	Not Defined	Not Defined	1	0	0

 Example

The number of geometrical isomers that can exist for square planar $[\text{Pt}(\text{Cl})(\text{py})(\text{NH}_3)(\text{NH}_2\text{OH})]^\oplus$ is (py = pyridine) :

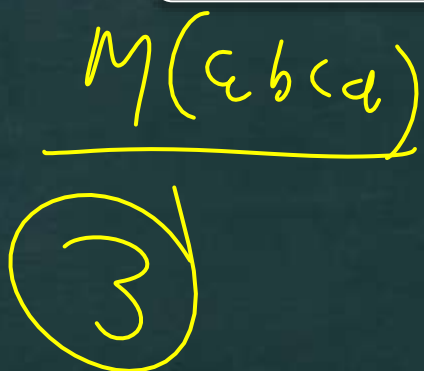
(1) 4

(2) 6

(3) 2

(4) 3

Solution



Ans. (4)

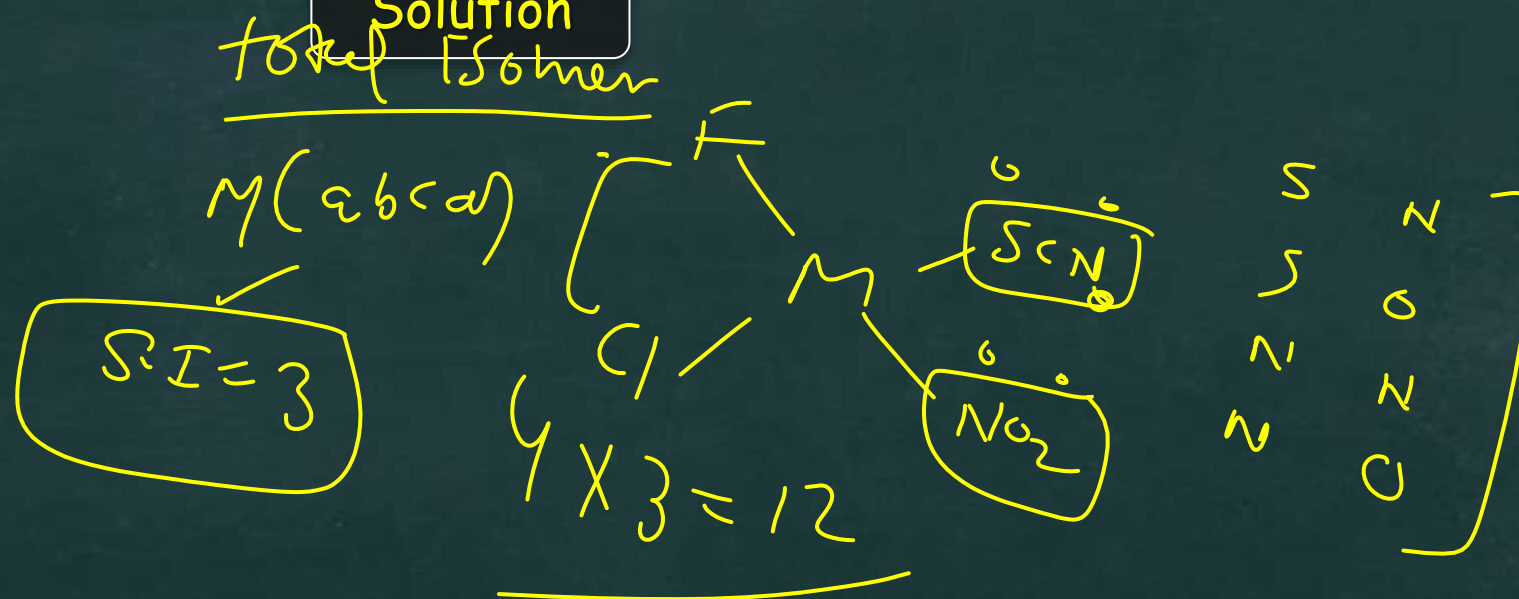
 Example

The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is :

- (1) 12 (2) 8 (3) 16 (4) 4

[JEE Mains-2019]

Solution



Ans. (1)

Concept :- 44

Topic Name :- Stereoisomerism for Tetrahedral

(i) $[Ma_4]$

(ii) $[Ma_3b]$

(iii) $[Ma_2b_2]$

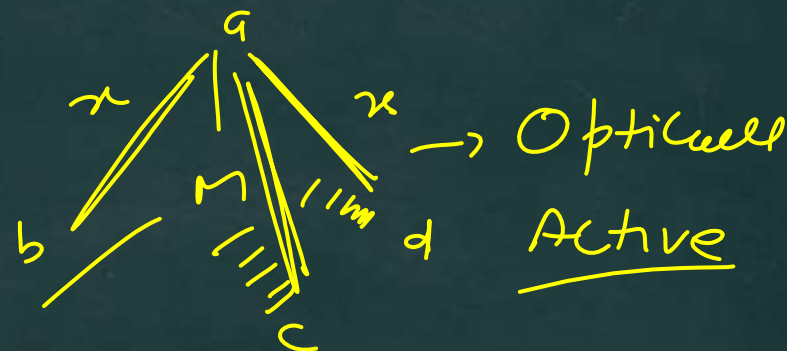
(iv) $[Ma_2bc]$

(v) $[M(AA)_2]$

(vi) $[M(AA)b_2]$

(vii) $[M(AA)bc]$

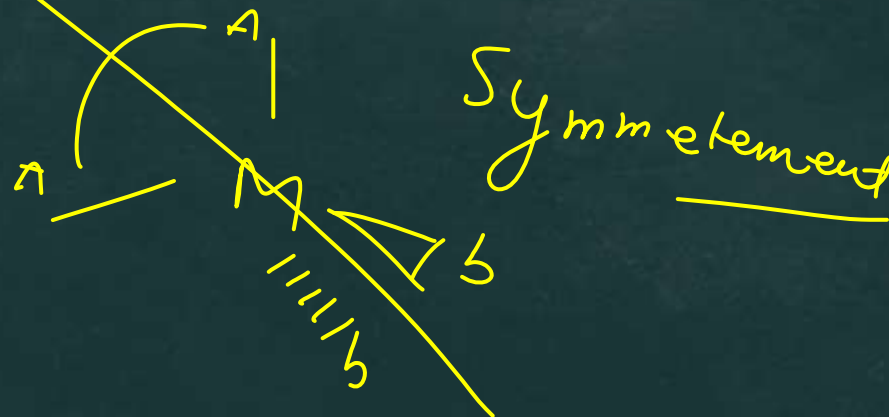
(viii) $[M(AB)c_2]$



Geometrical Isomerism = Not Defined

All are optically inactive

Stereoisomerism = not defined



Concept :- 44

Topic Name :- Stereoisomerism for Tetrahedral

Optically Active

(ix) [Mabcd]

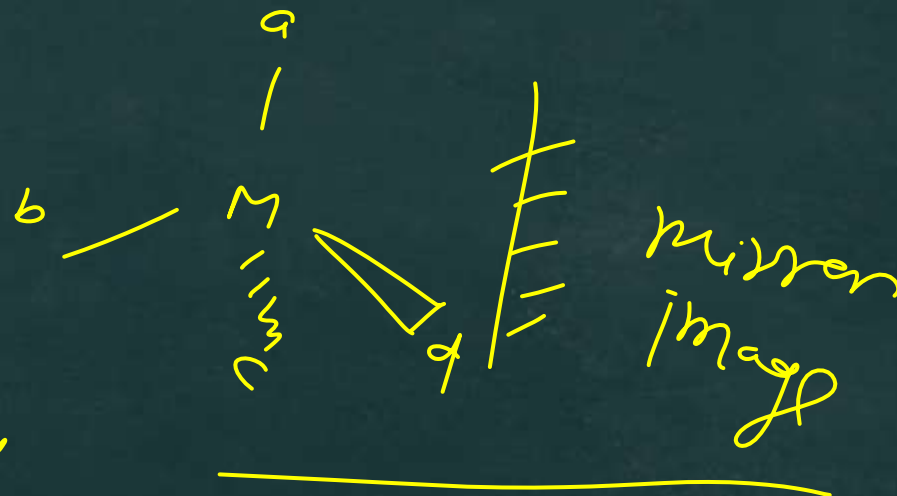
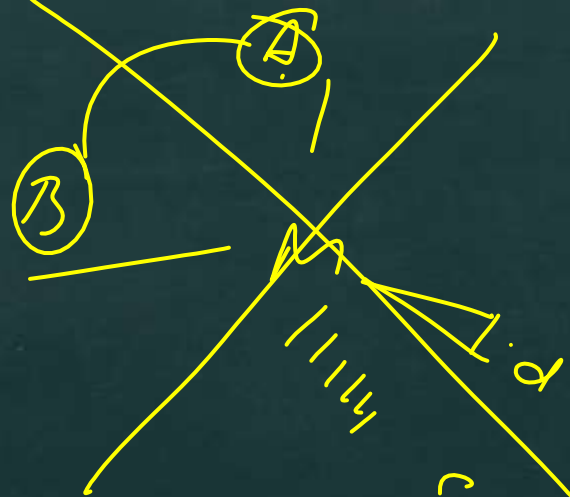
(x) [M(AB)₂]

(xi) [M(AB)cd]

Geometrical Isomerism = Not Defined

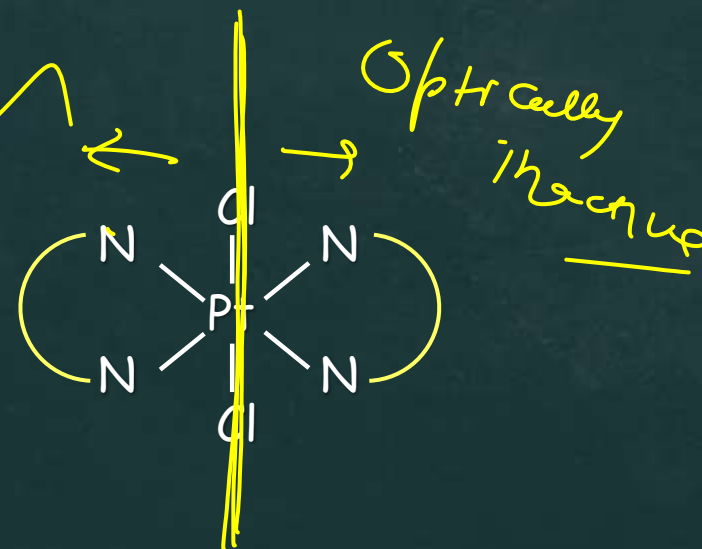
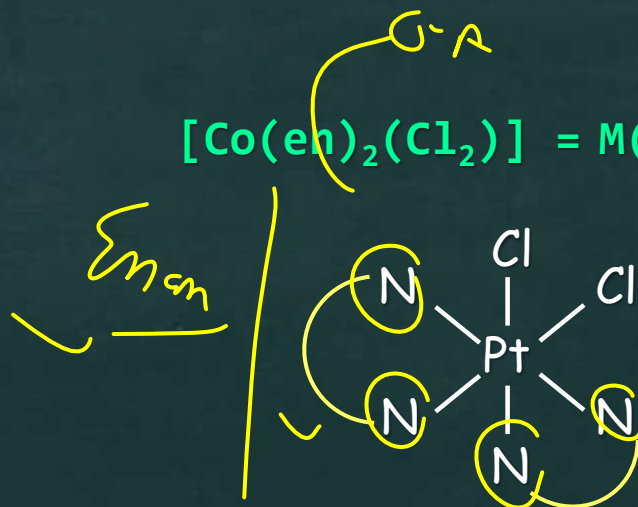
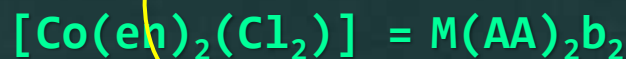
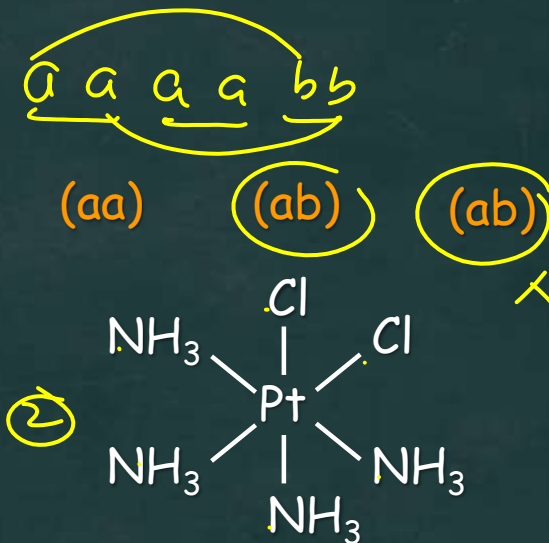
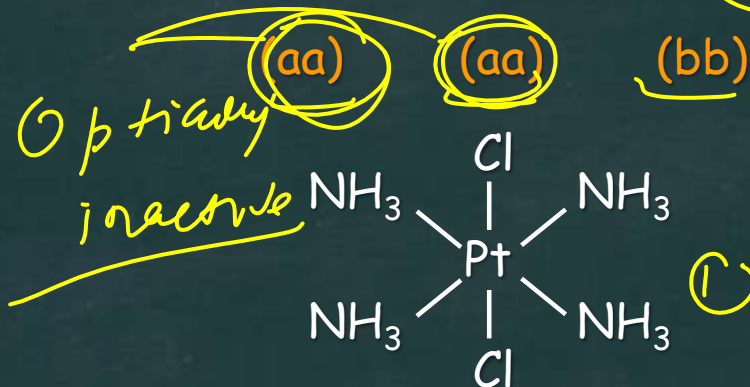
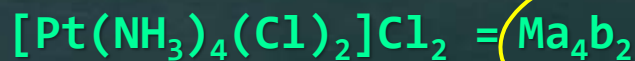
All are optically active

Stereoisomers = 2



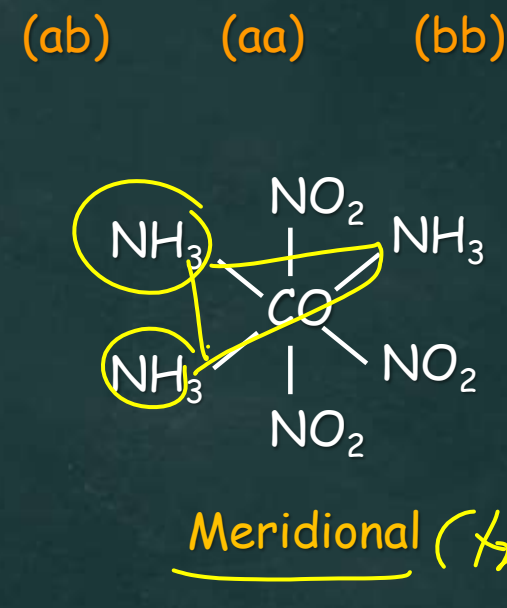
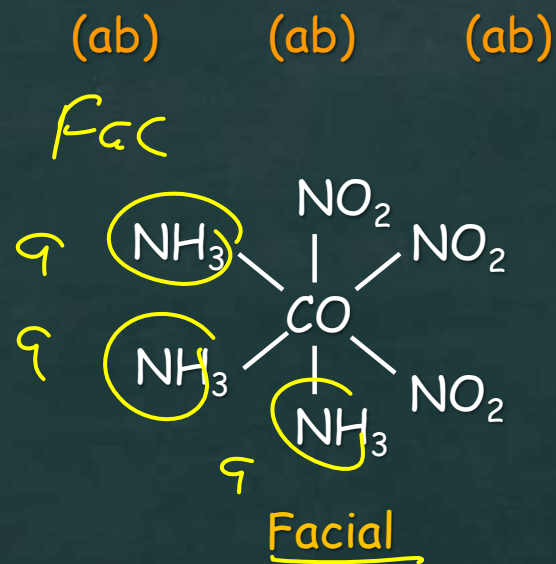
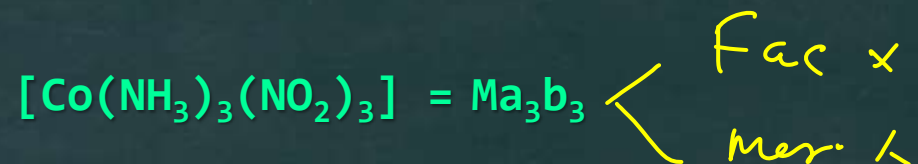
Concept :- 45

Topic Name :- Stereoisomerism for Octahedral



Concept :- 45

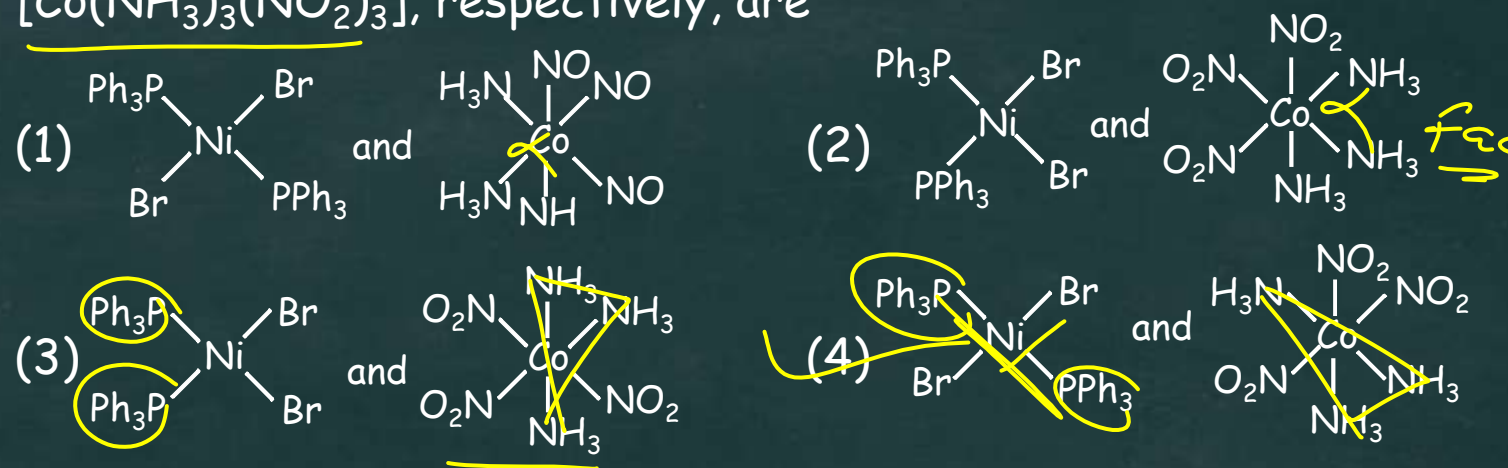
Topic Name :- Stereoisomerism for Octahedral





Example

The correct structures of $\text{trans-[NiBr}_2(\text{PPh}_3)_2]$ and meridional- $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$, respectively, are



[JEE Mains-2021]

Solution

Ans. (4)

Concept :- 45

Topic Name :- Stereoisomerism for Octahedral

[A] For Monodentate Ligands

Sr. No.	Complexes	Geometrical isomers	Stereo-isomers	Optically Inactive	Optically Active	E.P.
1.	$[Ma_6]$	Not Defined	Not Defined	1	0	0
2.	$[Ma_5b]$	Not Defined	Not Defined	1	0	0
3.	$[Ma_4b_2]$	2	2	2	0	0
4.	$[Ma_4bc]$	2	2	2	0	0
5.	$[Ma_3b_3]$	2	2	2	0	0
6.	$[Ma_3b_2c]$	3	3	3	0	0
7.	$[Ma_3bcd]$	4	5	3	2	1
8.	$[Ma_2b_2c_2]$	5	6	4	2	1

Concept :- 45

Topic Name :- Stereoisomerism for Octahedral

Sr. No.	Complexes	Geometrical isomers	Stereo-isomers	Optically Inactive	Optically Active	E.P.
9.	$[Ma_2b_2cd]$	6	8	4	4	2
10.	$[Ma_2bcde]$	9	15	3	12	6
11.	$[Mabcdef]$	15	30	0	30	15

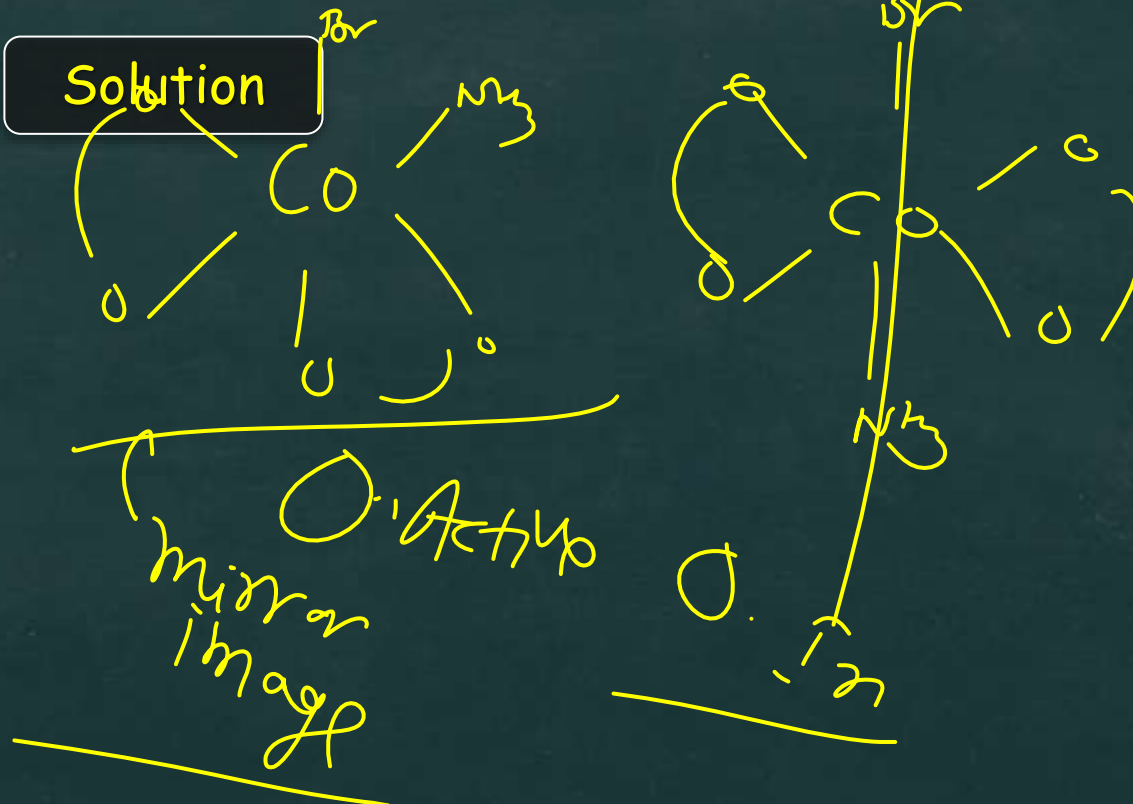
$Mabcdef$ or
 $5+4+3+2+1$

 Example

The number of stereoisomers possible for $[\text{Co}(\text{ox})_2(\text{Br})(\text{NH}_3)]^{2-}$ is
_____. [ox = oxalate]

[JEE Mains-2021]

Solution

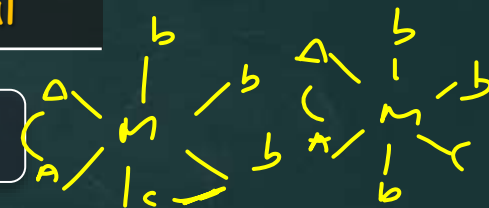


Ans. (3)

Concept :- 45

Topic Name :- Stereoisomerism for Octahedral

[B] For Symmetrical Bidentate Ligands



Sr. No.	Complexes	Geometrical isomers	Stereo-isomers	Optically Inactive	Optically Active	E.P.
1.	$[M(AA)_3]$	Not Defined	2	0	2	1
2.	$[M(AA)_2 b_2]$	2	3	1	2	1
3.	$[M(AA)_2 bc]$	2	3	1	2	1
4.	$[M(AA)b_4]$	Not Defined	Not Defined	1	0	0
5.	$[M(AA)b_3c]$	2	2	2	0	0
6.	$[M(AA)b_2c_2]$	3	4	2	2	1
7.	$[M(AA)b_2cd]$	4	6	2	4	2
8.	$[M(AA)bcde]$	6	12	0	12	6

Concept :- 45

Topic Name :- Stereoisomerism for Octahedral

[C] For Unsymmetrical Bidentate Ligands

Sr. No.	Complexes	Geometrical isomers	Stereo-isomers	Optically Inactive	Optically Active	E.P.
1.	$[M(AB)_3]$	2	4	0	4	2
2.	$[M(AB)_2C_2]$	5	8	2	6	3
3.	$[M(AB)_2cd]$	6	11	1	10	5
4.	$[M(AB)c_4]$	Not Defined	Not Defined	1	0	0
5.	$[M(AB)c_3d]$	3	4	2	2	1
6.	$[M(AB)c_2d_2]$	4	6	2	4	2
7.	$[M(AB)c_2de]$	7	12	2	10	5
8.	$[M(AB)cdef]$	12	24	0	24	12

Concept :- 46

Topic Name :- Bonding in Coordination Compounds



There are two theories to explain the nature of bonding in transition metal complexes.

1. Valence Bond Theory (VBT)

2. Crystal Field Theory (CFT)

WFL $\rightarrow$ pairing $\times$
SFL $\rightarrow$ pairing
if seqⁿ

Ligands strength order :-

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-}$

$< NH_3 < en < CN^- < CO$

SFL

WFL

Concept :- 46

Topic Name :- Bonding in Coordination Compounds

Octahedral Complexes



Example

Comment on hybridisation, geometry, and magnetic behaviour of the given complex : $[\text{Fe}(\text{CN})_6]^{4-}$

Solution

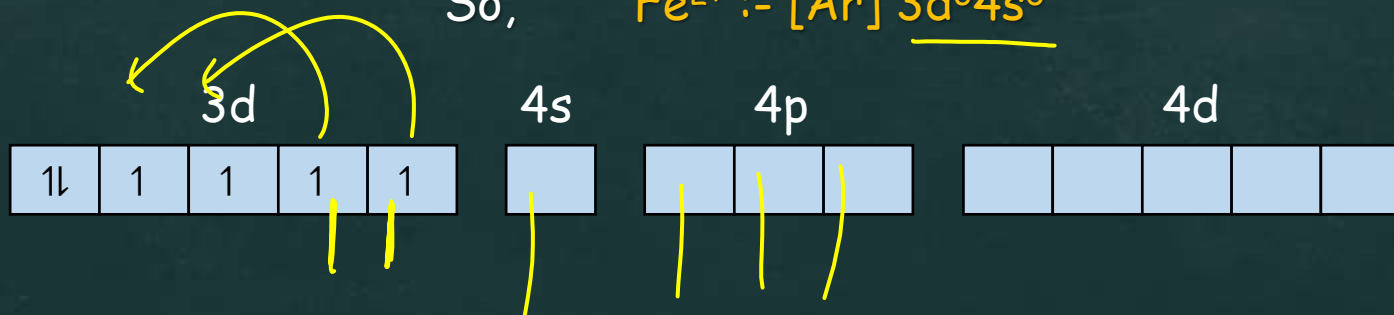
CN^- - SFL - Strong field ligand - inner d-orbitals

(CN^- is a monodentate ligand and strong field ligand)

Electronic configuration of



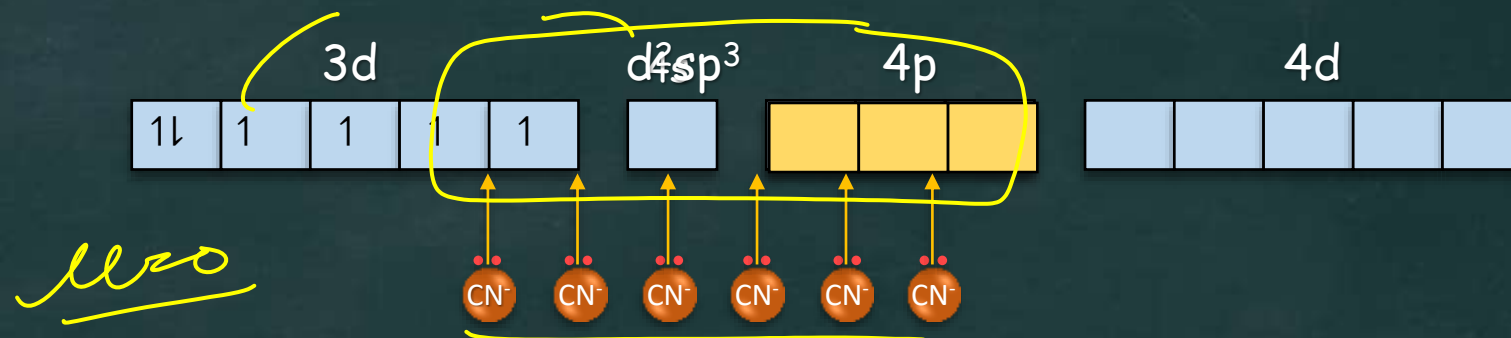
So,



Concept :- 46

Topic Name :- Bonding in Coordination Compounds

After pairing due to presence of strong field ligand (CN^-)



- Hybridisation (Fe) : d^2sp^3 *low spin*
- Geometry : Octahedral
- Unpaired electrons = 0
 - Diamagnetic
 - $\mu = \sqrt{n(n+2)} = \sqrt{0} = 0 \text{ B.M}$
- Inner orbital complex

Concept :- 46

Topic Name :- Bonding in Coordination Compounds



Example

Cl → w f l

Comment on hybridisation, geometry, and magnetic behaviour of the given complex : $[\text{FeCl}_6]^{4-}$

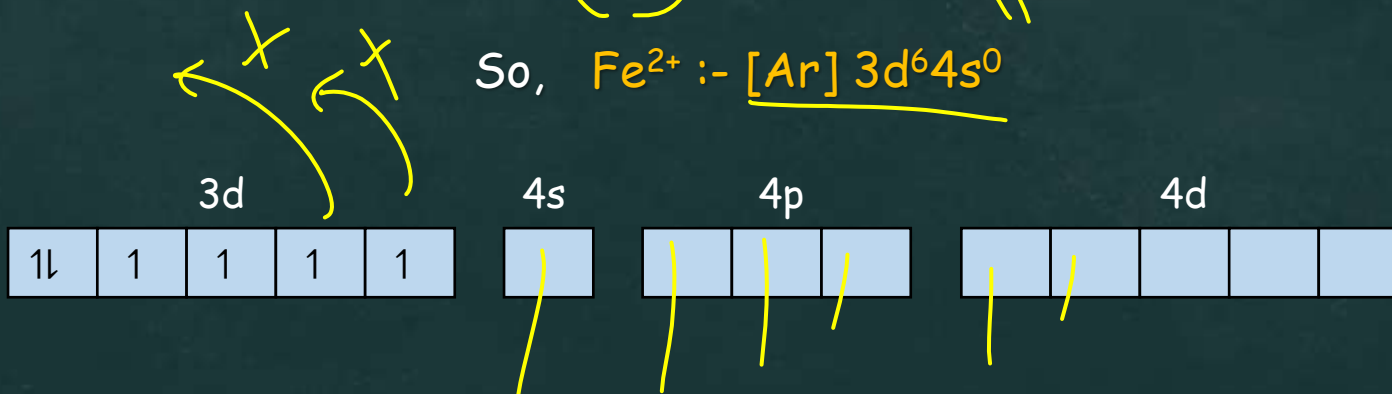
Solution

+2 6 Cl

(Cl⁻ is a monodentate ligand and weak field ligand)

Electronic configuration of $_{26}\text{Fe}$:- $[\text{Ar}] 3d^6 4s^2$

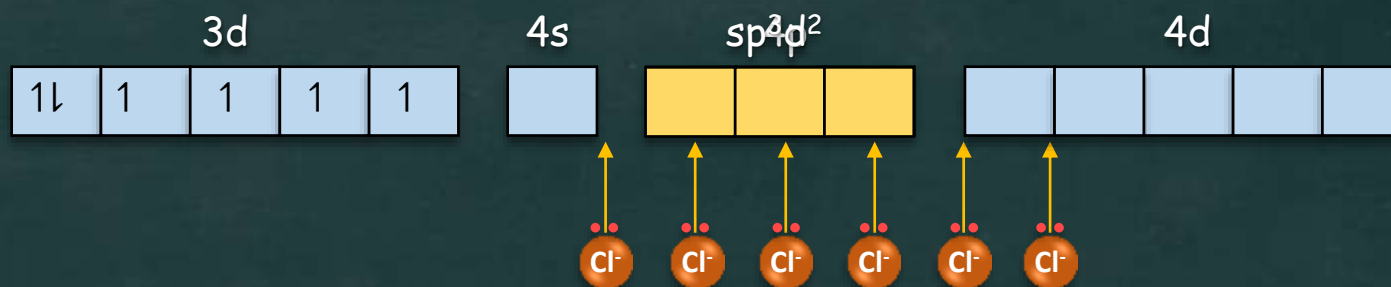
So, Fe^{2+} :- $[\text{Ar}] 3d^6 4s^0$



Concept :- 46

Topic Name :- Bonding in Coordination Compounds

No pairing occurs due to presence of weak field ligand (Cl^-)



$$Unpaired = 4$$

$$\mu = \sqrt{n(n+2)}$$

$$= \sqrt{4(4+2)} \text{ B.M.}$$

Paramagnetic

Concept :- 46

Topic Name :- Bonding in Coordination Compounds

Tetrahedral Complexes



Example

Sq planar
 dsp^2 $\leftarrow$ (sp^3)

Comment on hybridisation, geometry, and magnetic behaviour of the given complex : $[\text{NiCl}_4]^{2-}$

Solution

Ni^{+2}

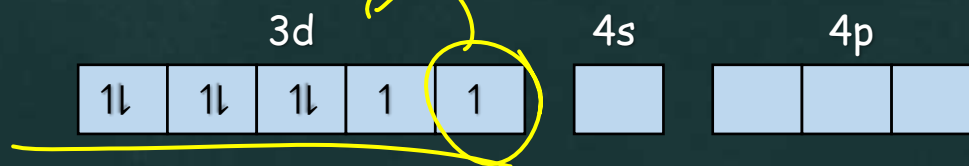
(Cl^- is a monodentate ligand and weak field ligand)

Electronic configuration of

$_{28}\text{Ni} :- [\text{Ar}] 3d^8 4s^2$

$_{28}\text{Ni}^{2+} :- [\text{Ar}] 3d^8 4s^0$

So,

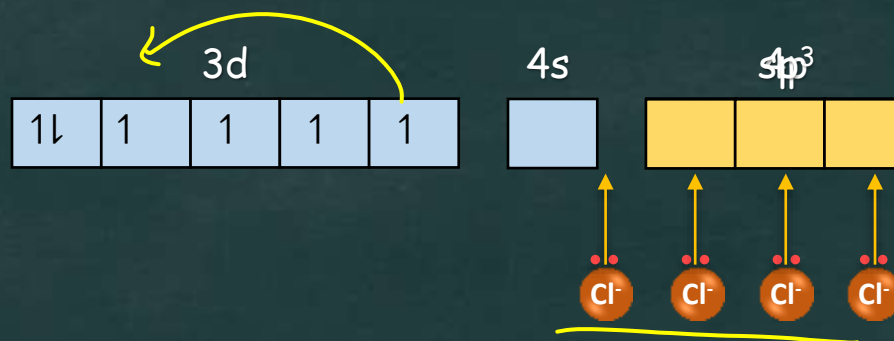


Concept :- 46

Topic Name :- Bonding in Coordination Compounds

SF L

No pairing occurs due to presence of weak field ligand (Cl^-)



- Hybridisation (Ni) : sp^3
- Geometry : Tetrahedral
- Unpaired electrons = 4
 - Paramagnetic
 - $\mu = \sqrt{n(n+2)} = \sqrt{24} = 4.89 \text{ B.M}$

Concept :- 46

Topic Name :- Bonding in Coordination Compounds

Square Planar Complexes

 Example

Comment on hybridisation, geometry, and magnetic behaviour of the given complex : $[\text{Ni}(\text{CN})_4]^{2-}$

Solution

$\text{CN}^- \rightarrow \text{strong field}$ sp^2

(CN^- is a monodentate ligand and strong field ligand)

Electronic configuration of $_{28}\text{Ni} :- [\text{Ar}] 3d^8 4s^2$

$\text{Ni}^{2+} :- [\text{Ar}] 3d^8 4s^0$

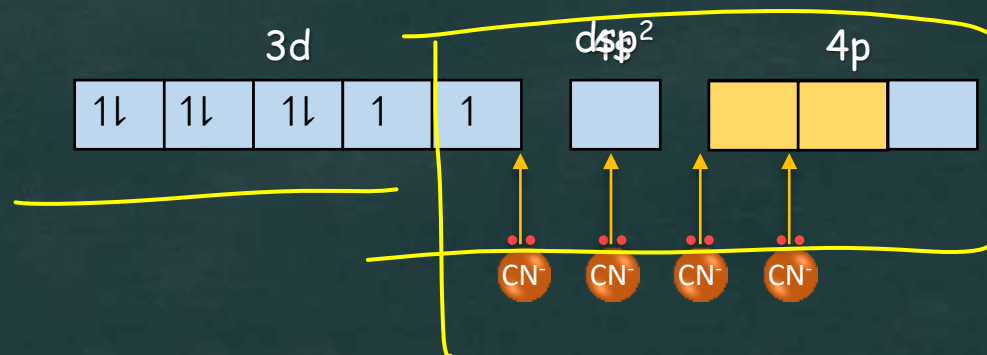
So,



Concept :- 46

Topic Name :- Bonding in Coordination Compounds

After pairing due to presence of strong field ligand (CN^-)



- Hybridisation (Ni) : dsp^2
- Geometry : Square planar
- Unpaired electrons = 0
 - Diamagnetic
 - $\mu = \sqrt{n(n+2)} = 0 = 0 \text{ B.M}$

 Example

[JEE Mains-2019]

The correct statements among I to III are :

✓ (I) Valence bond theory cannot explain the color exhibited by transition metal complexes.

(II) Valence bond theory can predict quantitatively the magnetic properties of transition metal complexes.

✓ (III) Valence bond theory cannot distinguish ligands as weak and strong field ones.

(1) (I) and (II) only

(2) (I), (II) and (III)

(3) (I) and (III) only

(4) (II) and (III) only

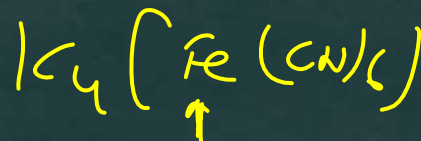
Solution

Ans. (3)

Concept :- 47

Topic Name :- Crystal Field Theory

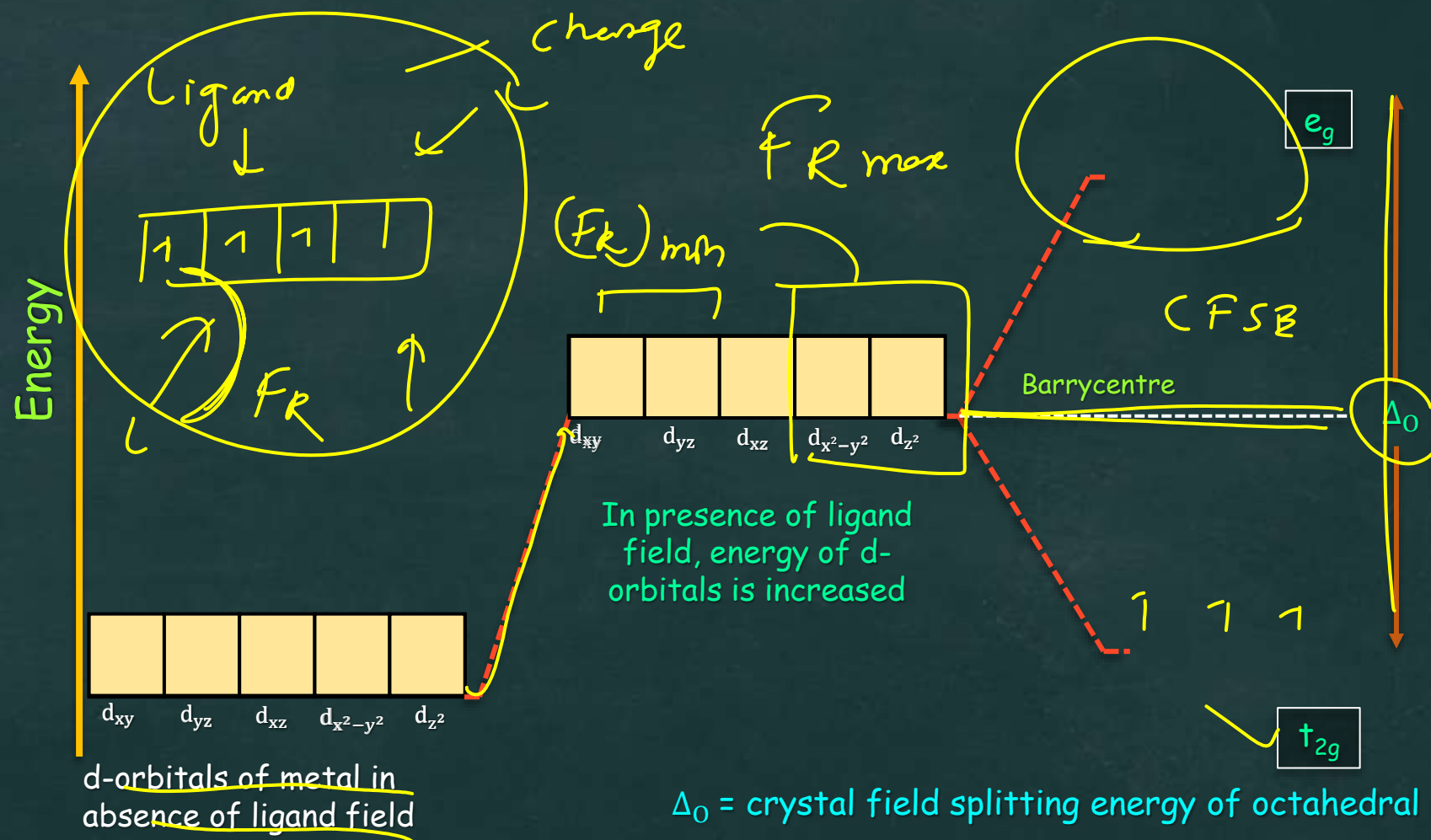
Assumptions



- 🧪 Ionic ligands such as Cl^- , OH^- , CN^- are regarded as negative point charges (or simply point charges) and the neutral ligands such as H_2O , NH_3 , py are regarded as point dipole
- 🧪 Central metal atom and ligands are treated as point charges.
- 🧪 There is purely ionic interaction between central metal atom and ligands

Concept :- 48

Topic Name :- CFT for Octahedral



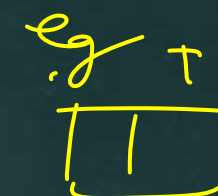
Concept :- 48

Topic Name :- CFT for Octahedral

Electron filling sequence in splitting diagram

d^n configuration	SFL ($\Delta_o > P$)	WFL ($\Delta_o < P$)
d^1	$t_{2g}^1 e_g^0$	$t_{2g}^1 e_g^0$
d^2	$t_{2g}^2 e_g^0$	$t_{2g}^2 e_g^0$
d^3	$t_{2g}^3 e_g^0$	$t_{2g}^3 e_g^0$
d^4	$t_{2g}^4 e_g^0$	$t_{2g}^3 e_g^1$
d^5	$t_{2g}^5 e_g^0$	$t_{2g}^3 e_g^2$
d^6	$t_{2g}^6 e_g^0$	$t_{2g}^4 e_g^2$
d^7	$t_{2g}^6 e_g^1$	$t_{2g}^5 e_g^2$
d^8	$t_{2g}^6 e_g^2$	$t_{2g}^6 e_g^2$
d^9	$t_{2g}^6 e_g^3$	$t_{2g}^6 e_g^3$
d^{10}	$t_{2g}^6 e_g^4$	$t_{2g}^6 e_g^4$

SFL
WFL



 Example

Comment on hybridization, number of unpaired electrons, outer or inner orbital complex for the given complex

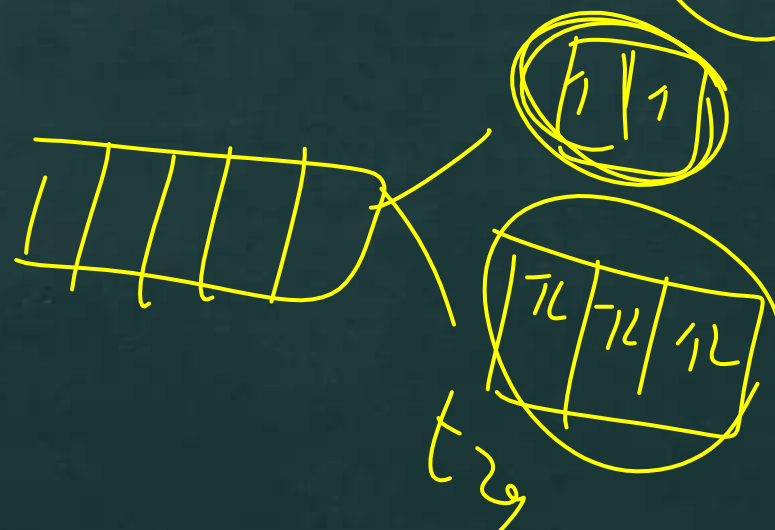


Solution

SFC

NH - $3d^8 4s^2$ 4p 4d

$\text{Ni}^{+2} \rightarrow 3d^8$



Ans.

- sp^3d^2 , Octahedral
- $n = 2$
- Outer orbital complex



Example

Spin only magnetic moment of an octahedral complex of Fe^{2+} in the presence of a strong field ligand in BM is : [JEE Mains-2021]

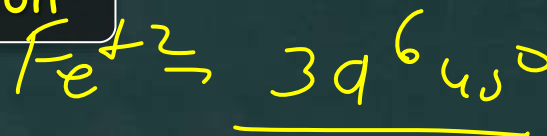
(1) 4.89

(2) 2.82

(3) 0

(4) 3.46

Solution

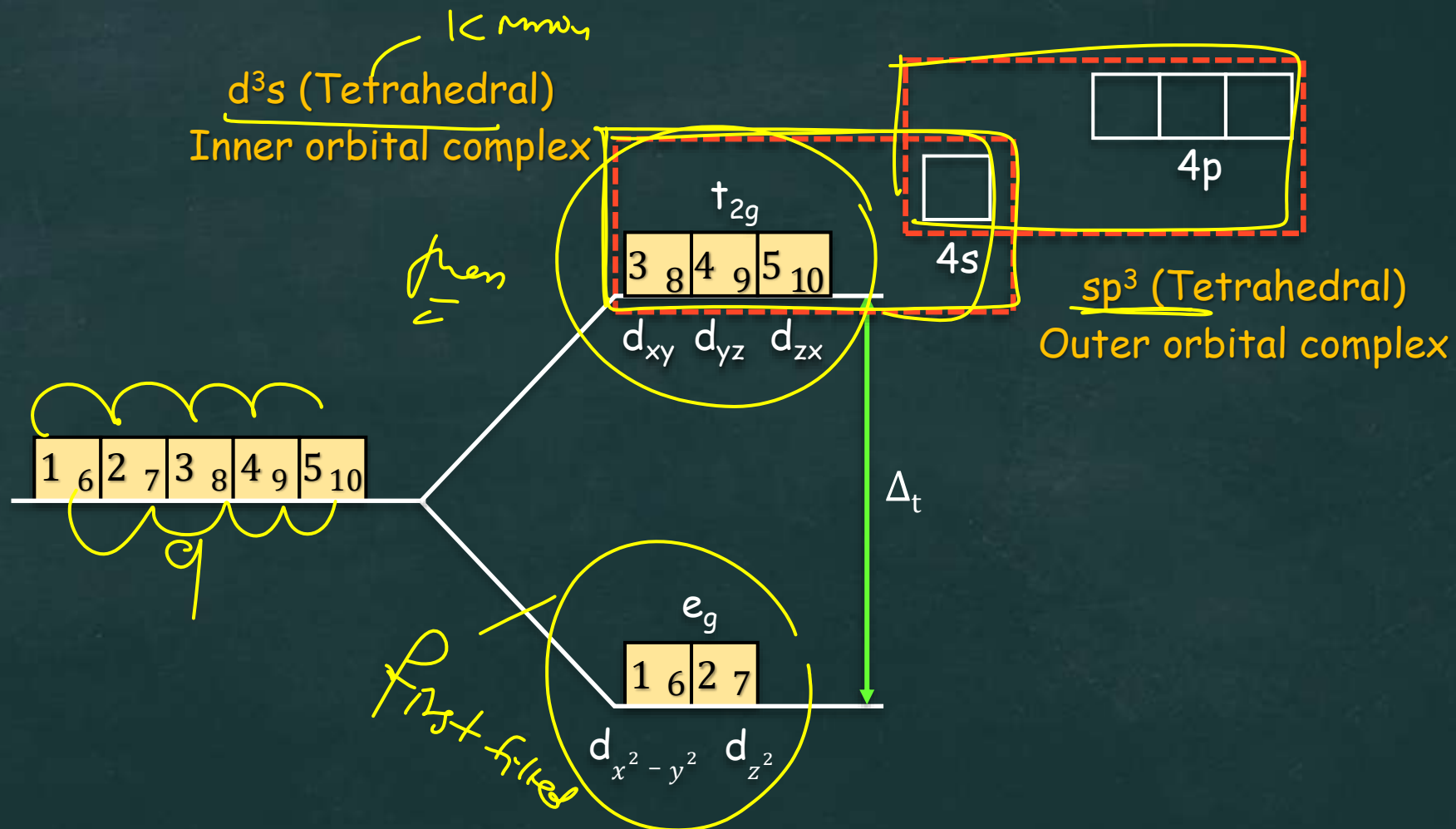


Ans. (3)

Concept :- 49

Topic Name :- CFT for Tetrahedral

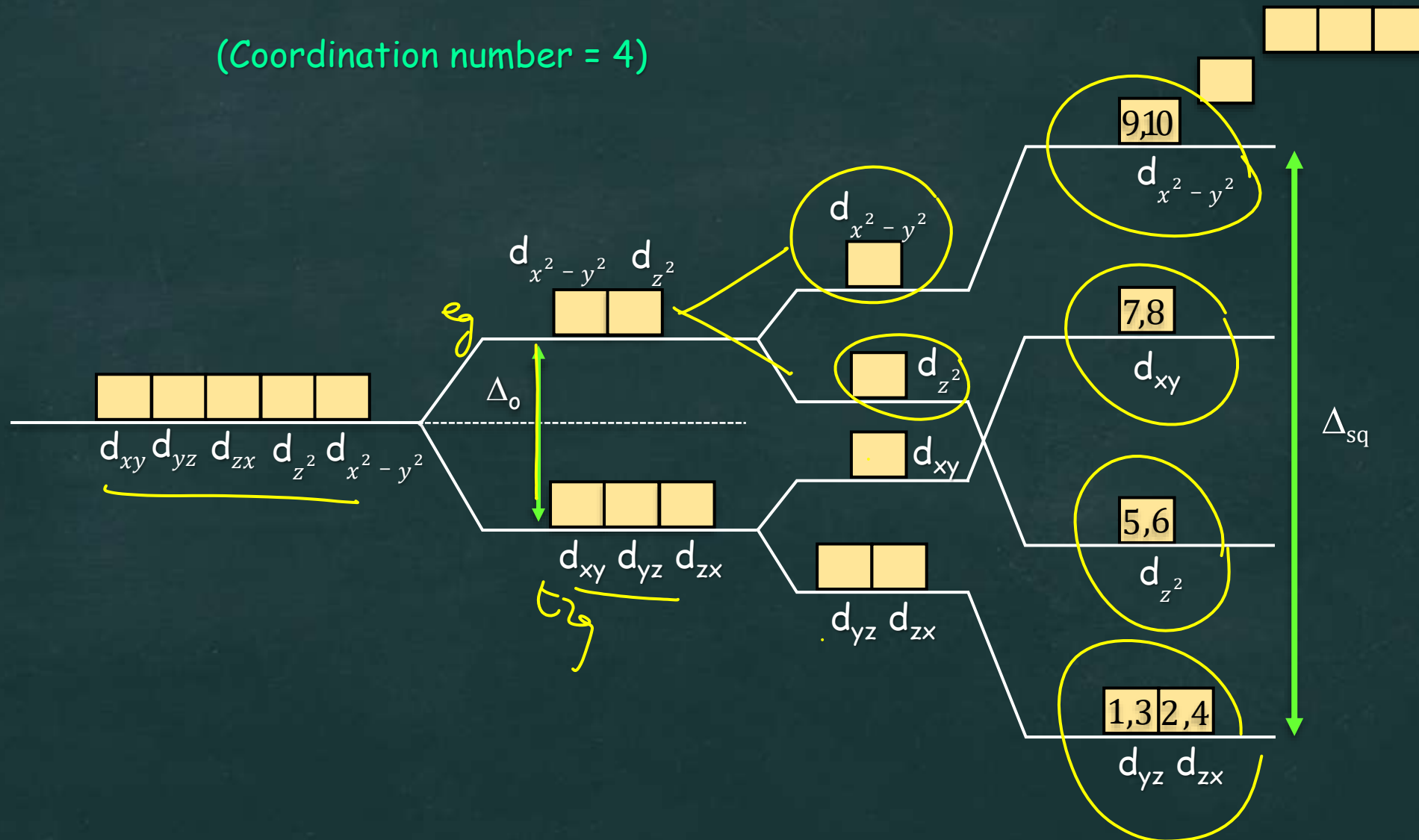
(Coordination number = 4)



Concept :- 50

Topic Name :- CFT for Square Planar

(Coordination number = 4)

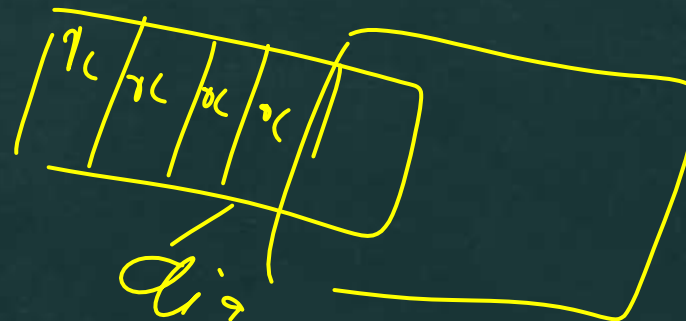
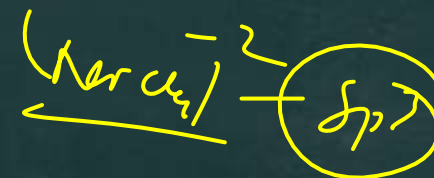
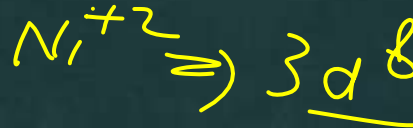


 Example

The correct combination is

- (1) $[\text{Ni}(\text{CN})_4]^{2-}$ - tetrahedral ; $[\text{Ni}(\text{CO})_4]$ - paramagnetic
- (2) $[\text{NiCl}_4]^{2-}$ - paramagnetic ; $[\text{Ni}(\text{CO})_4]$ - tetrahedral
- (3) $[\text{NiCl}_4]^{2-}$ - diamagnetic ; $[\text{Ni}(\text{CO})_4]$ - square-planar
- (4) $[\text{NiCl}_4]^{2-}$ - square-planar ; $[\text{Ni}(\text{CN})_4]^{2-}$ - paramagnetic

Solution



Ans: (2)

Concept :- 51

Topic Name :- Crystal Field Splitting Energy

Charge on metal ion

C.M $\rightarrow$ Chargeⁿ

CFSE $\uparrow \uparrow$

CFSE $\propto$ Charge on central atom with sign

Type of splitting

$$\Delta_{sq} > \Delta_o > \Delta_t$$

Note

Vibop

$$(i) \Delta_{sq} = 1.3 \Delta_o$$

$$(ii) \Delta_t = \frac{4}{9} \Delta_o$$

Concept :- 51

Topic Name :- Crystal Field Splitting Energy

Type of Metal Ion

Z_{eff}
5d - metal ion > 4d metal ion > 3d metal ion

REASON : Z_{eff} ↑ CFS E

In d - block, down the group, Z_{eff} increases due to increase in poor shielding electrons.

Type of Ligands

CFS E ↓ ↓

Strong field ligand > Weak field ligand

 Example

Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value.

Complexes : $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Co}(\text{en})_3]^{3+}$

A $+3$ B $+2$ C $+3$ D $+3$

Choose the correct option :

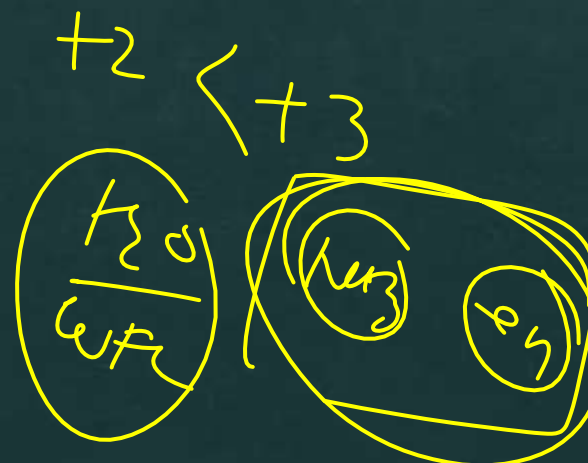
(1) $A < B < C < D$

(3) $B < C < D < A$

(2) $B < A < C < D$

(4) $C < D < B < A$

Solution

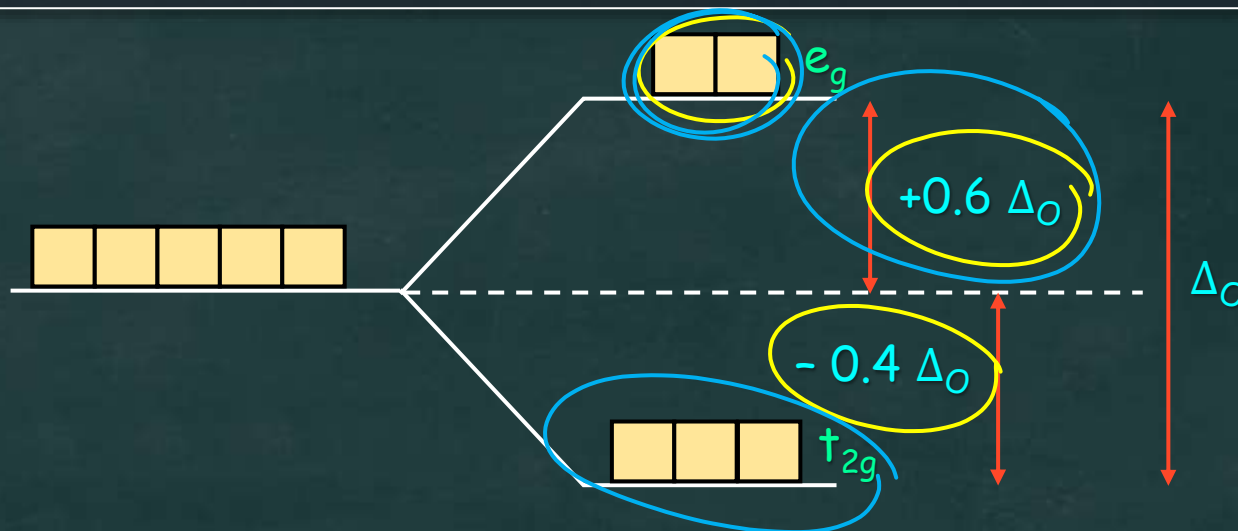


Ans: (2)

Concept :- 52

Topic Name :- Calculation of C.F.S.E

Calculation of crystal field stabilization energy for octahedral complexes



$$\boxed{CFSE = n_{t_{2g}} \times (-0.4 \Delta_o) + n_{e_g} \times (0.6 \Delta_o) + xP}$$

*Pairing
Energy*

where, $n_{t_{2g}} \rightarrow$ number of electrons in t_{2g}

$n_{e_g} \rightarrow$ number of electrons in e_g

$P \rightarrow$ Pairing energy

$x \rightarrow$ Number of electron pairs (forcefully)

 Example

Calculate the C.F.S.E. value for $[\text{CoF}_6]^{3-}$

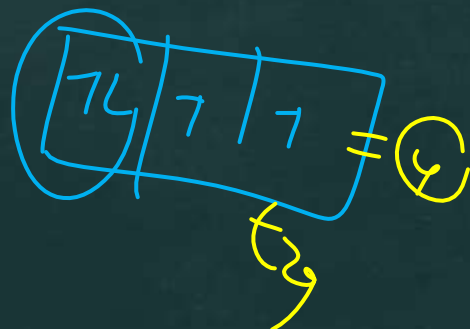
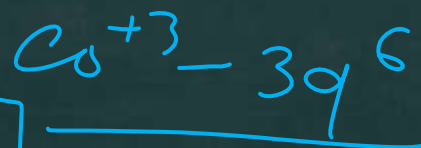
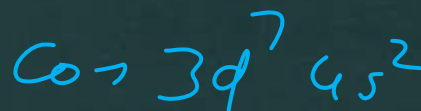
(A) $-2.4 \Delta_0 + 3P$

(B) $-0.4 \Delta_0 + P$

(C) $-0.4 \Delta_0$

(D) $-2.4 \Delta_0 + 2P$

Solution



$$\begin{aligned} \text{C.F.S.E.} &= 4 \times (-0.4 \Delta_0) \\ &\quad + 2 \times 0.6 \Delta_0 + 0P \\ &= -1.6 \Delta_0 + 1.2 \Delta_0 \\ &= -0.4 \Delta_0 \end{aligned}$$

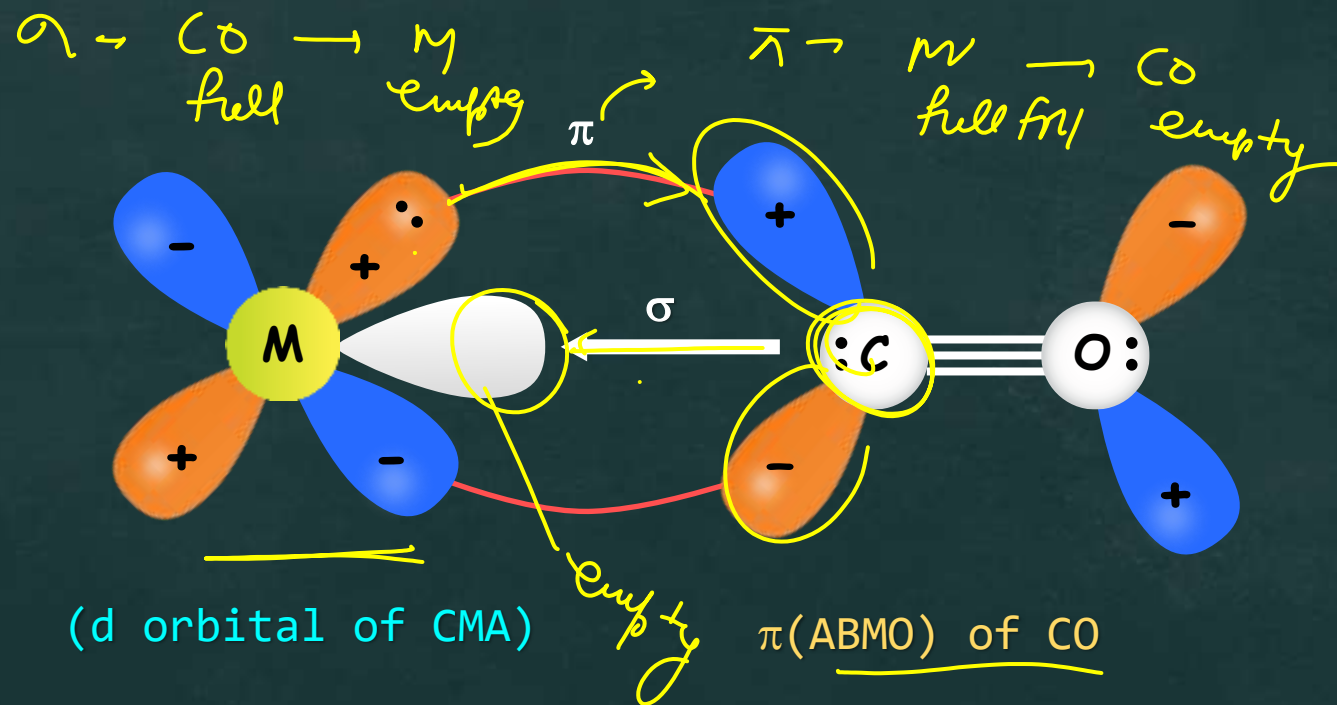
Ans: (C)

Concept :- 53

Topic Name :- Synergic Bonding



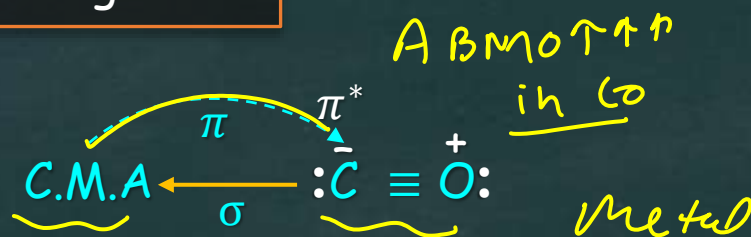
π - back bond from metal to vacant orbital of CO makes CO electron rich or makes it better lone pair donor. Similarly σ - donation from CO to metal makes metal electron rich so metal's lone pair donating tendency to CO increases. (synergising effect)



Concept :- 53

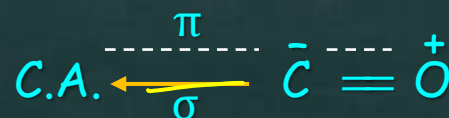
Topic Name :- Synergic Bonding

During S.B. :



$\pi^* \text{ABMO} \uparrow \uparrow$

After S.B. :



Bond Order $\uparrow$

Bond Length $\downarrow$

Bond Strength $\uparrow$

Bond Order $\downarrow$

Bond Length $\uparrow$

Bond Strength $\downarrow$

Carbonyl
 $\downarrow$
 BO $\downarrow$
 BL $\uparrow$
 BS $\downarrow$

Metal
 BO $\uparrow$
 Legend
 BO $\downarrow$

Concept :- 53

Topic Name :- Synergic Bonding

Bonding in Carbonyl Compounds (Synergic Bonding)



When negative charges on Central metal atom increases then tendency of CMA to donate electron pair increases. Thus extent of synergic bonding increases.

(M—C) :- Bond order (↑)

Bond strength (↑)

Bond length (↓)

(C—O) :- Bond order (↓)

Bond length (↑)

Bond strength (↓)

Concept :- 53

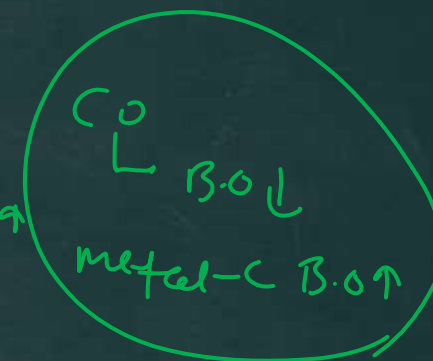
Topic Name :- Synergic Bonding

Example



Extent of synergic bonding (↑)

$e \uparrow \uparrow \uparrow$



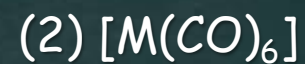
🧪 When we move from left to right electron density on CMA (↑)

🧪 Back donation (↑)

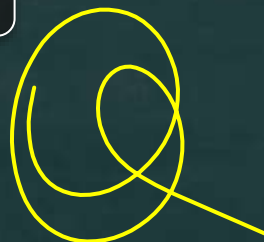
🧪 Extent of synergic bonding (↑)

 Example

In the following complexes, the maximum synergic bonding takes place in



Solution



Ans. (4)